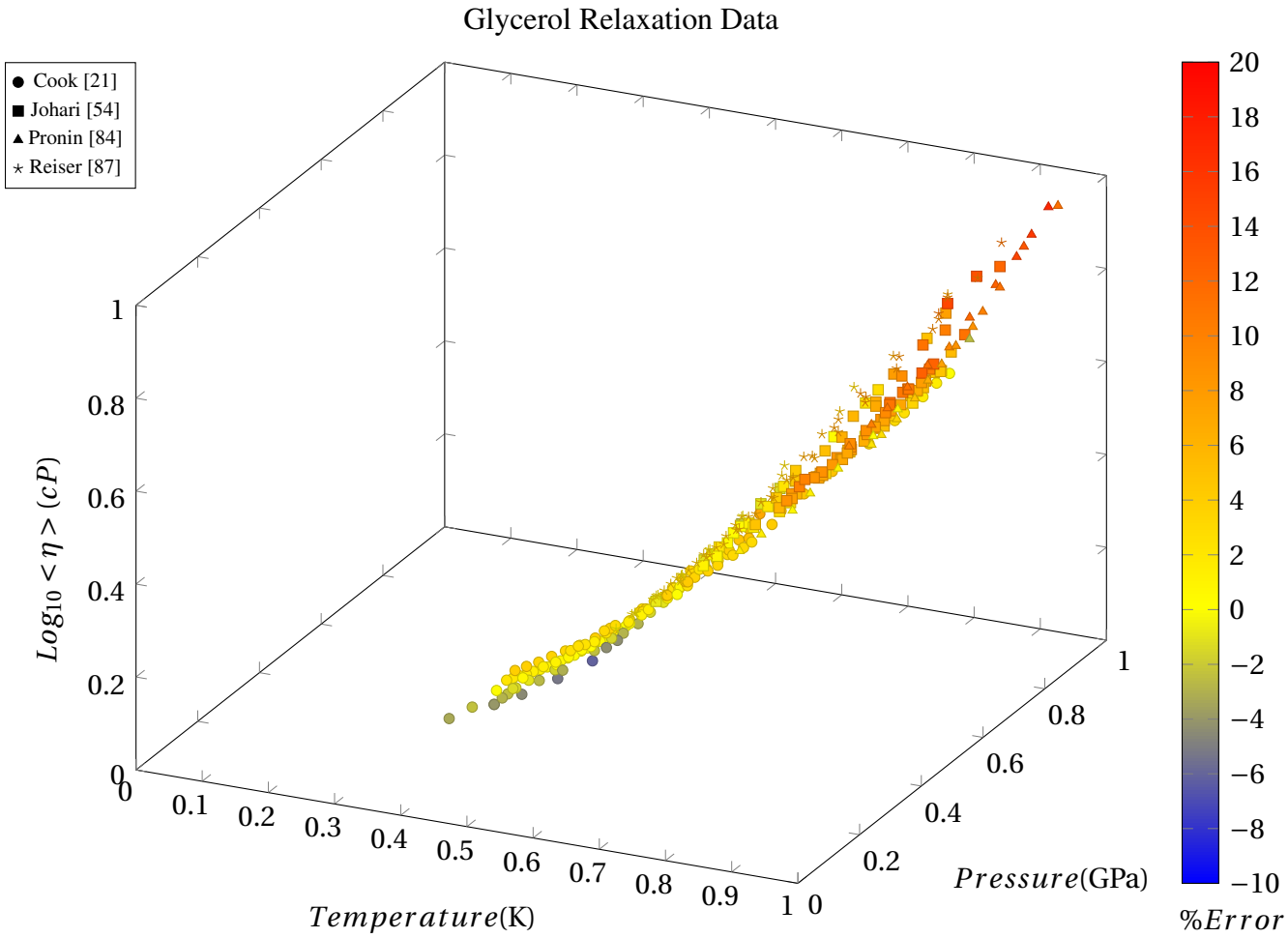


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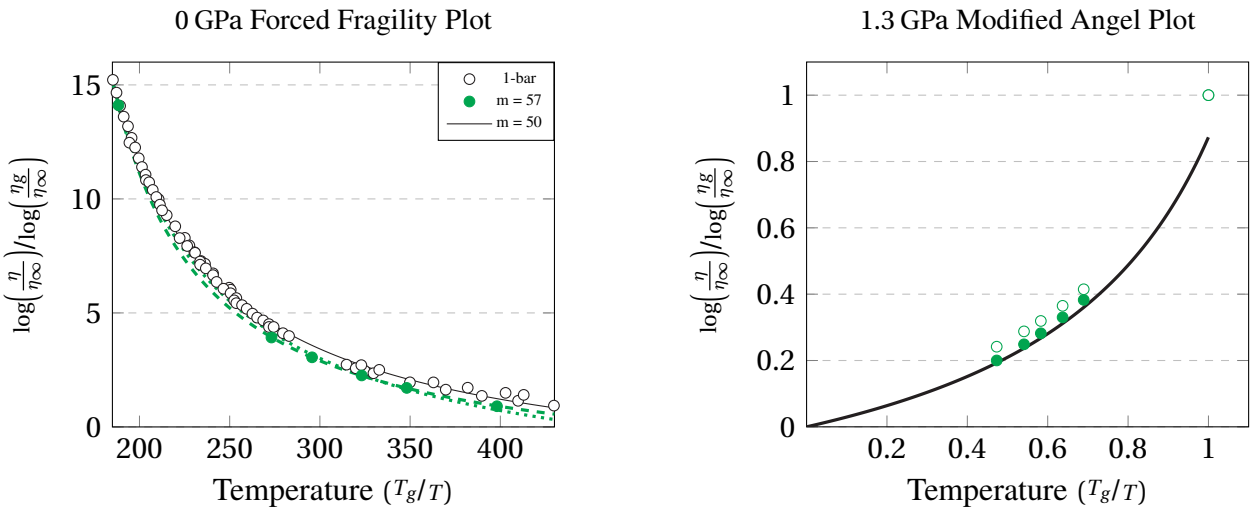


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Depolarized Photon Correlation Spectroscopic Study of the Glass-Forming Liquid Cumene at Very High Pressures

Depolarized Photon Correlation Spectroscopic Study of the Glass-Forming Liquid Cumene at Very High Pressures

A dissertation submitted in partial fulfillment
of the requirements for the degree of
Doctor of Philosophy in Physics

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Abstract

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DEDICATION

To the Smirfs whose blue shenanigans always kept Gargamel too busy to impose actual harm on the world. To God who chooses not to smite us even as we pray for his intervention at a football game knowing full well the world is full of starving children. To my inability to come up with an actually eloquent and appropriate dedication.

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alleviate the burden of formatting.

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PUBLICATIONS

"*Title of published manuscript 1*", Adequate citation style, *Published*.

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1 Introduction

A very recent article, published by one David Main in the Atlantic, asserts that "Glass has changed the world like no other substance" despite often being generally taken for granted [65]. Main does an excellent job delving into the history behind human adaptation of glass, from the pre-historic world where volcanic glasses were used as tools, to when it was likely first manufactured and spread through Mesopotamia more than four millennia ago, followed by the development of glass blowing arising in first century Jerusalem and the appearance of clear glass allowing for more functional, rather than solely ornamental, implementations. Though his title may be slightly hyperbolic, certainly many materials have had great impact on our technological as well as cultural development, it is fair to assert that the impact glass has had on society is at the very least on par with that of other materials such as metal alloys and petrochemicals or, in more recent times, semiconductors.

In modern times an astonishing variety of glasses are manufactured on an industrial scale. Silicate based glasses are doped with a variety of substances such as lead, germanium oxide, zirconium, titanium, or barium to control their refractive index. Control of refractive index has found both ornamental applications, the higher index of refraction increasing optical dispersion in a lead glass vase, as well practical applications—a fiber optic cable having a higher degree of flexibility while suppressing optical leakage via a high critical angle [37, 109, 51]. Other doping materials, such as boron trioxide, may be used to mitigate thermal expansion or aluminum oxide to increase tensile strength or specific heat capacity [5, 28]. Glasses find applications in waste disposal, for example the very promising possibility of immobilizing nuclear waste via confinement in a glassy

matrix which could alleviate any concern of the waste materials ever leaching into groundwater [83, 77].

Of course what constitutes a glass goes far beyond simple Silicates. Amorphous metallic alloys have been a hot topic of research since their emergence half a century ago as they exhibit many fascinating mechanical and magnetic properties [38, 5]. With regards to magnetic properties, spin glasses, or materials where magnetic domains undergo a glass transition that is very much analogous to that of structural glasses, have been heavily studied since at least the mid 1970's [9, 74, 92].

Some materials form glasses relatively easily while others tend to fall into a crystalline state more readily. However, just about any liquid that can form a crystal will form a glass under the right conditions. Water is a material that does not readily form a glass. This is due, in large part, to its exceedingly large heat capacity coupled with a very small molecular moment allowing for orientational relaxation on a timescale generally too rapid for a glass transition to occur—crystallization out-pacing vitrification during thermal quenching. In order to form a glass, water must be cooled to its glass transition temperature of around 137 K on a timescale of milliseconds to avoid spontaneous crystallization. However, glassy water has been produced in laboratory conditions and is found to have some unusual characteristics including a tendency to exhibit a transition directly from a glassy state into a crystalline state when heated[4]. Glassy water also has been found, predominately via neutron scattering experiments, to exhibit long range structural order but lack local order which is exceedingly atypical for a glass, as should be made clear in section1.1.1 that follows [67, 10]. This is not the only instance of atypical behavior. The melting point of water exhibits a negative slope above the triple point arising from its unusual lattice structure giving raise to a material that expands upon freezing. Even at exceedingly low temperatures, crystalline water

can be melted via compression if sufficiently high pressures are applied. It is this phenomenon that allows for the pressure amortization of crystalline ice first observed by Mishima *et al*. in the mid eighties[71]. Here a glass is formed via pressure induced melting of ice well below the glass transition temperature. In this instance, amorphization is dependent on that most usual property of water to melt under increasing pressure. However pressure induced amorphization has also been observed in crystalline silica—a more typical glass forming system [44]. In point of fact, any liquid exhibiting a more typical phase diagram that can be thermally quenched to form a glass can equivalently be "quenched" from a liquid state into a glass via sufficiently rapid compression. This will be central to the discussion of the TGP method of obtaining the glass transition temperature in Sec. 1.3.4 that follows.

It is, perhaps, strange to think of water as being a glass, yet evidently it can be and exhibits some remarkable properties when in a glassy state. As unusual a material as it is, water glass is not solely relegated to laboratory experiments. Water in the Kuiper belt has slowly nucleated over billions of years to form comets and other celestial bodies. This nucleation occurs so slowly, and at such low temperatures, that individual molecules find themselves lacking sufficient energy to allow them to reorient into a crystalline structure, thus forming, one mono-layer at a time, an amorphous glassy ice. A marvelous article on the topic of glassy water, published in Physics Today, asserts that "glassy water may be the most common form of water in the universe [24]." Indeed, given that water is the second most abundant molecule in the Universe following hydrogen, glassy water may well be the most common glass in the universe.

There also exist countless organic substances that readily form amorphous glassy solids. This work will focus predominately on two organic glass forming systems—glycerol in Chap. 3 and cumene in Chap. 4. However, examples of organic glasses permeate the literature. They find

applications in medicine, for example the use of glass scaffolds for drug delivery or even potentially taking advantage of the glassy state of a drug itself where the lower Gibbs energy associated with increased structural entropy may give rise to greater water solubility and thus increase the efficacy of delivery [98, 39]. Poly-methyl-methacrylate (PMMA), a strong thermoplastic often refereed to as an acrylic glass and marketed under the brand names Plexiglass, Acrylite, and Lucite among others, has found a wide variety of applications since its invention by Dupont in the early 1930's including automobile manufacture, aircraft windows, marker boards, and even the ceiling of the Houston Astrodome ballpark due to its high transparency, resistance to external elements such as UV light and water, and other beneficial properties [66]. Polylactic acid (PLA) is another glass forming organic thermoplastic which has found extensive application in rapid prototyping due to the ability of such materials to be softened in a controlled manner upon heating [97]. Metallic glasses also share this characteristic and thus are quickly capturing interest for their potential to allow for direct 3D printing of metallic parts [76].

The material qualities that define a glass are still somewhat contested today, however functional definitions do exist and will be the topic of the following section. Here it may suffice to say that the term glass describes a vast array of materials spanning countless applications from window panes to, perhaps somewhat surprisingly, even cotton candy—a sucrose glass [40]. In 2015 the Carnegie Science Center of Pittsburgh had a marvelous display spotlighting the indispensable role that glasses play in our day-to-day lives. It was somewhat disappointing to me that a prominent display case labeled "Is it Glass" did not contain any cotton candy—arguably almost as disappointing as the fact that such a fascinating topic as to what constitutes a glass, so poorly understood by the public, was relegated to a small non-imposing display case.



Figure 1.1: Exhibit on display at the Carnegie Science Center (2015)

Sucrose, or common sugar, is actually a very good glass forming material despite generally being purchased in its crystalline form. An excellent example of a sucrose glass, always worthy of mention as it constitutes for many a childhood memory of witnessing a material transition from a liquid to a glass, would be peanut brittle. During the process of making peanut brittle it is typical to draw out the molten sugar mixture that will ultimately become the "brittle" from the pot with a spoon or spatula allowing strings to form. This is often refereed to as "pulling" the brittle and is a means of determining when it is ready to be poured which, as anyone who has made peanut brittle will attest, is a very time critical part of the process. Pouring the mixture out of the pot too early will leave one with a hot, sticky, watery mess yet if the brittle is allowed to cool too much it will rapidly solidify within the pot and become impossible to pour. Pulling the mixture allows one to gage viscosity. When the brittle has cooled sufficiently to pour these strings will rapidly solidify

when drawn out of the pot and snap. This occurs very near the glass transition temperature. When you witness this rapid vitrification you are witnessing, in real time, the transition of a material from a liquid to a glass.

If you have never seen peanut brittle made, set this dissertation down for a spell and go make a batch, for in the sections that follow, we will be delving into the details of what constitutes a glass as well as the molecular dynamics of the transition from a liquid to a glassy state. As is the case with all dissertations, beyond this introduction the material will become less colloquial and more technical. Having a plate of peanut brittle to enjoy as you read could certainly do no harm and may even serve to bestow on you a certain experiential intuition regarding the glass transition process. Besides, as Pamela Travers would almost certainly assert, "A glass made of sugar helps the medicine go down [105]."

1.1 Amorphous Solids

Before one can have a meaningful discussion of glassy dynamics and the glass transition, one must define what constitutes a glass. Zarzycki offers two potential definitions of glass—one operational and the other structural [116]. From an operational standpoint, one might define a glass as a liquid rapidly thermally quenched into a non-crystalline solid phase. Alternatively, from a structural perspective, a glass may be defined more generally simply as a non-crystalline solid—ignoring the means by which this state was obtained. As Zarzycki delineates, both of these fall short of providing a sufficient definition as the simple structural perspective is simply too broad, including materials such as gels that do not share the physical attributes of a glass, yet his operational definition is two specific, omitting states that are structurally identical yet were arrived at via means other than thermal quenching such as compression, which will be central

to the topics of Chap. 3 and 4, pressure-induced amorphization such as that demonstrated by Mishima or Hemley *et al*. [44, 71], or cold deposition as previously discussed regarding water ice on comets formed via accretion [24]. His conclusion, and the general consensus today, is that the solution lies in a definition of the glassy state which incorporates the structural perspective with the added caveat that glasses constitute a non-equilibrium solid-state system—a merger of structural and dynamic descriptions.

1.1.1 Structural Description

Zarzycki's definition still requires further clarification. Is a glass a liquid, solid, or a separate phase of matter in its own right? One may be tempted to assert that glass is nothing more than a highly viscous liquid whose viscosity may be rapidly diverging however will still flow—albeit on a remarkably long timescale. Indeed one would not be too hard pressed to find, even among physicists, those who believe that cathedral stained glass is thicker on the bottom of the pane due to the glass having flowed over the centuries under the influence of gravity—a very common misconception. The best models of molecular dynamics in glassy systems must be heavily extrapolated here, yet extremely conservative estimates place the average timescale for these silicate glasses to flow appreciably at ambient temperatures to be several orders of magnitude greater than the age of the Universe [115]. In light of this, to assert that a material in a glassy state is 'fluid' in any colloquial sense may seem absurd. However one's incredulity does not a valid definition make, and one cannot dismiss the fact that a glass's radial distribution function is often indistinguishable from that of the liquid state as conceptually illustrated in Fig. 1.2.

The radial distribution function, alternatively referred to as the pairwise correlation function, is a model describing the expectation value of the particulate number density in a region as

a function of distance from an origin placed at any arbitrarily representative particle or molecule within a bulk volume of material. This is best conceptualized as the ensemble average of the distance between all other particles or molecules as measured from each individual particle or molecule within a bulk material.

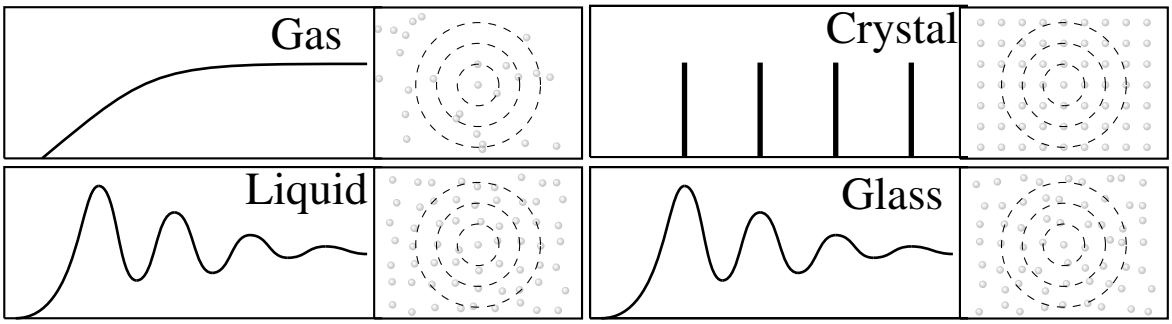


Figure 1.2: Typical radial correlation functions for a gas, liquid, glass, and crystal. This figure is derivative of that originally published by Johari. [52]

Significantly simplified two dimensional representations of radial distribution functions and the structures giving rise to them for a gas, liquid, crystal, and glass are provided in the before-mentioned Fig. 1.2. Note that the exact particle locations for this figure were determined implementing a random number generator and thus are not representative of the ensemble average resulting from any specific intermolecular potential. This figure is relevant to a discussion of the characteristics typical of pairwise correlations functions for these four material states only in the most general of terms. The two dimensional analog of a crystalline structure is shown in greater detail here in Fig. 1.3 where the characteristic radii between horizontally and vertically adjacent particles are labeled r_1. The characteristic radii corresponding to the nearest neighbor, next nearest neighbor, and next-next nearest neighbor are labeled r_1, r_2 and r_3 respectively. These will exhibit rapidly decreasing correlation amplitudes for higher order nearest neighbors. The corresponding radial distribution function exhibits very sharp periodic peaks associated with these infinitely

repeating lattice spacings. The exact appearance of a radial distribution function for any real crystalline bulk material can clearly be quite complex and will vary from one material to the next as it depends directly upon the crystal symmetry, however it will always consist of such well defined periodic peaks. The arguably oversimplified crystalline pair correlation spectra shown in Fig. 1.2 ignores all peaks associated with these off diagonal contributions from the lattice structure to provide an even further conceptually idealized spectra. Strictly speaking, such a simple correlation spectra could only ever arise from a one dimensional distribution of evenly spaced particles however it represents the behavior of a repetitive lattice structure sufficiently well for the purposes of this discussion.

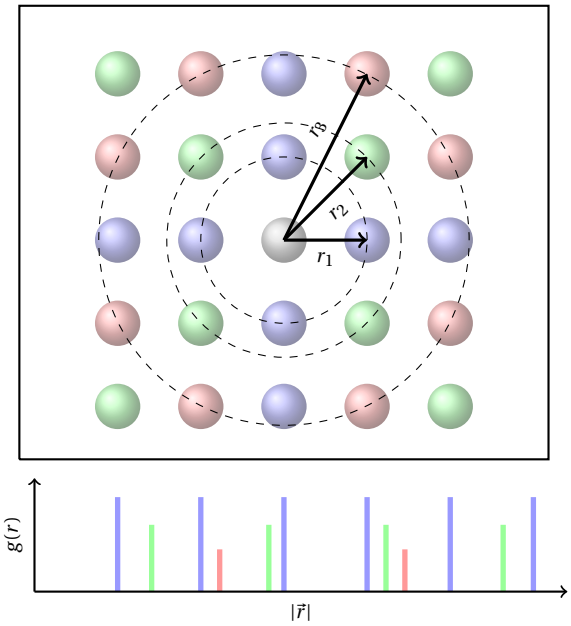


Figure 1.3: A crystalline solid exhibits one or more characteristic radii, for which adjacent individual particles or molecules appear at integer multiples. The corresponding radial distribution function for such a material will consist of sharp periodic peaks.

Conversely, gases exhibit no such periodic order. Due to the strong yet exceedingly short range coulombic repulsion, molecules will spend very little time in extremely close proximity, existing mostly in a free state where intermolecular interactions are negligible. As such, their

pairwise correlation function will very rapidly approach a uniform value corresponding to the average particle density. An idealized representation of this is provided in the top left window of figure1.2.

From a structural perspective, both liquids and glasses will exhibit radial distributions functions that appear almost as if they consisted of a damped superposition of the gas and crystalline states. This arises due to the fact that the intermolecular interactions between the constituent molecules of both liquids and glasses are not negligible. Short range order emerges as bonds between adjacent constituent molecules are formed. As this order is at the same time imperfect relative to the crystalline structure, short range, and in the case of a liquid also of short duration, it will manifest itself in the broadening and dampening of the periodic peaks of the corresponding pairwise correlation function. Representative idealized liquid and glass radial distribution functions are provided in Fig. 1.2 as well. Note the periodic peaks that correspond to an imperfect crystalline structure in short order giving way to an utter lack of long range order indicative of a gas—albeit at significantly higher density.

The similarities between the radial correlation functions for a liquid and a glass are worthy of note. Structurally they are identical. However, as will be discussed shortly, they differ vastly in their dynamics. For a liquid, this radial correlation function may be thought of as arising from either a time average of the constituent molecular positions or equivalently the ensemble average calculated from every individual particle at a single instant in time. However the dynamics of a glass, being an amorphous solid, are such that its pairwise radial correlation function is best thought of only in terms of an ensemble average as constituent molecules simply do not move appreciably on any meaningful timescale. Thus we have the very curious condition of two material states having identical structure yet arising from vastly differing dynamics. A great deal of sense can be

made of this apparent conundrum via a discussion of the dynamics of these states from an energy perspective as will now follow.

1.1.2 Dynamic Description

Molecules in a gas state have high enough energy on average that they do not really "feel" the potential energy landscape produced by their neighbors. They take on a relatively random and highly degenerate spacial distribution where intermolecular interactions are limited to the strong yet exceedingly short range Coulombic repulsion of their electronic shells. Thus the pairwise correlation function for a well thermalized gas will exhibit very little structure beyond a rapid increase in correlation, arising from this Coulombic repulsion repelling other molecules at close proximity, quickly leveling off to a region of constant number density, beyond the influence of this repulsive force, that will extend indefinitely as demonstrated in Fig. 1.2.

Gases and crystals are, in this sense, polar opposite scenarios in that the former has sufficient thermal energy that the potential landscape arising from the intermolecular forces is almost irrelevant as molecules spend very little time in close enough proximity to any other molecule for even the dominant Coulombic interaction to play a major role in their dynamics, yet the latter has sufficiently low thermal energy such that dynamics are dominated by intermolecular forces beyond Coulombic repulsion—higher order terms such as dipole interactions. Molecules in a crystal state certainly *feel* their potential energy landscape, yet are not free to explore it as most of that parameter space is energetically inaccessible. Thus they are locked into discrete, highly symmetric arrangements associated with the lowest energy structural state of the bulk system. These are the two extremes of which the liquid state is an intermediate.

Within the energetically intermediate liquid state, molecules or particles have sufficient

energy to explore their potential energy landscape yet still be heavily influenced, and to some extent confined, by said landscape. Short lived bonds between molecules form, increasing the average time in which they are in close proximity and giving rise to cooperative motion. However these nearest neighbor bonds *are* very short lived—the activation energy required to break the bonds being on the order of the thermal energy, $\kappa_B T$, of the material. The bonds conforming to the lowest energy orientations corresponding to the crystalline structure, though not exclusively occurring, will clearly be statistically favored. Thus liquids exhibit the rapid emergence, and subsequent disintegration, of short range order that may be described as spatially distorted crystalline arrangements. One would expect that the radial distribution function of a liquid will therefore have structures that resemble the crystalline peaks over a short range, only broadened due to spacial distortion associated with the accessibility of higher energy bond orientations. However, over very long ranges, these short lived local nucleations will bear no correlation from one location to the next and as such we expect the radial correlation function for a liquid to be indistinguishable from a gas for large radii—with the caveat of exhibiting significantly higher number densities. The radial correlation function for a liquid displayed in Fig. 1.2 shows what would be expected for a state that is intermediate between a gas and a crystalline state, *i.e.* broadened crystalline peaks at short range quickly washing out to a uniform average number density at larger radii.

Another way to describe the dynamics of a glassy system, beyond energy considerations, is to consider the timescales on which the potential energy landscape is being explored—*i.e.* the structural relaxation times. It should be noted that there are a variety of "relaxations" occurring in any glassy system, not all of which are directly associated with the bulk structure. These will include thermal relaxations as well as those arising from diffusion and orientation dynamics. The distribution of these relaxations will vastly differ in complexity depending, in large part, on the

internal degrees of freedom available to the molecules making up the bulk. However, even the simplest of glass formers will generally exhibit relaxation modes that collectively fall into two domains—fast β-relaxations and the much slower α-relaxations. It is the structural relaxation modes associated with the breaking of nearest neighbor cages, molecular reorientation, thermal diffusion, *etc.* , that will collectively constitute the α-relaxation and will be the focus of this dissertation.

In the following section we will discuss several probes of these relaxation dynamics, including Photon Correlation Spectroscopy in Sec. 1.3.2. The time-domain coloration spectra that are obtained from this process for glass forming systems are generalized here in Fig. 1.4 for the purposes of illustrating this dynamic behavior under changing thermodynamic conditions. Here, the correlation time, τ, is simply the average time required for a bulk material to relax having been subjected to some form of mechanical perturbation—inherently tied to the imaginary component of the bulk modulus. Generally speaking, the fast β-relaxation will be relatively unaffected by changing thermodynamic parameters. However, as temperature decreases, or equivalently pressure increases, the liquid will begin to vitrify and the characteristic timescales of the slower α-relaxations will begin to rapidly increase by orders of magnitude. This can happen over very moderate ranges of temperatures—on the order of tens of Kelvin. Understanding these dynamics, as they relate to the freedom to explore the energy landscape being restricted with the removal of thermal energy or under compression, is key to describing the nature of the glass transition.

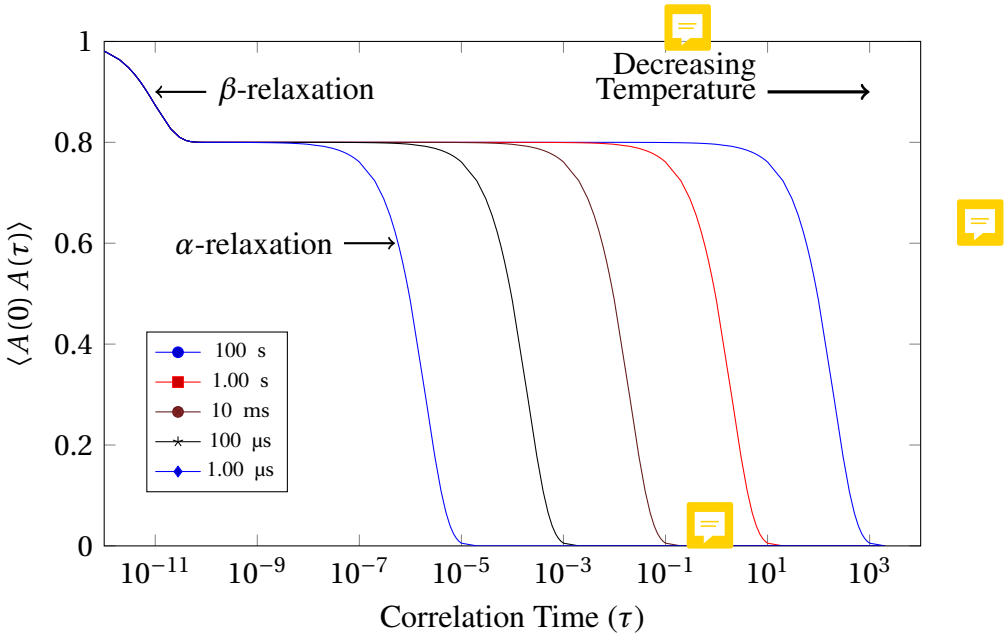


Figure 1.4: An idealized set of autocorrelation curves typical of what might be obtained for any given glass forming system. The characteristic structural relaxation times, τ_α, which will be discussed latter in greater detail, fall near the foot of each curve. As temperature is decrease or pressure is increased, the faster β-relaxation remains relatively fixed, however the α-relaxation shifts to the right as the rate of structural relaxations slow corresponding to more of the energy landscape becoming energetically inaccessible.

In the most simplistic of terms, glassy materials exhibit the rigidity of a crystalline state, having insufficient thermal energy to readily explore their potential energy landscape, yet maintain the semi-ordered structure of their liquid state. Their radial correlation functions, as characterized in Fig. 1.2, will be very similar in structure to that of the liquid state, often exhibiting short range crystalline structure albeit with copious irregularities, transitioning into rapidly diminishing structural correlation over longer distances. A very marked difference between a system in a liquid *vs*. a glassy state is the apparent violation of ergodicity. Although the radial correlation function of a glass may be similar to the liquid state in general structure, it may not be quite as identical as implied by Fig. 1.2. Glassy systems will exhibit mechanical hysteresis stemming from their exceedingly slow structural relaxations. This along with the lower thermal energy of these systems

results in intermolecular bonds being much longer lived than in the liquid state which may lead to semi-crystalline domains growing quite large. Cooperative motion of these domains will play a much greater role in the dynamics of glassy systems than in a liquid state. These considerations, among others, leave open the question as to whether glass exist as a separate phase of matter or as nothing more than a highly viscous fluid—a question we will now address.

1.2 The Nature of the Glass Transition

A material undergoing a change in phase will experience a dramatic and discontinuous shift in bulk properties such as density, specific heat, bulk modulus, thermal expansivity, *etc.*. Ehrenfest more precisely defined phase transitions as points in the pressure-temperature phase space of the material where discontinuities in the derivatives of the Gibbs free energy with respect to any external thermodynamic parameter emerge. Transition between any two states described in figure1.2 with the exception of the glass transition will constitute what Ehrenfest described as a first order phase transition in that it will involve discontinuities in first order derivatives of the free energy, or equivalently enthalpy, for a transition at constant pressure [50]. The glass transition, however, is distinct from the others in that no such first order discontinuity exists. Figure 1.5 shows the plot of enthalpy or specific volume as a function of temperature for an idealized glass forming system maintained at constant pressure. The enthalpy of fusion is non-trivial for most liquids and corresponds to the discontinuity in the graph of Fig. 1.5 arising at the melt temperature, T_m. In principle, any liquid capable of forming a crystalline structure, if brought to this temperature, provided with a single nucleation point, and allowed to cool slowly enough to maintain thermalization, will form a single crystal solid that constitutes the global minimum on the potential energy landscape as represented in Fig. 1.6. This conceptually idealized one dimensional energy landscape was borrowed

from the excellent review article by Debenedetti and Stillinger [25]. It is, obviously, not possible to aptly describe the potential landscape for any real system graphically as it will span a parameter space whose dimension will be on the order of the number of constituent molecules residing in the bulk multiplied by their individual degrees of freedom. However this overly simplified single dimensional representation is quite sufficient as a conceptual illustration.

If cooled beyond crystallization, enthalpy will continue to decrease with the removal of internal energy and any associated thermal contraction until, at absolute zero, only a single state is energetically accessible corresponding to zero entropy. This stands in stark contrast to a glassy state where a certain degree of entropy, associated with structural disorder, is locked in. This can occur when a liquid is thermally quenched very rapidly. Rapidly, in this sense, implies that thermal energy is removed quickly enough such that sufficient time is not allotted for the potential energy landscape to be throughly explored, thus there is a high probability of the system settling into local, rather than global, minimums. Fig. 1.6 has labeled one such local minimum as an "ideal glass," however it should be noted that, though there may well be a single unique crystalline state which will constitute the absolute global minimum on the energy landscape, this energy landscape is dependent on the location and orientation of every constituent particle or molecule and as such is a remarkably complex surface having innumerable local minimums in which a rapidly quenched system may settle into. Many of these local minimums are relatively shallow and are here labeled "Transition States" as any minor external perturbation to the system will likely be sufficient to overcome the potential barrier and invoke a transition. Super critical liquids exist in transition states. Most have seen an example of such a transition state in sodium acetate heat packs solidifying upon perturbation—an example of rapid crystallization. Some systems do not so readily crystallize and, upon rapid thermal quenching, settle into an amorphous solid locked

within a local energy minimum that rests above that of the crystalline state, but significantly below what would be considered a transition state. For such a system, the entropy does not approach zero as enthalpy is decreased due to the existence of other local minimums that are technically energetically accessible, however constituent molecules become locked into nearest neighbor cages that are insurmountable on any reasonable experimental timescale.

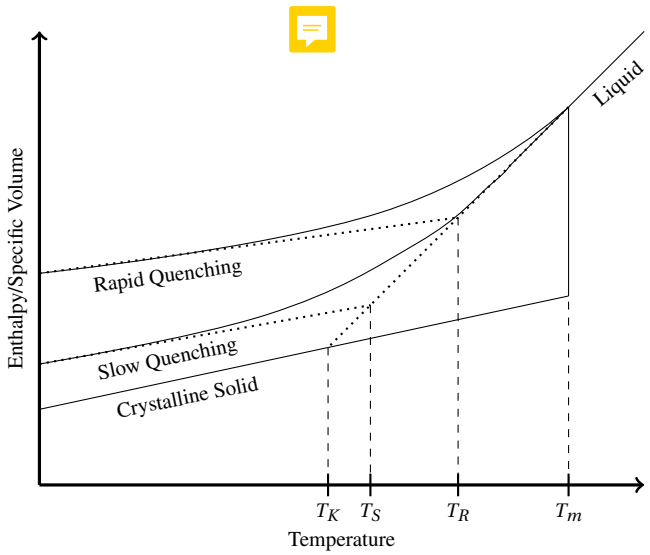


Figure 1.5: An idealized representation of a typical plot that may be obtained via the implementation of differential scanning calorimetry to determine the glass transition [78]. The smooth transition from the linear trend of the liquid state into that of the glassy state, exaggerated here, would be expected in the absence of any higher order Ehrenfest transition. The dashed lines indicate where the extrapolations from the liquid state at temperatures far above T_g intersect with similar extrapolations from an amorphous solid state at temperatures well below T_g for glasses which were formed via rapid and slow thermal quenching. The Kauzmann temperature for the limiting case is also shown as it provides an argument to why these smooth transitions are likely non-physical.

This is the glass transition and is perhaps well described again by Fig. 1.5 where the discontinuity in first order derivatives of the free energy are not present but rather a rapid yet smooth transition does occur as temperature decreases. The answer to the question as to the nature of the glass transition, it being a legitimate second order phase transition or simply rapid vitrification of the liquid state could, in principle, be experimentally reveled here. If this transition is indeed

smooth, lacking any discontinuity in higher order derivatives, then the glass "transition" is no more than a kinematic effect associated with a rapidly vitrifying liquid state. However if the smooth transition implied by Fig.1.5 is not reflective of an actual glass transition, but rather a discrete discontinuity in the slope exist, then the glass transition is more than a rapidly vitrifying liquid, but in fact a second order Ehrenfest transition into a distinct thermodynamic state. However, such measurements do not lend themselves well to the high precession necessary to resolve such a discontinuity sufficiently well to lay this question to rest. However there is a compelling argument for the necessity of a second order transition that appears consistent with the thermodynamic behavior of glass forming systems.

Assume for a moment that the glass "transition" is a purely kinematic vitrification process. As depicted in Fig. 1.5, upon super-cooling below the freezing point, T_m, enthalpy or specific volume will quickly approach a trend that tracks with that of the crystalline solid yet will remain systematically above the corresponding values of the crystal enthalpy as temperature continues to decrease. The exact values of enthalpy, specific volume, heat capacity, or many other bulk properties of the glassy material will depend on its thermal history including how rapidly it was quenched. Extremely rapid thermal quenching may result in a much less ordered glassy system as molecules have virtually no time to seek out even relatively pronounced local minimums on the potential surface. Slightly slower quenching may allow local order to arise thus giving bulk properties closer to that of the crystalline state. For a first order phase transition, such as from a liquid to a crystalline solid, there is a specific well defined temperature at which this transition occurs. It is desirable to define such a transition temperature for this glass transition as well. One such definition is demonstrated in Fig. 1.5 as the intersection of the extrapolations of the linear trends in Enthalpy or specific volume as a function of temperature from well within the glassy state

and from the supercooled liquid state. These extrapolations are shown as dotted lines in Fig. 1.5 and their intersections marked as vertical dashed lines corresponding to the characteristic transition temperatures, T_S and T_R, for a glass forming system undergoing slow and rapid thermal quenching respectively. These points of intersection clearly cannot suffice as a definition for the glass transition temperature, T_g, as they will inherently constitute a range of values strongly dependent upon the rate of cooling as, from an energy landscape perspective, the rate of cooling will determine to what extent the constituent molecules have time to explore the energy landscape before becoming energetically locked into a local minimum corresponding to a glassy state associated with a specific local minimum on the potential energy landscape. The more rapidly a system is quenched the more likely it is to be locked into a state that is energetically much further from the crystalline state as well as having a significantly higher residual entropy as temperature approaches absolute zero corresponding to the higher level of structural disorder. A standard metric experimental protocol must be agreed upon if one is to define a singular glass transition temperature, T_g. The most common method of defining T_g, at least at atmospheric pressure, has been via these calorimetric measurements which directly probe the heat capacity of the liquid as temperature is scanned at a controlled, generally agreed upon rate through the glass transition—typically on the order of tens of Kelvin per minute. At these rates a value of T_g is found consistently to correspond to viscosities on the order of 10^13 poise or an $\alpha-$relaxation time on the order of 100 seconds [100]. The temperatures associated with these viscosities or characteristic structural relaxation times have been generally accepted as the standard definition of the glass transition temperatures, T_g, throughout the literature and are those to which this work will conform. Further details of calorimetric methods will not be described here as these methods not prove practical at very high pressures and were not implemented to obtain any of the data appearing in this work. However, this thermodynamic

description of the glass transition remains pertinent as it illustrates, quite effectively, one argument for the glass transition being a second order phase transition, as opposed to a purely kinematic process of rapid vitrification as will now be discussed.

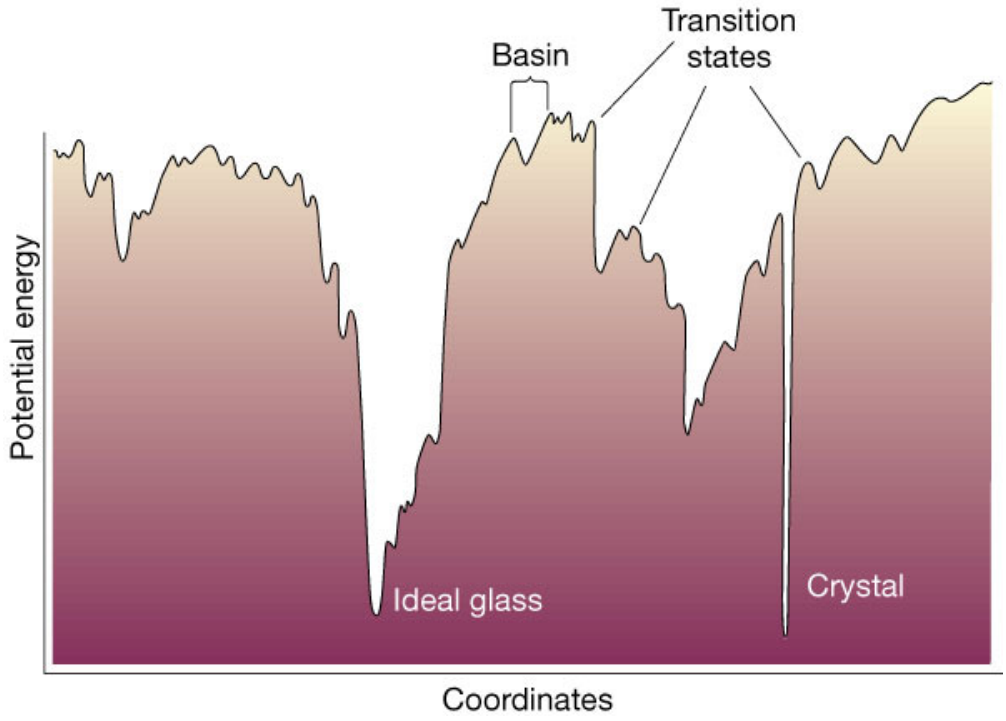


Figure 1.6: The two dimensional representation of an energy landscape for an idealized glass forming system provided by Debenedetti and Stillinger [25].

Consider that the curve labeled "Slow Quenching" in figure1.5 is in no sense representative of a limiting case for any given glass forming material. One may always thermally quench a material more slowly, at least theoretically. In practice, spontaneous nucleation events may invoke crystallization within a super-cooled liquid as has been observed in several glass forming systems such as propylene carbonate and salol [102]. However, for liquids that readily form glassy states, bulk crystallization may be avoided even for exceedingly slow rates of thermal quenching. Such a scenario may result in relatively large domains of local crystallization emerging randomly throughout the material. These local domains may ultimately limit the mobility of neighboring domains

and thus lock in the long range heterogeneity associated with a glassy state. The exact processes that frustrate large scale crystallization in these instances is still a topic of investigation [93]. However there does not seem to be a physical reason why such a glass forming system cannot form larger and larger local domains as the rate of cooling is continually decreased. As these local domains increase in size the bulk thermodynamic properties of the material must continue to approach those of the crystalline state. If such a glass forming liquid could be thermally quenched sufficiently slowly, while somehow avoiding a nucleation event leading to premature rapid crystallization, one might expect from the linear trend seen in Fig. 1.5 that the enthalpy or specific volume may continue to decrease below that of the crystalline state. However this would lead to a state having a lesser entropy than that of the crystalline state. This is known as the Kauzmann paradox [107]. As the Kauzmann paradox is nonsensical, clearly this approach to the properties of the crystalline state must be asymptotic or exhibit some form of phase transition to avoid the paradox. Kauzmann himself believed that crystallization could simply not be avoided and would always set in before the Kauzmann temperature, T_K, could be reached though he does not postulate a mechanism for this forced crystallization. In our previously described scenario, one could imaging all of the separate nucleation events giving rise to local domains that continually grow as temperature is lowered, ultimately reorienting and converging to the crystalline structure–a slow but steady stroll around the potential energy landscape that ultimately arrives at the global minimum, albeit with the absence of a first order phase transition, and would manifest in Fig. 1.5 as a stark discontinuity in the slope of enthalpy occurring at T_K—a true second order Ehrenfest transition. This was first proposed by Gibbs and Dimarzio though it has yet to be confirmed experimentally due to the before-mentioned issues with spontaneous crystallization occurring before T_K can be reached during exceedingly slow thermal quenching [36, 35].

1.3 Probes of Glassy Dynamics

Although the calorimetric method of defining the glass transition temperature, T_g, as described above is theoretically compelling and remains the standard at atmospheric pressure, it does require a large amount of data, taken with tight experimental control of the cooling/heating rates, in order to arrive at reproducible values for the glass transition temperature, T_g. As before mentioned, it has proven to be experimentally advantageous to define the glass transition temperature in terms of viscosity or, equivalently, the characteristic structural relaxation time. The characteristic timescales on which molecular dynamics evolve in a glass forming system vary from on the order of picoseconds for very high temperatures or in low pressure environments, to thousands of seconds at low temperatures or in high pressure environments. Due to this extreme dynamic range, a single experimental probe for tracking these molecular dynamics will likely never exist; and in fact, experimental gaps within the literature typically exist for dynamic data on any given glass-forming system. Be that as it may, with the aid of a proper mathematical model and the implementation of several different experimental probes, each within their applicable dynamic window, excellent descriptions of these dynamics can be ascertained over exceedingly large dynamic ranges encompassing the glass transition for many materials.

One question which may be plaguing the mind of the astute reader at this point is, quite simply, what are "these dynamics". Specifically, the study of the glass transition relies on experimental probes capable of tracking changes in molecular or atomic locations and orientations. Some may accomplish this directly, perhaps via some scattering mechanism that tracks thermal diffusion, or more indirectly via tracking changes to thermodynamic bulk properties such as viscosity which arises as a direct consequence of said diffusion. This work will exclusively focus on probes of

viscosity, η, and the characteristic structural α-relaxation time, τ_α. These two bulk properties are intrinsically related via the Maxwell relation which observes that, for small deformations or applied strain, $\tau_\alpha = \frac{\eta}{G_0}$ where G_0 is the elastic sheer modulus. Figure 1.7 illustrates how the stress resulting from an adiabatically applied strain relaxes over time due to viscous flow of the bulk material.

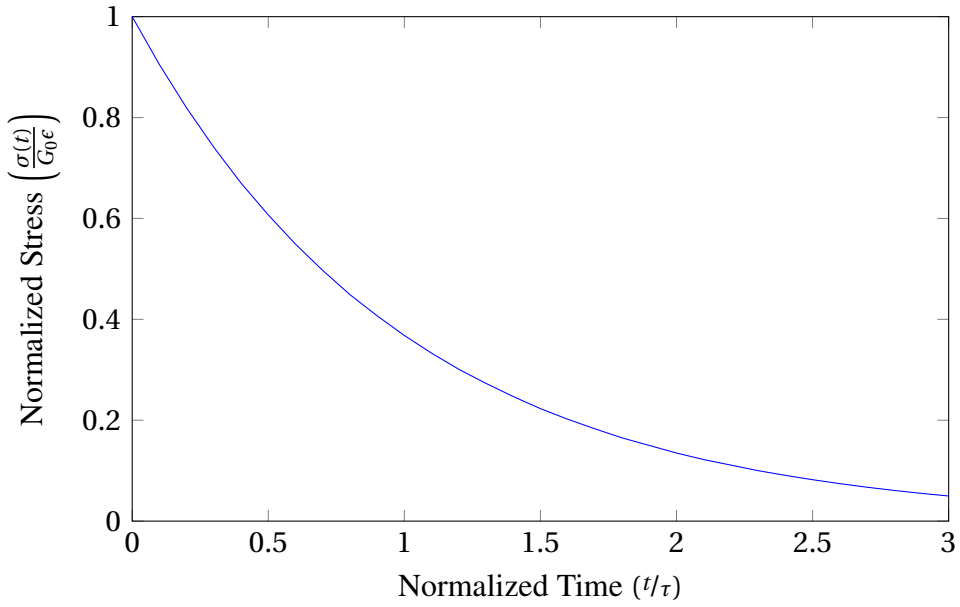


Figure 1.7: Here is provided an idealized time dependent decay of a stress, $\sigma(t)$, normalized to an adiabatically applied strain, ϵ, inducing it and made dimensionless via the elastic sheer modulus. The characteristic relaxation time over which viscous flow mediates the mechanical relaxation of stress is related to the elastic sheer modulus as $\tau = \eta/G_0$.

However direct measurements of structural relaxation and viscosity are certainly not the only window into glassy dynamics. Figure 1.5 describes a classical calorimetric determination of the glass transition temperature in which the underlying molecular dynamics are not directly probed but rather the behavior of enthalpy or specific volume as a function of temperature—thermodynamic bulk properties that arise stochastically from underlying molecular dynamics. Figure 1.2, though not comprised of actual data, is representative of that which has been obtained via direct struc-

tural probes such as neutron scattering. However neutron scattering has provided more than just structural snapshots. Though it will not be discussed in great detail here, as it is not relevant to any data implemented or referenced in this work, inelastic neutron scattering has also served as a probe of molecular dynamics within glassy systems and has certainly found implementation in recent structural and dynamic studies [96, 104, 55].

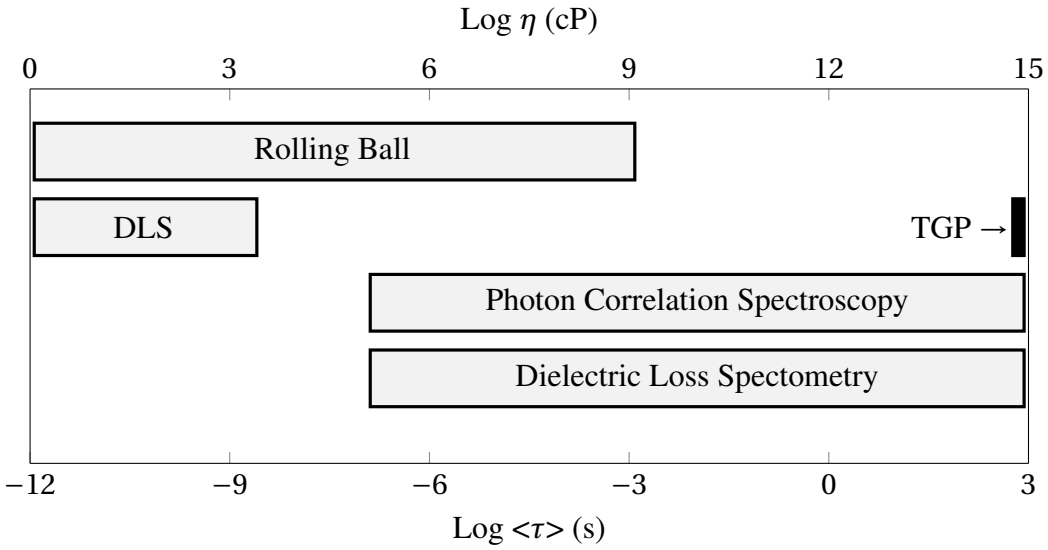


Figure 1.8: Accessible dynamic ranges for various probes of viscosity and structural relaxation time $<\tau>$ [19][112].

The experimental probes that will be referenced or implemented in this work are nicely summarized in Fig. 1.8 and include: dynamic light scattering, photon correlation spectroscopy, rolling ball viscometery, TGP, and dielectric loss spectrometry. The physical mechanism behind each of these probes will be described briefly below as well as their dynamic windows and other limitations to their applicability pertinent to this work. These descriptions will be kept intentionally brief as an entire dissertation could easily be written on any one of these probes alone, and indeed, chapter 3 is in effect a reanalysis of one such dissertation outlining a means of extending a previously existing probe, rolling ball viscometry, into a diamond anvil cell allowing for viscosity

measurements at higher pressures [20]. Following this, chapter 4 will provide a similar extension, this time of photon correlation spectroscopy, into the diamond anvil cell. Significantly more detail regarding these specific probes will be provided in their respective chapters.

1.3.1 Dynamic Light Scattering (DLS)

Electromagnetic fields and the radiation they give rise to have had a near monopoly on providing an experimental window into describing molecular structure and dynamics. Electromagnetic interactions with matter are at the heart of just about every probe, sometimes even neutron scattering—the neutron's appreciable magnetic moment providing a probe of ordered magnetic structures such as spin fluctuation in superconducting materials [114]. Light scattering methods focus on how condensed matter responds to the presence of electromagnetic radiation—specifically in giving rise to a scattered field which carries information about the structure and dynamics of the bulk material. A very generalized schematic for a light scattering experiment is shown in Fig. 1.9. Monochromatic light, generally exhibiting a spectral width of only a few MHz and often polarized, is collimated or focused through a sample volume. Some fraction of this light is scattered, collected, and then directed on to a spectral analyzer of some form, which in this dissertation will be either a triple-pass, tandem Fabry-Pérot interferometer or a photon correlation spectrometer.

A light source with exceedingly narrow spectral width is necessary for DLS as this probe makes use of very small spectral shifts in the frequency of scattered light due to a variety of dynamic processes that will be described in more detail within Sec. 2.2 and 2.3 of the following chapter. For now it will suffice to assert that light is scattered from the sample volume whenever there is a local temporal or spacial change in the dielectric susceptibility. This can be due to density fluctuations, local dipole induction, or orientational dynamics—anything that changes how the

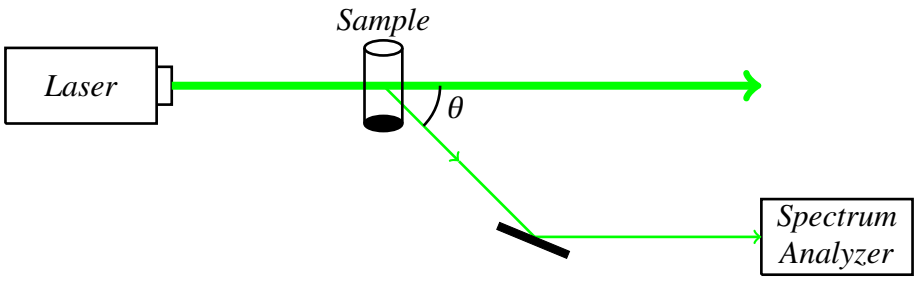


Figure 1.9: A simple forward scattering DLS experimental design in which light from a monochromatic source, such as a single mode locked AR^+ laser, is scattered from a sample and collected at a known angle. The collected light is then directed into a spectrum analyzer, such as a triple pass tandem Fabry-Pérot etalon.

charges in the medium react to an applied radiation field. If these changes are purely spatial, for example if a material has regions of variable densities frozen in place, the frequency of the scattered field will mirror the frequency of the incident beam, giving rise, *e.g.* , to elastic or Rayleigh scattering. However liquids and gases cannot maintain such static density gradations, thus density fluctuations in a liquid or gas are inherently time dependent and any temporal dependency of the dielectric susceptibility that arises from these fluctuations must impact the frequency distribution of the scattered field—an inevitable consequence of the field's mathematical Fourier transform. Even in a crystalline solid, such time dependent density fluctuations arise from vibrational modes within the lattice. Longitudinal and transverse acoustic modes, as well as higher frequency optical phonons, manifest themselves via frequency shifted peaks.

Dynamic Light Scattering, for the purposes of this work, will specifically refer to spectral analysis of *depolarized* scattered light arising from Brillouin scattering within a sample. Brillouin here is made distinct from Raman in that Raman spectroscopy involves higher energy phononic interactions, most often between nearest neighbor pairs of individual molecules, and thus the shift in photon energy upon scattering is much larger—being resolvable via a grating based spectrometer. Brillouin scattering, though also arising from acoustic scattering of light, arises from phonons

or elastic wavefronts whose wavelengths are very large relative to the average molecular spacing within the bulk material. Thus Brillouin scattering is a probe of larger scale cooperative dynamics within the bulk material. If one is to be strictly technical in our nomenclature, it is actually the Rayleigh scattered light arising from incoherent thermal fluctuation within the bulk material, rather than Brillouin acoustic modes, that we will focus on in this work, though it is common to refer to these processes collectively as Brillouin Light Scattering. The underlying physical mechanism and scale of these probes vary, however they all probe fluctuations in dielectric susceptibility within the bulk material. Exactly how the spectrum of scattered monochromatic light is impacted by the aforementioned fluctuations in susceptibility, and how information regarding structural relaxation is extracted from the spectral analysis of this scattered field, will be more thoroughly covered in Sec.2.3. However it does bear mention that this significantly limited definition of DLS, as it is applicable to the present work, provides a direct probe of structural alpha relaxation, albeit over a relatively limited dynamic range on the order of 10^{-12} s $< \tau_\alpha < 10^{-9}$ s as shown in Fig.1.8 above. There are broader definitions of DLS elsewhere in the literature, some of which make use of less direct probes of relaxation, such as spectral analysis of the width of lateral acoustic (LA) modes that arise from Brillouin scattering [43]. These probe a much broader dynamic range yet may be influenced by relaxation processes other than the α-structural relaxation alone. However, we must here adopt this definition of DLS specifically to distinguish the method from that of photon correlation spectroscopy, also implemented as a direct probe of the α-structural relaxation, yet obtained via analysis of temporal fluctuations in scattered light intensity and applicable over a very different dynamic window.

1.3.2 Photon Correlation Spectroscopy (PCS)

Photon Correlation spectroscopy (PCS) is a light scattering technique, similar to DLS described above, yet performed in the time domain. Both methods provide access to the α-structural relaxation dynamics within a material however cover a vastly different dynamic range. Whereas DLS lends itself to the probing of very fast relaxations, on the order of tens to thousands of picoseconds, photon correlation spectroscopy generally is applicable in the very slow to intermediate dynamic range, probing characteristic relaxation times on the order of tens of nanoseconds to thousands of seconds. A typical experimental setup for PCS is shown in Fig. 1.9 below—differing from DLS only in the form of analyzer implemented. In fact, data taken both for DLS and PCS, analyzed in Chap. 4, were all taken on a single experimental setup in which a flip mirror allowed for collected scattered light from a sample to be directed either into a tandem triple-pass Fabry-Perot interferometer for frequency analysis or, alternatively, a digital autocorrelator for temporal analysis.

There are two general experimental domains relevant to PCS—heterodyne and homodyne autocorrelation. It is important to define these terms here as there is some discrepancy within the literature regarding their use. For the purposes of this work, homodyne PCS involves temporal correlation analysis of light scattered from a sample in which great care has been taking to isolate only that light arising directly from scattering events within the sample itself. The intensity of this scattered light will fluctuate as a result of any temporal dependency associated with these dynamic processes. As such, this experimental arrangement is often refereed to as "self-beating", a labeling which only makes sense in light of the other common PCS experimental arrangement—heterodyne scattering.

In a heterodyne experimental arrangement, such as is represented in Fig. 1.10, a fraction of the source laser is split off before it is passed through the scattering sample and then re-mixed before being directed onto the detector for autocorrelation. In this arrangement, the fluctuations in detected intensity do not arise from the total number of scattering events, but rather from a shift in phase between light arising from scattering events and that of the local oscillator—a portion of the laser being mixed in providing a phase reference for the scattered light to beat against. Below is shown one possible heterodyne experimental arrangement. It is also common simply to collect light from the sample such that the specular reflection from the container can itself provide a local oscillator. All that is required to operate within a heterodyne regime is that the local oscillator intensity be significantly greater than that of the light scattered from the sample. Conversely, if one wishes to operate in a homodyne regime, one must assure that any specular reflections or light scattered from other static sources collected along with that scattered from dynamic processes within the sample volume is of negligible intensity. This often requires significant spacial filtering as well as exceedingly pure samples and clean experimental environments.

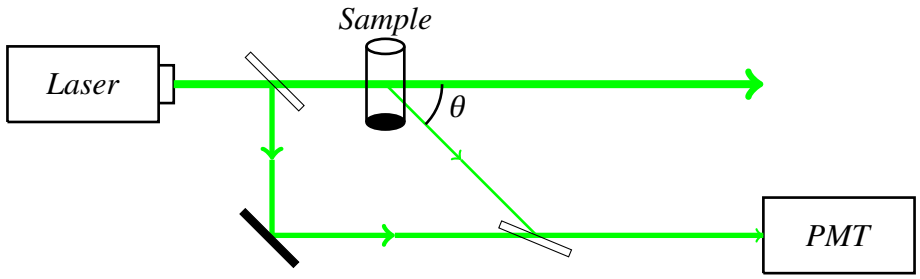


Figure 1.10: Heterodyne experimental arrangement for PCS. A local oscillator is mixed with the scattered signal before being redirected onto the photomultiplier or other detector.

In both experimental arrangements the intensity fluctuations at the detector will contain the same information regarding the molecular dynamics from which they arise. However, proper analysis of resulting data requires that one be firmly within one or the other experimental domain.

It is generally very undesirable to be within an intermediate homodyne/heterodyne scenario. If a homodyne experiment is desired one must often take great care to assure that only light arising from the scattering events of interest is collected. Dust and contaminates, both within the sample and on the sample container, or even undesired specular reflections may give rise to undesired heterodyne components which may make proper data analysis challenging or even impossible. These concerns are not always easily mitigated, great care being taken specifically to identify what regime one is operating in, as will be discussed in section 4.2.1 in which a means to extend PCS into high pressure will be described in detail.

1.3.3 Rolling Ball Viscometry

Rolling or falling Ball viscometry provides a means of direct mechanical measurement of viscosity. Many different types of high pressure vicometers and their range of application are described in more detail by Cook, Herbst, and King [19]. These methods generally implement a spherical ball as it moves through a viscous fluid under the influence of gravity or other forces. Placing this ball within a diamond anvil cell allows access to very high pressures however the viscosities that can be measured are obviously limited to those low enough for a macroscopic ball to progress through the material during the course of a reasonable experimental timescale. In practice this means that rolling and falling ball viscometry in a diamond anvil cell is limited to approximately 10^7 cP—eight orders of magnitude away from the glass transition. In the work previously cited, Cook *et al*. managed to extend this viscosity range an additional two and a half orders of magnitude, up to approximately $10^{9.5}$ cP, via placing the diamond anvil cell viscometer onto a centrifuge which greatly increases the g-forces driving the ball through the fluid. However this still limits the probe to a viscosity range many orders of magnitude away from the glass transition.

Though this probe cannot provide measurements in the vicinity of the glass transition it does, as shown in Fig. 1.8, provide access to a very large dynamic range of viscosities—over nine orders of magnitude. There are some experimental concerns even within its range of application. Specifically, rolling ball viscometry must rely on a correction or "wall factor" that accounts for dissipative losses as the ball rolls down the surface of the diamond cell. This correction factor is assumed to be constant as experimental variables such as pressure and temperature are changed. It is also generally assumed that the probe ball is rolling down the diamond cutlet without slipping. It is very difficult, if not impossible, to directly verify the degree to which these assumptions hold over the entire range of experimental parameters. However this work will show that data for the glass former glycerol obtained by Cook *et al*. via rolling ball viscometery appears to be reasonably consistent over its applicable range with data taken via other probes and non-related methods.

1.3.4 TGP

The TGP process is a very straight forward means of measuring the isochronal curve of the glass transition temperature T_g as a function of Pressure. A liquid is loaded into a diamond anvil cell and brought to a temperature and pressure such that it is well within the solid glass phase. It has been found advantageous to pre-indent the gasket, which hardens it, thus making higher pressures more achievable. Two or more rubies are arranged throughout the sample as shown in the far left DAC in Fig. 1.11. A pressure gradient is then applied to the DAC via asymmetric application of force as indicated by the dashed Isobaric Lines.

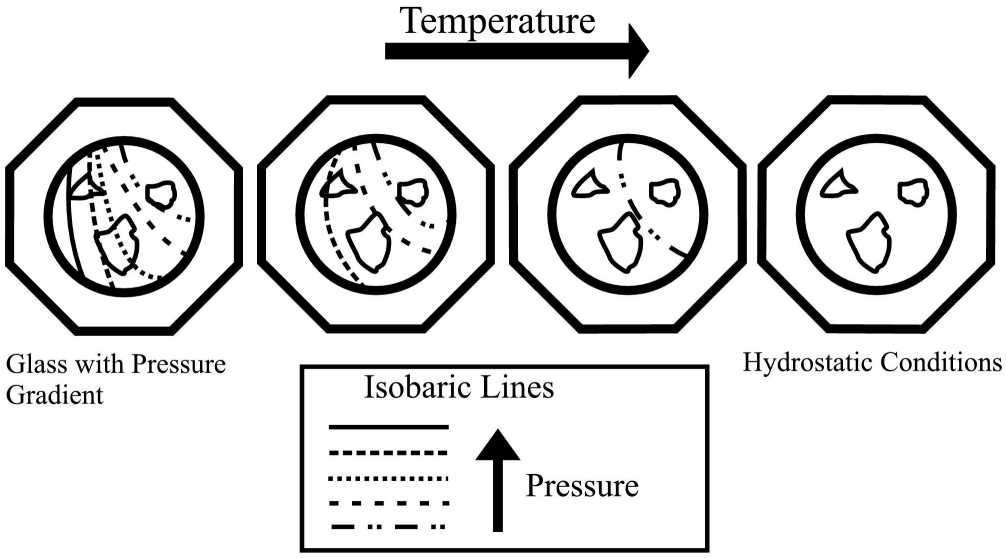


Figure 1.11: Established pressure gradients within a diamond anvil cell decrease as temperature increases and the system approaches the glass transition ultimately obtaining hydrostatic conditions. This figure is taken from Ref. [85].

Temperature is then increased incrementally, typically in 5 to 10 degree steps initially and then smaller steps as the glass transition is approached as evident via the pressure gradients giving way to hydrostatic conditions above T_g. At each temperature, pressure is measured from a consistent location on each ruby to again check for pressure gradients. Temperature is then increased to a new set point and a small set time is allowed for the sample to reestablish thermal equilibrium. A means of determining the approximate temperature at which the gradients dissolve and the material transitions into a liquid have been developed previously and described in Ref. [86, 85]. The values obtained for the glass transition temperatures as a function of pressure are fit to an Andersson-Andersson equation provided along with data for glycerol and cumene in App. B and C respectively[2]. The automation of this process will increase its efficacy in the future and may even allow it to provide more than a single isochronal curve near and around the glass transition. The automation of the TGP process is discussed in more detail in Sec. 6.2.

1.3.5 Dielectric Loss Spectrometry

Frequency Response Analysis includes several probes which collectively allow for measurements of molecular dynamics over a huge dynamic range. However the primary method of interest in this work will be dielectric spectroscopy (DS). This method takes advantage of molecular dipole moments within the material under study which ether exist permanently or are induced by an oscillating external field. As molecular orientations are driven by this oscillating external field, the frequency of the field is swept through frequencies from fractions of a hertz up to several GHz. Dielectric loss spectra are obtained and the frequencies of the peaks of these loss spectra at different temperatures and pressures can be used to determine molecular dynamics such as structural relaxation times and faster secondary relaxations.

True, the characteristic loss spectra or imaginary part of the dielectric constant as a function of frequency gives relaxation times typically correlated with the structural relaxation time, but the real part also gives valuable information including a measure of tau. Furthermore, combining the real and imaginary parts give rise to Cole-cole plots or Cole-Davidson plots, from which other information is extracted such as the stretching parameter beta. The point is that a slightly longer discussion of dielectric spectroscopy would be warranted in a dissertation on the glass transition.

2 Supporting Theory

This chapter will provide the working definition of fragility as originally defined by Austin Angell and is pervasive through the literature [3]. This classic definition of fragility will be implemented in this work to compare results with those previously obtained in other works. However a modified definition of fragility, arguably a more physically satisfying definition due to its being a dimensionless quantity among other considerations, will be introduced as well. This dimensionless fragility allows one to more readily compare fragilities obtained via different probes—some measuring viscosity, η, and others measuring the characteristic structural relaxation time, τ_α. Beyond this, focus will turn to a simple derivation of the scattering field as well as provide important theory coupling this scattered radiation field to the dynamics of the liquid under study. Next we will discuss the spectral structure of light scattered from a simple fluid and how this translates into the time domain in the case of intensity autocorrelation. Finally, mathematical models which describe these scattered fields and allow extraction of thermodynamic parameters of interest are provided.

2.1 Fragility

Fragility is a useful parameter for describing the dynamic behavior of a system as the glass transition is approached. Fragility describes how rapidly, not in the temporal sense but rather as a function of temperature, a system vitrifies as the glass transition is approached—specifically via thermal quenching. The viscosity and associated relaxation dynamics of a certain subset of glass forming liquids known as Arrhenius systems, which will be described in Sec. 2.1.1, exhibit a nearly exponential dependency on temperature. Other glass forming systems are found to be

exceedingly non Arrhenius—what will be referred to as fragile systems. In all cases, the viscosity and associated relaxation dynamics change by many orders of magnitude as a liquid is cooled or compressed, from very high temperatures or equivalently low pressure, into an amorphous solid state. For these reasons it makes sense to define fragility, as Angell originally did, in terms of the slope of the logarithm of viscosity, or structural relaxation time, as a function of dimensionless temperature at the glass transition as provided in Eq. 2.1.

$$m = \left(\frac{\partial \log_{10} \eta}{\partial \left(T_g / T \right)} \right)_{T=T_g} \tag{2.1}$$

To further understand the motivation behind this definition of fragility we introduce the Angell Plot as shown in Fig. 2.1 as applied to three typical glass forming liquids exhibiting varying degrees of fragility. The x-axis of a classic Angell plot is the inverse temperature normalized to the glass transition temperature, T_g, under isobaric conditions. This assures that all liquids, regardless of their individual glass transition temperatures, will converge at $T_g / T = 1$ on the right side of the Angell plot corresponding to a viscosity of $\eta_g = 10^{13}$ Poise or an alpha relaxation time of $\tau_g = 100$ seconds–the values for which the glass transition has been here defined.

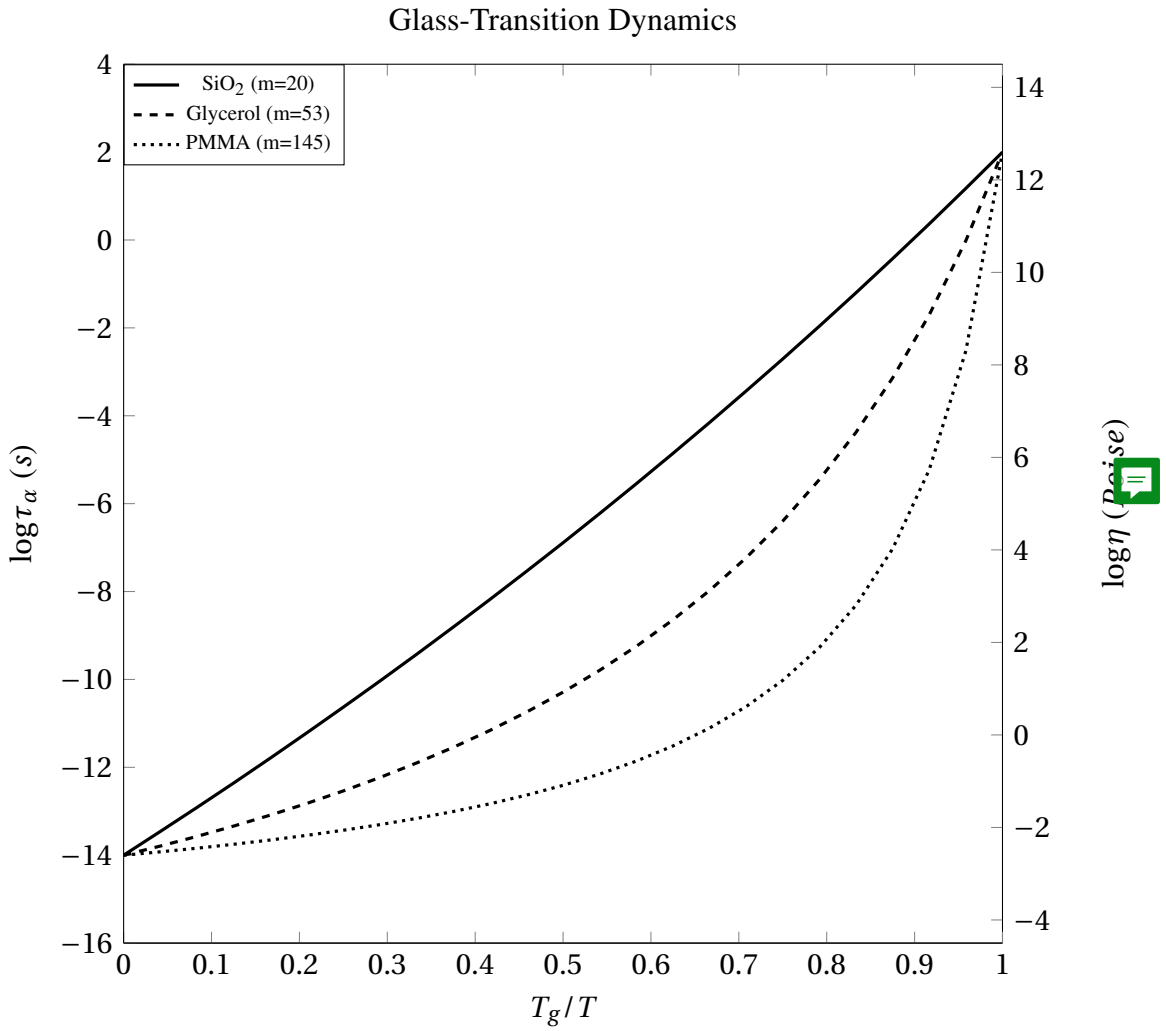


Figure 2.1: An example of an Arrhenius plot, commonly refereed to as the Angell plot, providing a non-ambiguous visual portrayal of the fragility of several glass forming systems.[3]

As temperature approaches infinity the average thermal energy of any given molecule is orders of magnitude higher than the potential felt from nearest neighbor interactions. Thus, from an energy perspective, these molecules are effectively free particles. Though they may yet lack the energy to leave the bulk, they are free to move about the energy landscape relatively unimpeded. Thus, in the high temperature limit, relaxation times will depend only on internal parameters such as molecular moments of inertia. Most organic molecules are on the order of a few angstroms. Macro molecules and long chain polymers may be much larger however even these have a general

"floppiness", or ability to fold and bend on the scale of several angstroms and thus, when talking about relaxation dynamics, all of these molecular relaxations are happening on a similar spatial scale. As such it is not surprising that all liquids in any given isobaric condition will have an infinite temperature relaxation time, τ_α on the same order of magnitude–from hundreds of femtoseconds to a few picoseconds.

Figure 2.1 shows the Angell plot for three very different glass forming systems. Silicon dioxide is shown as an example of a system in which strong covalent bonding dominate intermolecular interactions and is also a classic example of an Arrenius or "strong" system. Glycerol is provided as an example in which hydrogen bonding plays a major role in intermolecular interactions, at least at low pressures, and is an intermediately fragile, or non-Arrenius, system. The polymeric system PMMA is provided as an example of an exceedingly fragile glass forming system.

The Angell plot provides an effective means to describe and compare the dynamics of drastically differing glass forming systems on a single plot. It also very effectively allows for visualization and comparison of their fragilities. Note that, as defined in equation 2.1, the fragility of these liquids is geometrically represented as the slope of these Angell plots where $T_g/T = 1$ at the glass transition. Thus one can determine the relative fragilities of vastly different liquids, or for that matter a single liquid at various pressures as was the case for Cook *et al.*, easily at a glance.[21]

As the horizontal axis of a proper Angell plot will always progress from zero to one and most liquids will have a common infinite temperature viscosity on the order of 10^{-2} Poise, the minimum fragility possible will be the slope of a straight line stretching from the infinite temperature viscosity to that at which the glass transition is defined to occur. Generally $\eta_g \equiv 10^{13}$ Poise

thus the fragility of a purely Arrhenius liquid will be on the order of $m \approxeq 15$. This is an awkward minimum value for a parameter such as fragility. In hindsight it would have made more sense to define the Y-axis of the Angell plot as a dimensionless quantity and adjust our definition of fragility as follows:

$$m^* = \frac{1}{\log_{10}\left(\frac{\eta_g}{\eta_\infty}\right)} \left(\frac{\partial \log_{10}\left(\frac{\eta}{\eta_\infty}\right)}{\partial \left(T_g / T\right)} \right)_{T=T_g} \tag{2.2}$$

In practice, this redefinition only serves to scale the classical definition of fragility by a correction factor thus would not add any complexity to data analysis and yet would precisely constrain both the high and low temperature limits of the Angell plot, remove some minor ambiguity in the definition of fragility in that a purely Arrhenius system will exhibit a fragility of precisely $m^* = 1$, and involve only dimensionless quantities which would aid in comparisons between viscosity and structural relaxation data.

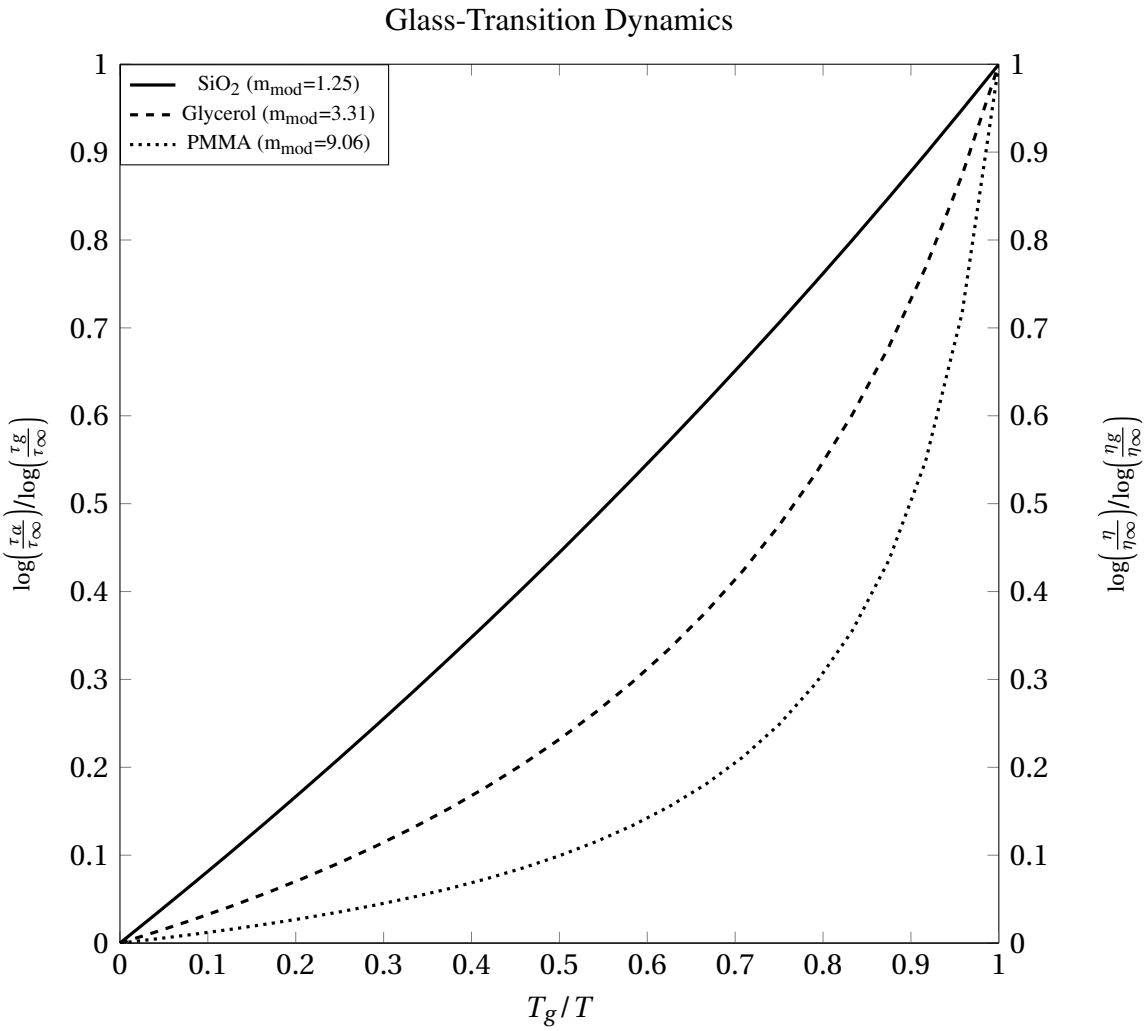


Figure 2.2: A modified (normalized) Angell plot shown here is constrained at the glass transition as well as in the high temperature limit and involves only dimensionless quantities.

The associated modified Angell plot provided in figure 2.2 is effectively identical to that in figure 2.1 however the infinite temperature relaxation times or viscosities in the classic Angell plot may not necessarily be identical and thus the plots do not converge to the same viscosity or relaxation times at high temperatures yet, in the modified Angell plot, the plots are all constrained to values between precisely zero and one. In this modified Angell plot, both viscosity and structural relaxation data share the same dimensionless scale.

2.1.1 Arrhenius Systems

Arrhenius or "strong" glass forming systems, in the most general terms, are those whose dynamic behavior upon approaching the glass transition appear to be governed by a single relaxation process. For a perfectly Arrhenius system, this process will have a single characteristic activation energy, or equivalently volume, and the probability distribution of molecules in the bulk material having sufficient energy to undergo a relaxation at a temperature, T, will be equivalent to the population distribution of any two state system having an energy gap of ΔE. Thus, as the characteristic structural relaxation time or viscosity are dependent upon these energy activated events, they are expected to be proportional to the Boltzmann distribution as follows.

$$\tau_\alpha = \tau_\infty \exp\left[\frac{\Delta E}{k_b T}\right] \tag{2.3}$$

Here τ_∞ is the infinite temperature relaxation time, previously discussed, in the limit where thermal energy is high enough that the potential energy landscape does not contributed appreciably to molecular motions. A classic example is silicon dioxide, or SiO_2, shown in the Angell plot of figure 2.1 above. Note the nearly linear relationship between viscosity or structural relaxation time and inverse temperature which is to be expected on a log plot for a strong glass former described by equation 2.3 above. It should be clear from either figure 2.1 or 2.2 that there is a minimum fragility possible which classically depends on how one defines the glass transition but generally will be between $m = 15$ and $m = 16$. Classically, as discussed above, this is not a dimensionless quantity. However, if one were to adopt the modified Angell plot and its associated definition of fragility, the minimum possible fragility for any glass forming system will be precisely $m^* = 1$ corresponding to perfectly Arrhenius dynamics.

2.1.2 Non-Arrhenius Systems

It should be clear, however, most systems do not exhibit purely Arrhenius behavior. Even the classic strong glass formers such as SiO_2 exhibit some level of curvature on an Angell plot and, sure enough, a fragility of $m \approx 20$ ($m^* \approx 1.25$) is significantly above that expected to arise from a single discrete activation energy described by equation 2.3 above. Realistically, any molecule or particle that does not possess perfect spherical symmetry is likely to exhibit more than one mode of relaxation giving rise to the deviations from Arrhenius dynamics seen in figures 2.1 and 2.2. Glass forming systems that deviate greatly from Arrhenius behavior are referred to as "fragile" glass formers to reference to their rapid rate of vitrification approaching the glass transition—rapid in this case again referring, not to the rate of thermal quenching, but rather to the temperature derivative of viscosity. PMMA is a classic example of a fragile glass forming system whereas glycerol is typically describe as being of intermediate fragility. Fragility in non-Arrhenius systems may have pressure dependency. Glycerol is a prime example as its fragility increases with increasing pressure–the result of hydrogen bonding being suppressed. Chapter 3 will explore this pressure dependency of glycerol fragility at some length. However the isobaric dynamics of such systems are "super-exponential" in that their rate of vitrification is more rapid than that which would be predicted by a pure exponential model. These dynamics are classically modeled via the Vogel-Tamman-Fulcher(VTF) equation first proposed by Tammann and Hesse as far back as 1929 to describe the temperature dependence of viscosity in polar liquids.[103] The common modern form of this equation most prevalent in the literature is as follows:

$$\tau_\alpha = \tau_\infty \exp\left[\frac{DT_0}{T - T_0}\right] \tag{2.4}$$

Here T_0 is often referred to as the zero mobility temperature at which constituent molecules or particles lack sufficient energy to navigate their potential landscape—structural relaxation time, τ_α, diverging to infinity. The parameter D is referred to as the fragility parameter as it is closely related to the fragility and, in fact, is often directly used as another means of defining fragility.

2.2 The Scattered Field

Any time light travels through a transparent medium the electromagnetic fields will couple with the charges within the medium inducing electronic moments. This interaction can be quite complicated resulting in a scattered field that requires a multi-pole expansion to properly describe. However, in the case where the wavelength is large relative to the dimensions of the scattering bodies, the electric dipole term generally dominates and induced dipole moments can be calculated assuming static field conditions. The general form of the scattered field, though technically a quantum effect, can be derived via classical electrodynamic theory and is very well described in general terms by both Jackson and Lipschitz while Berne and Pecora offer a more practical rendition of Landau's derivation specifically in the context of dynamic light scattering[61][49][8]. In all cases, an incident monochromatic plane wave is assumed to propagate through a scattering medium as depicted in figure 2.3 below.

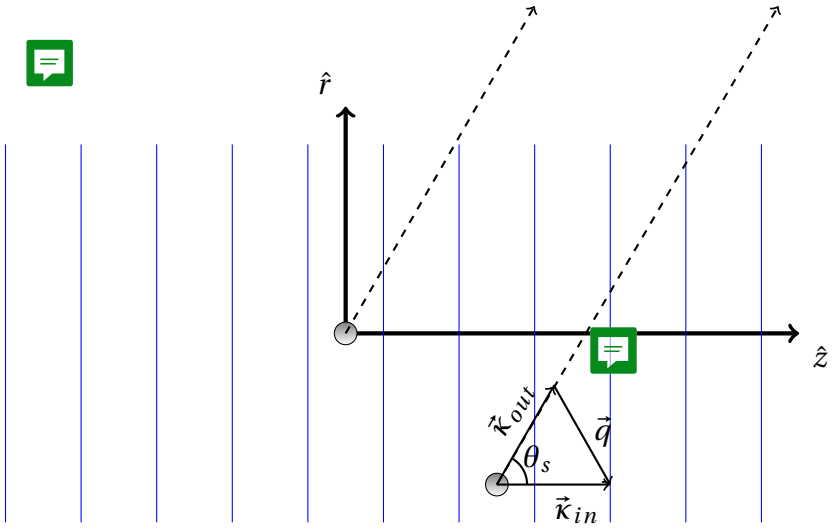


Figure 2.3: Vertical lines represent a plane wave, propagating in the $\hat{z}$ direction, into a region of disperse scattering bodies. The time dependent dipole moments induced in each molecule or particle acts as an independent source of a scattered field. Here we assume a diffuse scattering medium such that secondary scattering events constitute a negligible contribution to the net scattered field.[8]

It is important to its applicability as a structural probe that the scattering medium be sufficiently diffuse such that secondary scattering events are rare as their prevalence can lead to a situation where data analysis is prohibitively complicated. However, in this diffuse limit, light scattered off the $\hat{z}$-axis, at some arbitrary angle θ_s, can be assumed to arise from a single scattering event. As the incident field's wave vector, $\vec{k}_{in}$, is well defined, and the outgoing wave vector at which scattered light is collected, $\vec{k}_{out}$, can be precisely controlled, the scattering vector, $\vec{q}$, can be well defined. This scattering vector will be important shortly.

To understand the light scattering process, consider a system consisting of spherical particles suspended in a liquid medium. The number density of these particles in any given region will randomly vary in time due to their Brownian motion. The same will be true for a pure liquid as molecules move about and create short lived local density fluctuations within the bulk. Even in a sold crystalline lattice, vibrational modes will induce local density fluctuations. Amorphous solids

will also exhibit spatial fluctuations in local density. Thus to describe how light is scattered from a bulk medium one must recognize that the dielectric permittivity of any real material will fluctuate, potentially both temporally and spatially, due to the random thermal motions of the constituent particles or molecules as well as any structural inhomogeneities and can be modeled as a static "average" permittivity around which local fluctuations occur as follows:

$$\epsilon(\vec{r}, t) = \epsilon_0 I + \delta\epsilon(\vec{r}, t) \tag{2.5}$$

Here ϵ_0 is the average scalar dielectric permittivity within the bulk material, notably not the permittivity of free space, and $\delta\epsilon$ describes the deviations from this constant which will, in general, be time and space dependent and may include off diagonal elements thus providing a completely general description of local permittivity for any transparent medium.

From this point we will derive the wave equation of the scattered field to provide an understanding of what precisely gives rise to the scattering field. Landau and lifshitz as well as ~~David~~ Jackson provide excellent classical derivations of the scattering field however we will follow the more simplified derivation of Berne and Pecora though all three will serve as excellent references if one wishes to understand the process of light scattering in greater depth ~~than will be provided here~~ [61][49][8]. As illustrated in figure 2.3 above, if a plane wave traveling through a medium is to produce a scattered field, the resulting field in the medium will be the superposition of the

incident field and the scattered field as follows:

$$E = E_i + E_s \tag{2.6}$$

$$D = D_i + D_s \tag{2.7}$$

$$H = H_i + H_s \tag{2.8}$$

Whatever form the scattering fields assume, they must satisfy Maxwell's equations as follows:

$$\nabla \times E_s = -\frac{1}{c}\frac{\partial H_s}{\partial t} \tag{2.9}$$

$$\nabla \times H_s = \frac{1}{c}\frac{\partial D_s}{\partial t} \tag{2.10}$$

$$\nabla \cdot H_s = 0 \tag{2.11}$$

$$\nabla \cdot D_s = 0 \tag{2.12}$$

Making use of equations 2.9 and 2.10 one can obtain the following expression:

$$\nabla \times \nabla \times E_s = -\frac{1}{c^2}\frac{\partial^2 D_s}{\partial t^2} \tag{2.13}$$

The net displacement vector from equation 2.7 is related to the electric field vector from equation 2.6 by the dielectric tensor from equation 2.5 and can be expressed as follows:

$$D_i + D_s = (\epsilon_0 I + \delta\epsilon(\vec{r}, t)) \cdot (E_i + E_s) \tag{2.14}$$

Solving this for the scattering field vector, E_s and ignoring the term $\delta\epsilon \cdot E_s$ that would give rise to

secondary scattering events of significantly lesser magnitude, we arrive at the following expression:

$$E_s = \frac{1}{\epsilon_0} D_s + \frac{1}{\epsilon_0} (\delta\epsilon \cdot E_i) \tag{2.15}$$

This scattering field vector must satisfy equation 2.13 above and thus we have the following:

$$\nabla \times \nabla \times \frac{D_s}{\epsilon_0} + \nabla \times \nabla \times \frac{(\delta\epsilon \cdot E_i)}{\epsilon_0} = -\frac{\epsilon_0}{c^2} \frac{\partial^2 D_s}{\partial t^2} \tag{2.16}$$

Utilizing Lagrange's vector cross-product identity, this may be simplified as follows:

$$\nabla^2 D_s - \frac{\epsilon_0}{c^2} \frac{\partial^2 D_s}{\partial t^2} = -\nabla \times \nabla \times (\delta\epsilon \cdot E_i) \tag{2.17}$$

Bernie and Pecora go on to simplify this expression further however it is unnecessary for our purposes here. Equation 2.17 is a simple wave equation for the scattering displacement vector, D_s, with a source term proportional to $\delta\epsilon \cdot E_i$. This driven wave equation was derived in the most general of terms, assuming only that all propagating fields are consistent with Maxwell's equations and that the dielectric permittivity can be described by the second rank tenser quantity of equation 2.5. Yet, from equation 2.17, much can be said regarding what processes can give rise to a scattered field. Consider a perfect crystalline lattice, in the absence of any inclusions or impurities, maintained at a temperature of absolute zero. For such a material, the second term of the dielectric permittivity provided by equation 2.5 will have no spatial dependence, thus the spatial derivatives associated with the cross products in the source term of the wave equation for the scattered Displacement vector will always be zero, in which case no light scattering will occur in the medium. Equation 2.17 demonstrates that it is the deviations of the dielectric permittivity

from the average value within the medium which give rise to light scattering.

However, if this crystal lattice contains inclusions or impurities, the dielectric permittivity will exhibit non-zero partial derivatives at the locations of these imperfections which will correlate to a non-zero source term and give rise to scattered light. Similarly, consider now an amorphous glassy material at a temperature of absolute zero. Though no molecular motions exist, the precise density at any specific location within the material may well have spatial dependency. As density varies it stands to reason that dielectric permittivity will vary accordingly. Thus the cross products in the source term of equation 2.17 will, again, likely be nonzero. This will also give rise to light scattering from within the material. It should also be noted that, as these materials exhibit no motion at absolute zero temperature, the dielectric permittivity can only vary as a function of location. Though off diagonal components of the second rank tensor, $\delta\epsilon$, may give rise to a scattered field with a polarization other than that of the incident field, E_i, that scattering field must have the same frequency components as the incident field. Any material at absolute zero would be expected to exhibit purely elastic scattering.

Of course, in general, materials are never at a temperature of absolute zero. Even perfect crystalline lattices, which at absolute zero would not be expected to give rise to scattered light, do so at elevated temperatures. Vibrational modes within a crystal lattice will give rise to time dependent density fluctuations within the material—thus why the dielectric permittivity of equation 2.5 is modeled as having temporal and spatial dependency. In a perfect crystalline material above absolute zero light will scatter from phonons. In a liquid or glassy system one cannot speak meaningfully of light scattering from phonons as there is no well defined lattice however density fluctuations still give rise to a temporal and spatial dependency in the dielectric permittivity which will, in turn, give rise to scattered light. One can do well to conceptualize this as light scattering

from elastic wavefronts–still just propagating sound waves but now within a liquid or amorphous solid. However, due to the time dependency in the dielectric permittivity, the frequency transform of the source term in the scattering wave equation will necessitate that the scattering fields contain additional frequency components not present in the incident field, E_i. As this temporal dependence is driven by density fluctuations, which are in turn directly correlated to molecular motions, it is not surprising in the least that the spectrum of light scattered from a material contains information regarding molecular dynamics within the material. In the following section we will derive a model for the spectrum of light scattered from a simple fluid implementing some basic fluid dynamics. From such models one may obtain bulk properties of a material including, but not limited to, the speed of sound, bulk modulus, and of particular interest for this work, the characteristic structural relaxation time.

2.3 Spectrum of a Simple Fluid

Until this point, the scattering field has been discussed in very general terms. Moving forward the discussion will be focused on the application at hand–light scattering as a probe of molecular dynamics. As such, the incident field, E_i, will be assumed to be a monochromatic constant intensity light source passed through a transparent fluid from which scattered light will be collected along a specific scattering vector as shown below.

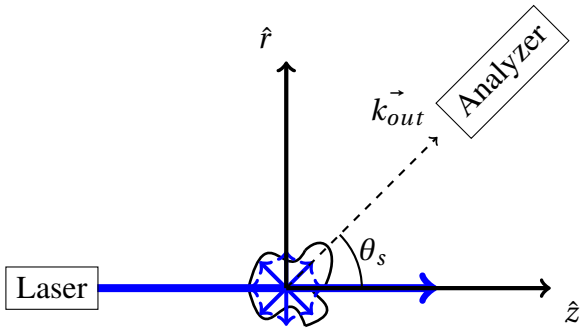


Figure 2.4: Generalized light scattering geometry

It is also important to note that, within the context of dynamic light scattering, the wavelength of the incident field, E_i, will always be very large when compared to intermolecular spacings within the scattering region. This was implicitly assumed in the derivation of equation 2.17 previously as scattering events ~~that~~ arising from dynamic variations in dielectric susceptibility emerge stochastically from the independent motions of large numbers of constituent molecules or particles. Within any fluid above absolute zero there will be local regions where bulk thermodynamic properties such as entropy, temperature, or pressure will fluctuate about the average value. It is these fluctuations, as alluded to previously, that give rise to the scattering field. Thus it is reasonable to invoke hydrodynamic and thermodynamic theory to arrive at a model for the spectrum of light scattered from a simple fluid. Such a model was first derived by Komarov and Fisher in their 1962 publication. In this work Komarov adapted neutron scattering models derived by Leon Van Hove in 1954 to describe light scattering spectra. It should be noted that this derivation is also provided in excellent detail by Bernie and Pecora.[5][7][8] The resulting spectra of this scattered light was found by Komarov to be the spatial and temporal Fourier transform of the time dependent

radial distribution functions described in section 1.1.1 and is modeled as follows:

$$I'(\vec{r},\omega) = I_0 N \omega^4 \sin^2\gamma S(\vec{q},\omega) \tag{2.18}$$

This result is, of course, quite similar to the result for Reyleigh scattering from a defuse gas with the addition of the structure factor, $S(\vec{q},\omega)$, which contains information regarding molecular dynamics of the system. Mountain later took Komarov and Fisher's results and, via application of the linearized hydrodynamic equations, the Navelei-Stokes equation, and the energy transport equation, derived a model for the frequency dependent structure factor arising from a classical fluid comprised of spherically symmetric molecule as follows:

$$S(q,\omega) = \frac{c_p - c_v}{c_p}\frac{2\lambda q^2/\rho_0 c_p}{\left(\lambda q^2/\rho_0 c_p\right)^2 + \omega^2} + \frac{c_v}{c_p}\left[\frac{\Gamma q^2}{\left(\Gamma q^2\right)^2 + \left(\omega + C_0 q\right)^2} + \frac{\Gamma q^2}{\left(\Gamma q^2\right)^2 + \left(\omega - C_0 q\right)^2}\right] \tag{2.19}$$

Here p_0, c_p, c_v, C_0, and λ are the equilibrium density, specific heat at constant pressure, specific heat at constant volume, low frequency limit of the sound velocity, and the thermal conductivity respectively.[73] The first, left-most term of equation 2.19 represents the broadened central peak of light scattering from entropy fluctuations which occur at constant pressure. In a simple fluid, as there are no internal degrees of freedom, this takes the form of diffusion or breaking of nearest neighbor cages–structural relaxation. The second, or right-most term of equation 2.19 represents light scattered ~~off of~~ propagating acoustic waves. As this is an adiabatic process, this occurs at constant entropy. This term forms two peaks symmetrically located around the central peak, the frequency of which is shifted from that of E_i proportionally to the speed of sound. These are often referred to as the longitudinal acoustic modes. This model is sufficient to describe the

dynamics of simple systems with no internal degrees of freedom ~~such as nano spheres suspended in a viscous fluid~~. However, your average glass forming system will have additional modes of structural relaxation.

Many glass forming systems are significantly more complex, exhibiting internal degrees of freedom, intermolecular hydrogen bonding, and other additional considerations. Polymer systems can fold and flex. Even simple molecular systems limited to Van Der Waals interactions, such as cumene, have rotational degrees of freedom that give rise to additional relaxation modes. Such considerations will add to the complexity of this structure factor. Mountain followed this previous work with a derivation of a structure factor accounting for internal degrees of freedom weakly coupled to the originally considered translational degrees of freedom via a single representative orientational relaxation time, τ as follows:

$$
\begin{aligned}
S(q,\omega) = \frac{c_p - c_v}{c_p}\frac{2\lambda q^2/\rho_0 c_p}{\left(\lambda q^2/\rho_0 c_p\right)^2 + \omega^2} + \\
\left[\frac{\left(c_\infty^2 - c_0^2\right)k^2 - \left(v^2/c_0^2 - 1\right)\left(c_0^4/v^4\tau^2 + c_0^2 q^2\left(\frac{c_p - c_v}{c_p}\right)\right)}{c_0^4/v^4\tau^2 + v^2 q^2}\right]\left[\frac{\frac{2c_0^2}{v^2\tau}}{\frac{c_0^4}{v^4\tau^2} + \omega^2}\right] + \\
\left[\frac{\left[1 - \frac{c_0^2}{v^2}\left(\frac{c_p - c_v}{c_p}\right)\right]\left[v^2 q^2 + \frac{c_0^2}{v^2\tau^2}\right] - \left(c_\infty^2 - c_0^2\right)q^2}{\frac{c_0^4}{v^4\tau^2} + \frac{v^2}{q^2}}\right] \times \\
\left[\frac{\Gamma_B}{\Gamma_B^2 + \left(\omega - vq\right)^2} + \frac{\Gamma_B}{\Gamma_B^2 + \left(\omega + vq\right)^2}\right]
\end{aligned}
\tag{2.20}
$$

Here the longitudinal acoustic modes represented by the third term are structurally indistinguishable from the second term of equation 2.19 though now the coefficient is τ dependent.[72] Likewise the first term, associated with translational diffusion, remains unchanged from before. However, most notably, there now emerges a secondary central feature symmetric about the incident fre-

quency, associated with this internal relaxation mode. Thus the orientational dynamics directly effect the structure of this central feature. There is a vast amount of insight provided by this model. However the primary consideration relevant to this work lies in this additional central term of equation 2.20 above. Consider the following simplified expression of this structure.

$$f(\omega) = \left(\frac{2\Gamma}{\Gamma^2 + \omega^2} \right) \tag{2.21}$$

This function, as before mentioned, will be symmetric about the axis $\omega = 0$ and will have a characteristic half width half maximum value of Γ as shown in figure 2.3 below.

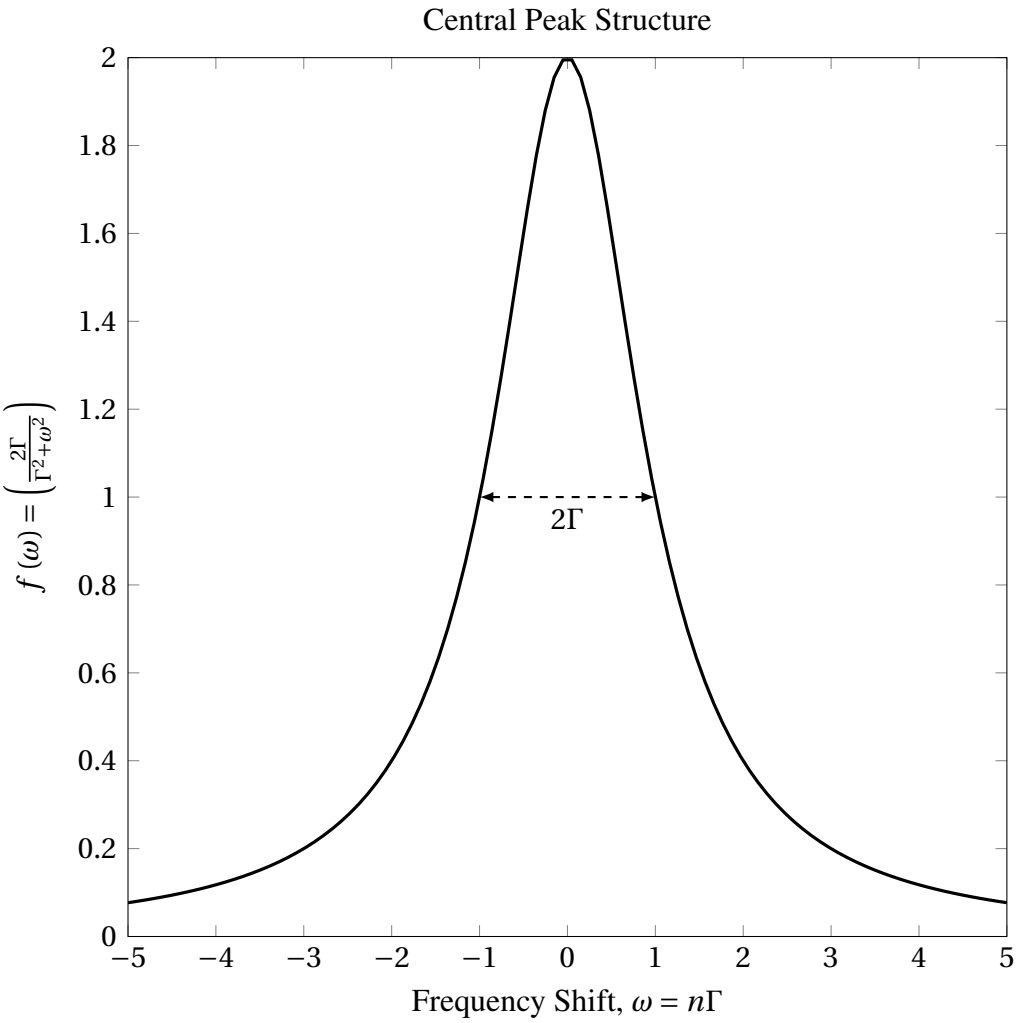


Figure 2.5: Here we have the generalized structure of the central features from equations 2.19 and 2.20. The x-axis here is in terms of multiples of the generalized parameter Γ from equation 2.21 which also defines the half width-half maximum value for this feature.

Placing aside the specific values of any particular bulk thermodynamic property, both the first and second term of equation 2.20 have the same basic structure in this model. Generally speaking, the structural relaxation associated with translational and orientational diffusion can occur on vastly different timescales however both will contribute to the width of this central feature. It bears mention that, for a simple mono-atomic fluid, only the translational diffusion term contributes to this central feature and $\Gamma = \frac{\lambda q^2}{\rho_0 c_p} q^2 = D_T q^2$ where D_T is the classic Stokes-Einstein coefficient of

self diffusion and there exist a dependency on the scattering vector. However, if the orientational diffusion contribution dominates the width of the central feature then $\Gamma = \frac{{c_0}^2}{v^2\tau} = \frac{D_\tau}{\tau}$ and there is no dependence on the scattering vector.

**This is where Demoulin gets further discussed.

2.4 Mathematical Models

–The Debye Relaxation

–Cole-Cole / Cole-Davidson

–Stretched Exponential

–The connection between time and frequency domain

3 The α-Relaxation Dynamics of Glycerol

Glycerol is perhaps one of the most widely studied glass forming systems. As a hydrogen bonding liquid, it exhibits more complicated behavior than a simple van der Waals liquid such as cumene which will be discussed in Ch. 4. A topic of significant interest within the literature over the past twenty five years has been the behavior of fragility as a function of pressure in such hydrogen bonding systems. Whereas most molecular liquid systems exhibit either a decreasing fragility as pressure increases or very little pressure dependence at all, glycerol and other hydrogen bonding liquids have been observed to show an increase in fragility which has been attributed to the suppression of hydrogen bonding under pressure [15, 80, 84]. This chapter will explore the current state of the literature on this topic, apply a new model to the dynamics of glycerol over a wide range of pressure and temperature with the aid of literature data obtained through multiple methods, reassess the efficacy of viscosity data obtained by Cook *et al*. that has come under some scrutiny, further compare resulting thermodynamic scaling regimens with previous literature results, as well as provide a more precise measure of the pressure dependence of fragility in glycerol.

3.1 Earlier Works Regarding Glycerol Dynamics

As previously asserted, glycerol is one of the most widely studied glass forming systems, with the temperature dependence of its viscosity being investigated as far back as 1926 in the work of Tammann and Hesse [103]. However the modern history of glycerol, as relevant to this work, arguably begins with Davidson and Cole's investigation of the temperature dependence of the

dielectric relaxation in glycerol circa 1951 [23]. Davidson and Cole found that not all associated liquids can be modeled with a simple single-relaxation or Debye equation as shown in Eq. 3.1 [26].

$$\epsilon^* = \epsilon_\infty + \frac{\epsilon_0 - \epsilon_\infty}{1 + i\omega\tau_0}, \tag{3.1}$$

where ϵ_∞ and ϵ_0 are the susceptibilities in the infinite-frequency and steady-state limits respectively. Though 1-propanol was found to be well described by a simple Debye relationship, glycerol exhibited a higher degree of dispersion as vitrification accelerates and required unrealistically small values for its molecular volume in order for the observed relaxation time to conform to that predicted by the Debye hydrodynamic model for hard spheres. Davidson and Cole correctly surmised that this was due to the existence of additional modes of relaxation available to glycerol that simply cannot exist for the simpler and more rigid 1-propanol molecule. To deal with this anomalous dielectric dispersion, Davidson and Cole suggested modeling such associated systems with an empirically derived "skewed" Debye like model ~~as follows in Eq. 3.2.~~

$$\epsilon^* = \epsilon_\infty + \frac{(\epsilon_0 - \epsilon_\infty)}{(1 + i\omega\tau_0)^\beta}, \tag{3.2}$$

where β is the "skewing parameter" and reduces to Eq. 3.1 when $\beta = 1$. Starting with Cole and Cole, it became common in the literature of this time to plot the real *vs*. imaginary components of this empirical expression as would have been common for the analysis of an analogous RC circuit. With regards to complex susceptibility, in the context of dielectric spectroscopy, the real part of Eq. 3.2 corresponds to the non-dissipative component and the imaginary part to the dissipative component. This can be modeled, for a simple Debye relaxation where $\beta = 1$, in terms of a three component RC circuit as follows:

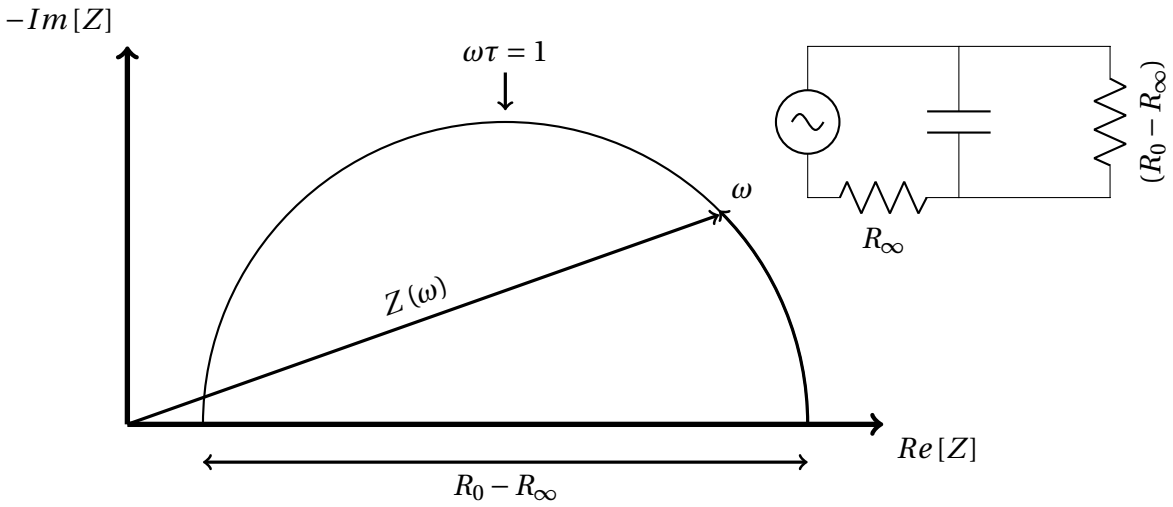


Figure 3.1: A system with a frequency dependent susceptibility and a single characteristic relaxation time τ driven at a given frequency ω can be modeled as an RC circuit as show here in the upper right corner. The complex impedance will be of the form $Z(\omega) = R_\infty + \frac{R_0 - R_\infty}{1+i\omega\tau}$ which is functionally identical to the dielectric susceptibility of the simple Debye relaxation of Eq. 3.1. Though the component which plays the role of irreversible dissipation is reversed in the case of the RC circuit above, both complex dielectric relaxation and this analogous electrical impedance will exhibit maximum dissipation at resonance where $\omega\tau = 1$.

Note that the magnitude of impedance will progress from R_0 to R_∞ as the driving frequency is increased from $\omega = 0$ to $\omega \to \infty$. The impedance of the two limiting cases, R_0 and R_∞, are analogous to the limiting values of the dielectric susceptibility ϵ_0 and ϵ_∞ from Eq. 3.1 respectively.

As previously mentioned, the Cole-Davidson model described by Eq. 3.2 above may be described as a "skewed" Debye model when $0 < \beta < 1$ as it results in a distortion of the corresponding Cole-Cole plot. An example of such a skewed plot of glycerol where $\beta = 0.6$ and n-propanol where $\beta \approx 1$, as obtained by Davidsion, is shown below. More importantly, Davidson provides a closed form expression for this distribution of simple Debye relaxations, first calculated by Fuoss and Kirkwood to describe dipole moments in polymers, which give rise to the Cole-Davidson model of Eq. 3.2 and this will be described in more detail in Ch. 4 [33]. For now, we assert that this skewing of the dielectric dispersion, increasing as the value of β decreases, corresponds to a broader

distribution of Debye-like relaxation processes.

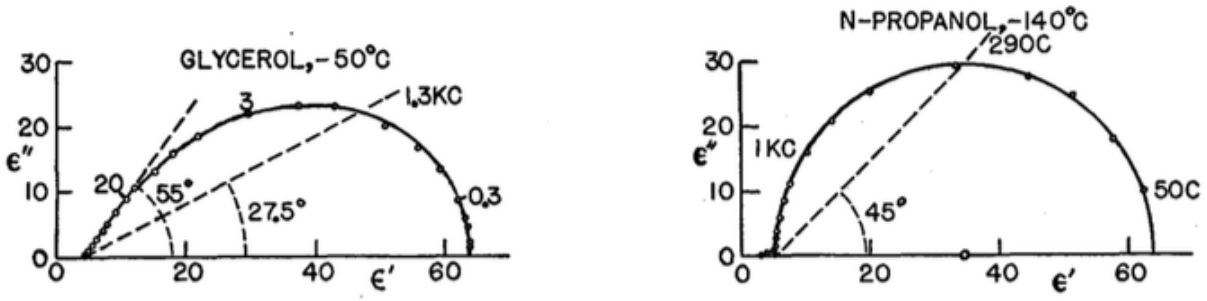


Figure 3.2: Cole-Cole plots of glycerol and n-propanol data (points). Solid lines represent fits to the Cole-Davidson model of Eq. 3.2 with beta-values of 0.6 and ≈ 1 respectively [23].

Davidson and Cole also noted that the value of β is essentially constant at ambient pressure over a temperature range from 40 °C to 75 °C. The implications of this being that the general structure of the distribution of relaxation modes is constant over a range of temperatures for which the dynamics of said relaxations are changing by at least six orders of magnitude. Similar results have been found for many systems over the years and for such systems, this is known as time-temperature superposition. More recently, the same has been found to hold over significant pressure ranges as well, in which case it is referred to as time-temperature-pressure superposition. We found such results for Cumene, which will be described in the next chapter—β being essentially constant over a wide range of temperature and pressure.

Though Davidson and Cole only considered the isobaric behavior of glycerol relaxation as a function of temperature, their contribution here provided some of the first data that could be used to estimate fragility—at least at atmospheric pressure. They demonstrated that the strong temperature dependence of glycerol relaxation is poorly described, even over this moderate temperature range, by a classical Arrhenius rate equation but was well fit, within experimental uncertainty, by the more empirical Vogel-Tammann-Fulcher (VTF) equation previously described and

first proposed by Tammann and Hesse to describe glycerol viscosity as a function of temperature [103]. Furthermore, they provided a closed form expression for the implied distribution of simple Debye relaxations underlying Cole-Davidson Eq. 3.2 above. ~~And~~ finally, they noted that the average molecular relaxation time of this distribution for glycerol would correspond to "hard-sphere" molecular dimensions, as provided by the classic Debye hydrodynamic theory, that are far too small to be physically realistic for glycerol, *e.g.* 1×10^{-25} cm^3, implying the existence of additional relaxation modes involving reorientation of molecular sub-units. It really is this observation, above all other contributions, that lead to the development and application of free-volume models for viscosity and structural relaxation upon which much of this chapter will rely. Due to the importance of free-volume in this chapter and throughout the literature, particularly with regards to modeling viscosity of systems beyond ambient pressure, a discussion and derivation of the free-volume model will be provided in Sec. 3.2.

Throughout the 1960s, work began to emerge within the literature probing viscosity and structural relaxation of glycerol beyond ambient pressure. One of the first to appear was published in 1964 by McDuffie and Kelly where data for both viscosity, *viā* a rolling ball viscometer, and relaxation data, *viā* dielectric loss spectroscopy, were obtained. These methods were described previously in Sec. 1.3.3and 1.3.5, respectively. McDuffie obtained viscosity data over three separate isotherms and dielectric relaxation data over a fourth which collectively covered a very small temperature range of $\Delta T = 20$ °C however he extended these isotherms over what was, at the time, a very large range of pressure from ambient up to 4 kbar. Beyond simply publishing some of the first high pressure data for glycerol viscosity, McDuffie's major contribution in this work was to show that neither the pressure nor temperature dependence of viscosity for glycerol was well described by the Eyring rate theory commonly used at the time and which will be described in the following

section. Also of considerable interest was that the free-volume model, also described in Sec. 3.2 below, was shown by McDuffie to yield a compressibility for glycerol that was not congruent with other high pressure values derived from ultrasonic probes at the time and also that the implementation of pressure-independent thermal expansion coefficients within the free-volume model leads to an over-estimation of the pressure dependence of viscosity. However these discrepancies were quickly addressed the following year, in 1966, by Matheson [68]. His arguably obvious approach was simply to apply a quadratic model to account for the pressure dependence of v_m in the free-volume expression (Eq. 3.29) from Sec. 3.2.1. An identical approach was applied for the temperature dependence for several other liquids however the limited temperature range in McDuffie's data was insufficient to meaningfully model a quadratic temperature dependence for v_m in glycerol. Still, in effectively addressing the shortcomings of early free-volume models exposed by McDufffie *viā* extending the model beyond ambient pressure, Matheson's contribution was very significant. What Matheson was in want of, and what was conspicuously lacking in the literature up to this point, would be carefully collected data for glycerol viscosity or structural relaxation spanning a large temperature and pressure range.

This data vacuum was quickly filled by Johari and Whalley in the early 70's when they published a very heavily sighted paper containing dielectric susceptibility measurements of glycerol spanning a dynamic range of six decades. These data also contained a much larger pressure and temperature range than any previous work—ambient to 53 kbar and −55 °C to 84 °C respectively [54]. At lower temperatures Johari and Whalley were able to probe the relaxation dynamics of glycerol very near the glass transition. At higher temperatures, above $T = 250$ K, their data falls around three decades short of approaching the glass transition. However their data has proven to be of superb quality and opened up the door to investigations of the pressure dependence of such

parameters as β, defined in Eq. 3.2, and much latter fragility. Similar contributions were made for several other glass-forming systems and many citations exist in the literature including this work, particularly to assess the pressure dependence of glycerol's fragility [53, 79, 31, 82].

Around a decade latter, in 1986, Forsman *et al*. published a paper also probing dielectric relaxation of glycerol [31]. They had the advantage of nearly a decade of advancement in instrumentation and thus were able to probe significantly faster relaxations, up to 13 MHz. However they did not provide a great deal of data and the pressure and temperature range was already well covered by Johari and Wally's previous work, particularly with regards to investigations into fragility as the faster relaxation times do not significantly add to the efficacy of fragility calculations. Though Forsman was focused more on the broadening of the loss peak as a function of pressure and furthers the discussion of many-body relaxation theory to account for deviations of the loss spectra from Debye theory, this work is worthy of mention here as it references several other papers providing data for glycerol relaxation collected over several decades.

Another work worthy of mention would be the work published by Demoulin in the early 70's [27]. Though data was only taken at ambient pressure and low temperatures, Demoulin probed glycerol relaxations directly in the time domain *viā* photon correlation spectroscopy as described in Sec. 1.3.2. Though his data was confined to a relatively limited dynamic range, as autocorrelators of this era were among the first and significantly slower than those currently commercially available, with the aid of ultrasonic measurements a reasonable dynamic range was obtained. The aim of Demoulin was not to extend the dynamic range of data available for glycerol relaxation, but rather to prove the efficacy of digital photon correlation spectroscopy as a probe of relaxation dynamics over what was an exceedingly large dynamic range for the time. In Ch. 4, a method for extending this probe into the range of very high pressures will be presented. It is interesting here

to note that, though the data obtained *viā* photon correlation spectroscopy is in the time domain, the Cole-Davidson distribution, being used pervasively throughout the literature to describe the dynamics of structural relaxation probed *viā* dielectric spectroscopy and ultrasonics methods, is most naturally expressed in terms of frequency. This work was published before that of Lindsey and Patterson, which worked out a simple means of translating between the Williams-Watts distribution, expressed most naturally in the time domain and commonly used to model data obtained *viā* photon correlation spectroscopy, and the Cole-Davidson distribution of Eq. 3.2 [63]. Thus it is not surprising that Demoulin chose to impart significant effort to fit his time domain data to the more common, yet inconvenient, Cole-Davidson distribution so that it may be directly compared to data within the literature. His method for accomplishing this will be extended in Ch. 4, taking advantage of modern computational tools, for exactly the same reasons.

Demoulin's method of PCS will later be well established to constitute a direct measurement of structural α-relaxation within a glass forming system. However another direct probe, this time of viscosity, was introduced before the end of the decade by Geils and Keezer [34]. The rolling ball viscometer as a means of directly measuring viscosity *viā* tracking the terminal velocity of a metal ball rolling down an incline, as previously described in Sec. 1.3.3, certainly predated the work of Geils and Keezer, however early implementations placed great restrictions on the geometry of the viscometer requiring that the ratio of the rolling ball diameter to that of the viscometer remain in one or the other limiting extremes—either rolling without slipping down a surface of the viscometer or lacking any significant interaction with the confining surfaces [101, 48]. Geils and Keezer worked out the means of accounting for the correction factors associated with wall effects for this method, allowing for greater flexibility in experimental design to allow for temperature and potentially pressure control, drastically increasing its efficacy as a probe for viscosity of glass

forming systems.

3.2 Free-Volume Theory

The concept of free-volume as a mechanism to explain viscosity was first postulated as far back as 1913 when Batschinski proposed a linear relationship between thermal expansion and viscosity [7]. Though this linear model will obviously suffice over any sufficiently small thermodynamic range, it fails spectacularly in the description of data from high temperatures down through a glass transition. However the general idea of describing viscosity in terms of solid molecular units passing through regions of free-volume is certainly conceptually appealing and was gaining more traction in the literature in the early fifties. Though Batschinski's model was known by this time to be insufficient to describe viscosity behavior beyond a first approximation, and certainly would only be applicable far from any glass transition where viscosity is rapidly diverging, there was no reason to believe the concept of free-volume was invalid. The main competing theory of viscosity at the time was the rate theory proposed by Henry Eyring, which one may assert is physically equivalent to free-volume theory though couched in terms of the energy fluctuations associated with creating that free-volume—in the simplest terms, pushing nearest neighbors out of the way [30]. Eyring proposed that viscosity of any non-associated liquid, to the extent that structural relaxation is assumed to be an energetic process which involves breaking free of potential wells created by neighboring molecules, may be treated in much the same way as any chemical process activated by thermal energy fluctuations and should be well described by a Boltzman-like distribution involving a single activation energy, E_0, and of the following form:

$$\eta = \frac{Nd}{V}(2\pi m k_b T)^{\frac{1}{2}} e^{\frac{E_0}{k_b T}} \tag{3.3}$$

It is not surprising that an associated liquid such as glycerol might fail to be well described by this Arrhenius model. However even most non-associated fragile glass forming systems also conform well to the phenomenological VTF equation. Thus if one is to overcome the apparent discrepancy between Eyring's rate theory of Eq. 3.3 above and the VTF equation, the functional dependence of the activation energy on temperature must be of the form $E(T) \propto \frac{T}{T-T_\infty}$, where T_∞ is the zero mobility temperature at which viscosity becomes infinite. This requirement rather diminishes the physical grounding of the Eyring rate theory as it offers no explanation as to why the temperature dependence of activation energy should take this form. Some have tried to link the temperature dependence of activation energy with local entropy fluctuations and met with some level of success [32]. In fact such a rate theory is the foundation for the Adam-Gibbs equation which has been heavily implemented to describe enthalpy relaxation in many glass forming systems [1, 46]. However the conceptual simplicity and elegance of a free-volume description may suggest that a non-linear free-volume equation, beyond that suggested by Batschinski, may be the most physically meaningful approach to describing viscosity of these non-Arrhenius systems. It was during the same year of 1951 that Doolittle published his work describing viscosity of n-paraffin hydrocarbons *viā* an exponential free-volume model, similar in form the the Eyring rate theory, yet dependent upon the relative free-space, $\frac{v_f}{v_0}$ within a bulk liquid rather than an activation energy,

$$\eta = Ae^{B\left(\frac{v_f}{v_0}\right)}, \tag{3.4}$$

where v_f is the volume of free-space per unit of liquid and v_0 is that free-volume density specifically evaluated at some reference temperature—often absolute zero [29].

This is useful as it circumvents the need to explain any functional dependence of a potentially complicated distribution of activation energies and attacks the problem from an arguably conceptually simpler standpoint of free-volume—is there sufficient free-volume, on average, for molecular mobility to progress at a given rate. Only thermal expansion needs to be considered as all other parameters are temperature independent. The Doolittle expression for free-volume of Eq. 3.4 above also provided the groundwork for Williams, Landel, and Ferry to explain the temperature dependence of kinetic constants as a function of temperature near the glass transition [111]. It was a major contribution that launched the free-volume model into common implementation. However Eyring's rate theory explanation was certainly not supplanted by a free-volume approach. Again, structural fluctuations that give rise to local free-volume, *i.e*. the opening of a path through which a caged molecule might diffuse, must be energetically accessible as must be the diffusion process itself. It should be unsurprising that a model relying on a temperature independent single activation energy to describe what, even in a simple liquids, amounts to several independent modes of relaxation all of which may have strong temperature dependence, especially as the glass transition is approached, would prove to be every bit as insufficient as Batschinski's linear free-volume model. Eyring's model failed to sufficiently describe diffusion even in some simple crystalline systems [94].

Consider again the potential energy landscape of Fig. 1.6 way back in Ch. 1. Though, to some extent, structural relaxation may be entirely entropy driven as regions held at constant energy may still explore equipotential surfaces, moving from one local minimum within the energy landscape to another will be an energy activated process. Particularly as one approaches a state

here described as an ideal glass, the activation energy associated with hopping between adjacent potential wells will be strongly temperature dependent and the complexity of the parameter space implies there will be a near uniform distribution of potential paths through the parameter space corresponding to physical modes of structural relaxation. Thus Henry Eyring's description of viscosity as an energy activated process underlies the concept of free-volume. Bueche makes note, in his contemporary work on the segmental mobility of polymers, that one is left free to conceptualize relaxation processes in terms of the activation energy, ϵ, needed to break free of nearest neighbor cages or equivalently in terms of a activation volume, ΔV, needed for a molecular unit, or sub-molecular in the case of polymer chains, to pass through [12]. Bueche also importantly notes that, as previously alluded to by Davidson and Cole regarding the failure of the Debye hydrodynamic model to sufficiently explain the relaxation processes of glycerol, an activation volume, ΔV, need not be on the order of the molecular volume as relaxation modes may involve cooperative motions of sub-molecular units.

Bueche's desire to connect free-volume theory, which since the contribution of Doolittle had been shown to aptly model viscous behavior of many liquids over a much larger range of temperatures than when first proposed as a linear model by Batschinski, with the Eyring rate theory is quite understandable. Eyring's rate theory of viscosity, though experimentally found wanting, was strongly rooted in a first-principles derivation and shares an elegant symmetry with the well established thermodynamic theory of chemical reaction rates. If viscosity is, in fact, an energy activated process it should be governed by similar expressions to those governing chemical reactions. Though it may be convenient to model viscosity in terms of free-volume, Bueche argues that the creation of this free-volume comes at an energetic cost. The free-volume model proposed by Doolittle, though experimentally successful, falls short of the Eyring rate theory in its lack of any

theoretical explanation—the exponential form of Eq. 3.4 being entirely phenomenological.

A first-principles theoretical derivation of Doolittle's free-volume model was finally provided in 1959 *viā* the work of Cohen and Turnbull [18]. To accomplish this, Cohen envisioned an entropic rather than energetic activation behind structural relaxation. Consider again the parameter space idealized by Fig. 1.6 as the n-dimensional surface that would exist for any real liquid. The energetic processes described by Bueche and Eyring may not be necessary to navigate a sufficiently complex parameter space. Couched in terms of the energy landscape picture, Cohen might suggest that one consider that what perhaps appears to be an insurmountable energy barrier between two adjacent potential wells in Fig. 1.6 in the absence of some introduced activation energy, ϵ, may in fact be accessible *viā* multiple paths along equipotential surfaces in higher dimensional parameter space. Evolution along these equipotential surfaces correspond to pure entropy fluctuations at constant energy and can by physically interpreted as a rearrangement of existing free-volume within the bulk material.

Cohen and Turnbull do well to point out that the diffusion constant for spherical particles dispersed in a viscous medium is inversely related to viscosity *viā* the Stokes-Einstein relation,

$$D = \frac{k_b T}{3\pi a_0 \eta}, \tag{3.5}$$

where a_0 is the hard sphere cross-section. Though this is derived in the context of low number density macroscopic spherical particles suspended in a viscous fluid, rather than the self diffusion of a pure liquid, it is still contextually relevant to the latter. In fact the use of nano-particles suspended in a liquid to serve as a probe of the molecular dynamics of the host liquid certainly appears in the literature and was considered as a possible method for extending visible photon

correlation spectroscopy into the diamond anvil cell, the results of which are described in the following chapter. Suffice it to say for now that there was concern regarding the light arising from density fluctuations within a pure liquid from the very small volume of a diamond anvil cell being sufficient in intensity for photon correlation spectroscopy. As such it was considered that sufficiently small probe particles, perhaps within an order of magnitude of the molecular size of the liquid under study, might provide a sufficient scattering intensity and their diffusion would approximate, or at least track with, that of the molecular self diffusion for the host liquid.

This idea was in part inspired by the work of Caronna *et al*. who implemented x-ray scattering from suspended nano-particles to probe the dynamics of a glass forming liquid [14]. They found that at higher temperatures the particles undergo Brownian motion as predicated by Eq. 3.5 above however, as the glass transition is approached, the probe particles exhibit what the authors referred to as "hyperdiffusive behavior" which they suggest is due to the emergence of cooperative motions of the host liquid. It may be further suggested that, while approaching the glass transition, the effective size of the molecular units involved in structural relaxation increase accordingly as cooperative domains of the host liquid increase in size. This hyperdiffusive behavior will emerge as the size of these cooperative domains approach that of the probe particles.

The point to be had here is that, to the extent that a pure liquid can be modeled as consisting of hard spheres of radius a_0, self diffusion within a pure liquid should be well modeled by the Stokes-Eienstein relationship. Cohen and Turnbull assumed that this would hold true through the glass transition, however, as found much later by Caronna *et al.* , this will depend on the emergence of cooperative domains that likely have the effect of imposing a strong temperature dependence on the effective size of the relaxing units, which may range from sub-molecular to macroscopic as the glass transition is approached. The emergence of cooperative domains and their roll regarding

structural relaxation within glass forming systems was not so well understood at this time. None the less, in the case of self diffusion from the perspective of free-volume, the effective size of the relaxing units, weather they be sub-molecular or large cooperative domains, will be intrinsically associated with the viscosity η. It is reasonable to assert that, no matter the functional form, to first approximation self diffusion within a pure liquid should be inversely proportional to the viscosity of the liquid.

The question that remains is weather this viscosity and diffusion result from energy activated processes or pure entropy fluctuations involving a rearrangement of free-volume within the bulk material. Cohen assumes the latter. What will follow here is Cohen's derivation of Doolittle's free-volume formula from first principles that exactly correlates with that of the energy activated rate theory of Eyring—the only difference being that energy constraints and statistical distributions are replaced with equivalent free-volume constraints and distributions. However it should perhaps be pointed out that, though Cohen here describes the process of diffusion, or equivalently viscosity, as arising from pure entropy fluctuations, the derivation that follows does not preclude free-volume being created or destroyed as the result of energetic processes. Generally speaking, a substance whose constituent molecules have sufficient average energy, $k_b T$, to exist in a liquid state will manifest local thermal fluctuations on the order of the heat of fusion giving rise to the creation of vacancies within the liquid—*i.e*. free-volume. Thus, arguably, one should not view free-volume theory as an alternative to an energy activated rate theory, but rather simply a more natural means of expressing such a theory *viā* circumventing the need to parameterize the complicated distribution of temperature dependent activated processes that may give rise to the creation of free-volume.

The total free-volume of a bulk liquid, having average thermal energy $k_b T$, will undoubt-

edly fluctuate, however the expectation value of that total free-volume will be as solidly defined as any thermodynamic parameter. Cohen defines the average free-volume per molecule, analogous to the average thermal energy per molecule defined by Eyring, as $v_f = V_f/N$, where V_f is the total free-volume and N is the total number of particles in the bulk. Though it was reasonable for Eyring to assume the total energy of an isolated system is constant, there certainly will be fluctuations in the total free-volume, V_f, even in an isolated system, however such fluctuations will be of negligible magnitude and thus would likely prove inconsequential to any derivation. In any case such fluctuations remain irrelevant as V_f is either assumed to be the time-average value of the total free-volume or, assuming ergodicity, v_f is simply defined as the ensemble average of local free-volume over the entirety of the bulk at any given instant. All of that being sufficiently addressed, we can move on to the derivation of Doolittle's free-volume model derived in a similar fashion as by Cohen.

3.2.1 Derivation of a Free Volume model

Let us consider a bulk fluid of finite volume. In general, there may be an infinite number of ways in which nearest neighbors to any particular molecule may be arranged within the fluid. However there is going to be a net free-volume, V_f, within the bulk at any given time which can be thought to be divided among N constituent molecular (or sub-molecular) units. Consider a set of small yet discrete increments of free-volume, Δv, that may be attributed to the ith molecule within the bulk as illustrated in Fig. 3.3.

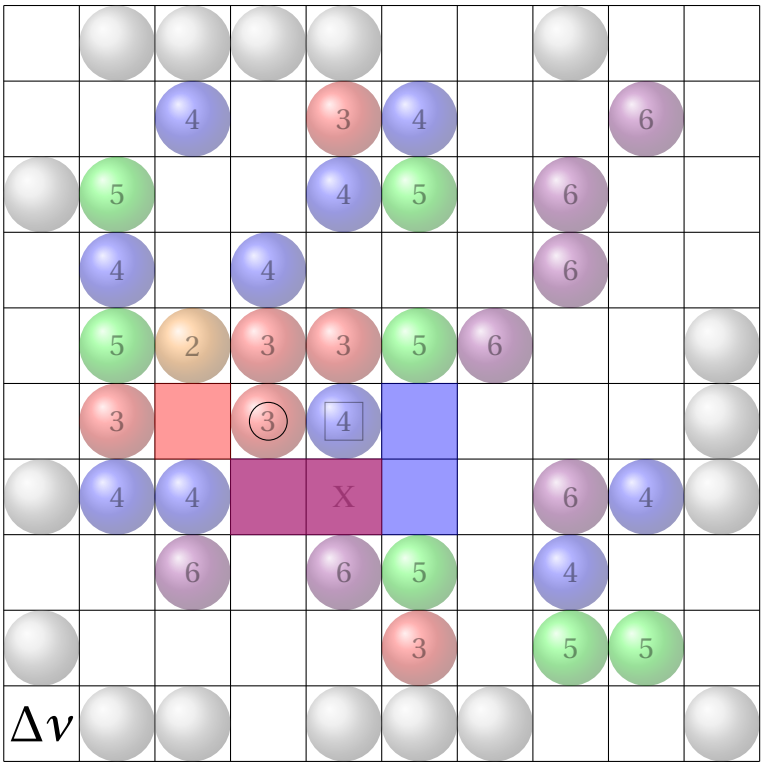


Figure 3.3: Here we have depicted a two dimensional conceptual illustration of free-volume in which the space is divided into small discrete volumes, Δv. Each particle is labeled with its number of associated free-volume increments, i, and color is used to indicate particles which share a common free-volume, v_i. The free-volumes associated with the circled and squared particles is indicated above *viā* red and blue shading respectively where $v_\circ = 3\Delta v$ and $v_\square = 4\Delta v$. Thus, as asserted in Eq. 3.6, the total free-volume within the bulk will be given as $V_f = \gamma \sum n_i v_i$ where we are summing over all possible values of v_i, in this case $0 \leq v_i \leq 8\Delta v$. Note the need for the γ-factor to account for the overlapping free-volume indicated by the magenta region above. In this instance, $\gamma \approx 1/4$ implying that any given Δv will be shared with four adjacent particles on average, as happens to be the case for the magenta square marked with an X.

The ith molecule within the bulk may have an associated free-volume $v_i = i\Delta v$, where i is just an integer coefficient, such that $0 \leq v_i \leq V_f$ subject to the following constraints:

$$\begin{aligned} N &= \sum_i n_i \\ V_f &= \gamma \sum_i n_i v_i \end{aligned} \tag{3.6}$$

In the sum over all possible values of free-volume, v_i is the number of molecules having an as-

sociated free-volume of v_i, and $0.5 \leq \gamma \leq 1$ is a geometric factor to account for overlapping free-volume attributed to adjacent molecules. It should be noted that the lower limit of $\gamma = 0.5$ arises from the fact that a typical liquid will only share, on average, no more than half of its surrounding free-volume with at most one adjacent molecule. However, the illustration provided in Fig. 3.3 would, being excessively tightly packed, have a value of γ significantly less than one half. There is no physical reason why γ can not be less than one half and, in fact, for tightly packed granular systems almost certainly will be.

Getting back to our derivation, from here we will assume, similarly to Boltzmann, that any arrangement of free-volume among constituent molecules within the bulk is equally probable so long as the constraints of Eq. 3.6 above are met. Thus the odds of finding the free-volume distributed in any particular arrangement among N molecules such that n_i molecules have an associated free-volume v_i will be given by the total number of permutations of n_i as follows:

$$W = \frac{N!}{\prod_i (n_i!)} \tag{3.7}$$

From here Lagrange multipliers are implemented to find the most probable permutation. Just as is typical to derive the classic Boltzmann distribution we will, for mathematical convenience, maximize $\ln W$ invoking the Stirling approximation.

$$N! \approx \left(\frac{N}{e}\right)^N \tag{3.8}$$

Thus:

$$\begin{aligned} \ln[W] &= \ln[N!] - \sum_i \ln[n_i!] \\ &\approx N(\ln[N] - 1) - \sum_i \ln[n_i!] \end{aligned} \tag{3.9}$$

Note that we can now invoke the Stirling approximation on the second term of Eq. 3.9 resulting from the denominator in Eq. 3.7 and this will be done shortly. One might argue that there is no assurance the number of molecules, n_i, within the bulk having any arbitrary associated free-volume, v_i, is always sufficiently large as to justify the Stirling approximation. However, as is so often the case with thermodynamic quantities, the distribution of volumes, v_i, are distributed sharply around their average value. Therefor, though there certainly will be small values of n_i, neglecting them entirely will have negligible effect on the total free-volume of the system. Now, to maximize Eq. 3.9, we implement the Lagrange multipliers λ and μ as follows:

$$\delta \ln[W] - \lambda \delta N - \mu \delta V_f = 0 \tag{3.10}$$

First we consider $\delta \ln[W]$:

$$\begin{aligned} \delta \ln[W] &= \delta[N(\ln[N] - 1)] - \sum_i \delta[n_i(\ln[n_i] - 1)] \\ &= 0 - \sum_i \ln[n_i]\, \delta n_i - \sum_i \frac{n_i}{n_i} \delta n_i + \sum_i \delta n_i \\ &= -\sum_i \ln[n_i]\, \delta n_i \end{aligned} \tag{3.11}$$

Next we consider the constraints $\lambda\delta N$ and $\mu\delta V_f$:

$$\begin{aligned}\lambda\delta N &= \lambda\sum_i \delta n_i \\ \mu\delta V_f &= \mu\gamma\sum_i v_i\delta n_i\end{aligned} \tag{3.12}$$

Eq. 3.10 then becomes:

$$-\sum_i \left(\ln[n_i] + \lambda + \mu\gamma v_i\right)\delta n_i = 0 \tag{3.13}$$

As λ and μ are arbitrary Lagrange multipliers, asserting the sum to be zero implies that each term must be zero. From this we find the distribution of the number of particles, n_i, having associated free-volume, v_i which maximize the permutations of Eq. 3.7 corresponding the most probable distribution of free-volume within the bulk liquid as follows:

$$\begin{aligned}\ln[n_i] &= -\lambda - \mu\gamma v_i \\ n_i &= e^{-(\lambda+\mu\gamma v_i)}\end{aligned} \tag{3.14}$$

Thus the maximum permutations of Eq. 3.9 is found to be as follows:

$$\begin{aligned}\ln[W_{max}] &\approx N(\ln[N]-1) - \sum_i n_i(\ln[n_i]-1) \\ &= N(\ln[N]-1) - \sum_i e^{-(\lambda+\mu\gamma v_i)}\left(-\lambda-\mu\gamma v_i - 1\right) \\ &= N\ln[N] - N + \lambda\sum_i n_i + \mu\gamma\sum_i n_i v_i + \sum_i n_i \\ &= N\ln[N] + \lambda\sum_i n_i + \mu\gamma\sum_i n_i v_i\end{aligned} \tag{3.15}$$

Returning to the results of Eq. 3.14 and the constraints of Eq. 3.6 we can now express the total free-volume within a bulk liquid as follows:

$$\begin{aligned} V_f &= \gamma \sum_i n_i v_i \\ &= \gamma \sum_i v_i e^{-(\lambda + \mu\gamma v_i)} \end{aligned} \tag{3.16}$$

In the limit where the difference in free-volume among the discrete bins, thus far enumerated as v_i, becomes infinitesimal we may express the total free-volume of the bulk, V_f, as a continuous integral over an infinite number of infinitesimal bins.

$$V_f = \gamma \int_0^\infty v e^{-(\lambda + \mu\gamma v)} \delta v = \frac{e^{-\lambda}}{\mu^2 \gamma} \tag{3.17}$$

Solving this for the Lagrange multiplier, λ, yields the following:

$$\lambda = -\ln\left[\mu^2 \gamma V_f\right] \tag{3.18}$$

The constraints of Eq. 3.6 on the total number of molecules in the bulk can similarly be assessed.

$$\begin{aligned} N &= \sum_i n_i = \sum_i e^{\lambda + \mu\gamma v_i} \\ &= \int_0^\infty e^{-(\lambda + \mu\gamma v)} \delta v = \frac{e^{-\lambda}}{\mu\gamma} \end{aligned} \tag{3.19}$$

From here the average free energy per unit molecule, or the ensemble average $< v_f >$, is found to be as follows:

$$v_f = \frac{V_f}{N} = \frac{\left(\frac{e^{-\lambda}}{\mu^2\gamma}\right)}{\left(\frac{e^{-\lambda}}{\mu\gamma}\right)} = \frac{1}{\mu} \tag{3.20}$$

Returning to our previous solution for λ in Eq. 3.18 above we find the following:

$$\begin{aligned}\lambda &= -\ln\left[\mu^2 \gamma V_f\right] = -\ln\left[\frac{\gamma V_f}{v_f^2}\right] = -\ln\left[\frac{\gamma v_f N}{v_f^2}\right] \\ &= -\ln\left[\frac{\gamma}{v_f}\right] - \ln[N]\end{aligned} \tag{3.21}$$

In the limit where the difference in free-volume among the discrete bins becomes infinitesimal, we may express Eq. 3.14 as follows:

$$\begin{aligned}\delta n &= e^{-(\lambda+\mu\gamma v)} \\ &= \exp\left[-\left(-\ln\left[\frac{\gamma}{v_f}\right] - \ln[N] + \frac{\gamma v}{v_f}\right)\right] \\ &= N\left(\frac{\gamma}{v_f}\right) e^{\frac{-\gamma v}{v_f}}\end{aligned} \tag{3.22}$$

Thus we can express the number density of particles, $\rho(v)$, having an associated free-volume, v to be described as follows:

$$\rho(v) = \frac{\delta n}{N} = \left(\frac{\gamma}{v_f}\right) e^{-\frac{\gamma v}{v_f}} \tag{3.23}$$

where the normalization condition $\int_0^\infty \rho(v)\,\delta v = 1$ is naturally met. As previously mentioned, Cohen and Turnbull approaches this derivation assuming that self diffusion is not an energetically activated process, but rather is a purely structurally entropic process in which free-volume within the bulk can be thought of as being randomly redistributed *viā* Brownian type motion along equipotential paths through the energy landscape. This is actually a reasonable assumption to the extent that a liquid is well approximated by a spherically symmetric force field where the intermolecular potential landscape is found to be relatively flat [106]. This assumption will hold true for almost any liquid in the high temperature limit where the potential energy landscape is not really felt by

the constituent molecules. At lower temperatures, certainly as one approaches the glass transition, energy activated process may play a larger role in the active creation of free-volume however the considerations of Cohen and Turnbull remain valid. A diffusion event is going to be inherently dependent on the existence of free-volume sufficient for molecular egress regardless of how that free-volume comes to be present. This makes free-volume arguably a more fundamental consideration in regards to molecular self diffusion than is any energetic rate theory. Thus ignoring for the moment how a constituent particle might find itself in the presence of free-volume v, it may be considered to be moving across a cage of nearest neighbors having a diameter of $\alpha(v)$ that is a function of its associated free-volume, v, and traveling with a velocity u. Cohen and Turnbull then reasonably assert that diffusion proceeds in much the same way as it does within crystals and any molecule's contribution to the diffusion constant will be given as follows [88]:

$$D(v) = \zeta u \alpha(v), \tag{3.24}$$

where ζ is just a geometric factor. As the distribution of free-volume is provided by the results of Eq. 3.23, the sum of all contributions will yield the following constant of self diffusion:

$$D = \int_{v_m}^{\infty} D(v)\, \rho(v)\, \delta v, \tag{3.25}$$

where v_m is the minimum value of free-volume needed to allow for any molecular egress and for which any contribution to diffusion can occur. The value of v_m is generally going to be at least an order of magnitude greater than the average available free energy per molecule, $< v_f >$. As such, Eq. 3.24 is going to be a slowly evolving function on the timescales of any particular diffusion

event. Thus Eq. 3.25 can be simplified as follows:

$$\begin{aligned} D &= \zeta u \alpha(v_m) \int_{v_m}^{\infty} \left(\frac{\gamma}{v_f}\right) e^{-\frac{\gamma v}{v_f}} \delta v \\ &= \zeta u \alpha(v_m) e^{-\frac{\gamma v_m}{v_f}} \end{aligned} \tag{3.26}$$

As Cohen points out, this is equivalent in form to the phenomenological model proposed by Doolittle [29]. In the case of "hard sphere" liquids, $\alpha(v_m)$ will be on the order of the spherical diameter and $u \propto \sqrt{k_b T}$ implies that even the simplest liquids must undergo a glass transition if cooled sufficiently in the absence of crystallization. Of course this model is certainly not limited to this $T^{1/2}$ temperature dependence. Eyring's general concept of a rate theory of diffusion, even extended to include any arbitrary temperature dependence or distribution of activation energies suggested by the broad applicability of the VTF equation, will here give rise to a temperature and perhaps pressure dependence of the average free-volume, v_f, related to the coefficient of thermal expansion. However, though energy activated processes may determine the average free-volume within a bulk, it is the existence of this free-volume, not the energy associated with its creation, that is fundamental to diffusion, viscosity, and relaxation processes. Thus Eq. 3.26 can be generalized as follows:

$$D(T,P) = \zeta u \alpha(v_m) e^{\frac{-\gamma v_m}{v(T,P) - v_m}} \tag{3.27}$$

The diffusion of spherical particles dispersed in a viscous medium described by Einstein and Stokes *viā* Eq. 3.5 provided above can, to first approximation, be assumed to apply to self-diffusion of non-associated pure liquids well described by the flat potential energy landscape of hard spheres. It is physically reasonable to assume that, even when the potential energy landscape is significantly more complex, the diffusion rate will still remain proportional to fluidity or, equivalently, inversely

proportional to the viscosity. Thus one may arrive at the following simplified model for viscosity as a function of free-volume:

$$\eta(T,P) = C(T)\, e^{\frac{\gamma v_m}{v(T,P)-v_m}} \tag{3.28}$$

Here $C(T)$ is just some temperature dependent constant that will correspond to the viscosity at infinite volume which will be an exceedingly small value. It is worthy of note that viscosity really should trend to zero as free-volume approaches infinity. The fact that this model does not predict this implies that it cannot accurately describe viscosity over an infinite dynamic range regardless of the functional dependence of the parameters. However it does show remarkable agreement for most liquids over a very large dynamic range. As such it is common to expand this model around its value at ambient pressure as follows:

$$\eta(T,P) = \eta_0(T)\, e^{\frac{\gamma v_m}{v_m - v_0(T)}} e^{\frac{\gamma v_m}{v(T,P)-v_m}} \tag{3.29}$$

Here $\eta_0(T)$ and $v_0(T)$ are the temperature dependent viscosity and specific volume respectively at ambient pressure. As this model is generally applied to systems at or above ambient pressure, the physical incongruity regarding the non-zero infinite free-volume viscosity is not of significant concern. As implied by the super-exponential form of Eq. 3.29, these viscosities will increase by many orders of magnitude quite rapidly as either pressure is increased or temperature decreased. As such, it is very common to express Eq. 3.29 in logarithmic form as follows:

$$\ln\left[\frac{\eta(T,P)}{\eta_0(T)}\right] = \gamma v_m \left(\frac{1}{v(T,P)-v_m} - \frac{1}{v_0(T)-v_m}\right) \tag{3.30}$$

Some variation of Eq. 3.30 has been used extensively, with various functional dependencies of

parameters on temperature and pressure, throughout the literature to fit a large variety of associated and non-associated systems, including polymers, with high efficacy for viscosities ranging over several orders of magnitude [19, 109, 113, 69].

3.3 Reduction of extrapolation error in fragility measurements from pressure-limited data *viā* implementation of TGP

Viscosity measurements at elevated pressure for glycerol up to about 4 kPa, first obtained by McDuffie and Kelly, was reported in the Journal of Chemical Physics in 1964 [70]. This pressure range was significantly extended by just under an order of magnitude *viā* the work of Cook *et al*. in 1992 implementing a diamond anvil cell viscometer attached to a centrifuge [19, 21]. The data of Cook *et al*. were then used to investigate the pressure dependence of fragility over this domain and appeared to show a trend of steadily increasing fragility from approximately $m \approx 55$ ($m^* = 3.2$) at ambient pressure up to $m \approx 170$ ($m^* = 13.6$)) at 3 GPa as shown in Fig. 3.4. Note that, though the definition of fragility provided by Eq. 2.2 is arguably more physically meaningful, it may be convenient for the sake of literature comparisons to express some fragilities in terms of the more classical definition of Eq. 2.1. Thus to alleviate any confusion we will adopt a consistent notation of m and m^* to indicate the classical fragility of Eq. 2.1 and the modified fragility of Eq. 2.2 respectively. Returning now to Fig. 3.4, note the excessive error associated with these fragility calculations arising from the necessity to extrapolate the VTF model over several orders of magnitude up to the glass transition. The fact that this trend towards extreme fragility is not in satisfactory agreement with subsequent publications, particularly at elevated pressures, and the excessive error implying that the fragility measurements at high pressure are essentially meaningless has lead to the Cook *et al*. data being generally under-emphasized. Paluch *et al*. drew attention

to these concerns as well as the fact that several of the viscosity measurements were made near the melt temperature, T_m, where VTF model extrapolations are known to be unreliable [79]. The fundamental limitation for Cook *et al*. was that rolling ball viscometery, even when implemented within a centrifuge, can only measure viscosities up to on the order of $10^9 cP$—*i.e*. as much as six orders of magnitude away from the glass transition at which fragility is assessed. The data of Cook *et al*. , even with these concerns, does show strong support for an increasing trend in fragility with pressure up to around 2 GPa, which is not unexpected for a hydrogen bonding liquid where increasing fragility can be explained as arising from the diminished free-volume inhibiting hydrogen bonding. Beyond approximately 2 GPa, the minimal uncertainty in these fragilities obtained from propagating the errors in the fit parameters of the VTF equation seems to render the actual values of fragility statistically meaningless. Though the assessment of Cook *et al*. , that the fragility continues to increase with pressure beyond this point, my have been somewhat compelling given that the measurements continue along the same well defined linear regression and their standard Angel plots, shown here as modified Angel plots and provided on the right side of Fig. 3.4, seem to consistently show increasing curvature as pressure increases, the higher pressure data is dubious to say the least, not just due to the excessive error but also in that, having reached a pressure of 3.0 GPa, glycerol has become about as fragile as a liquid can become. This linear trend in fragility certainly cannot continue indefinitely yet, at a pressure of 3 GPa, there is no indication of departing from said trend. Furthermore, it is worth noting that the 3 GPa fragility shown in Fig. 3.4 is based on only four data points and, though it is less evident in our modified Angel plot of Fig. 3.4 than it would be in a classical Angel plot, does not converge to a similar infinite temperature viscosity as do data obtained at lower pressures. It is therefor not surprising that this data might yield an excessively high fragility while having error bars that include negative values of fragility—statistically

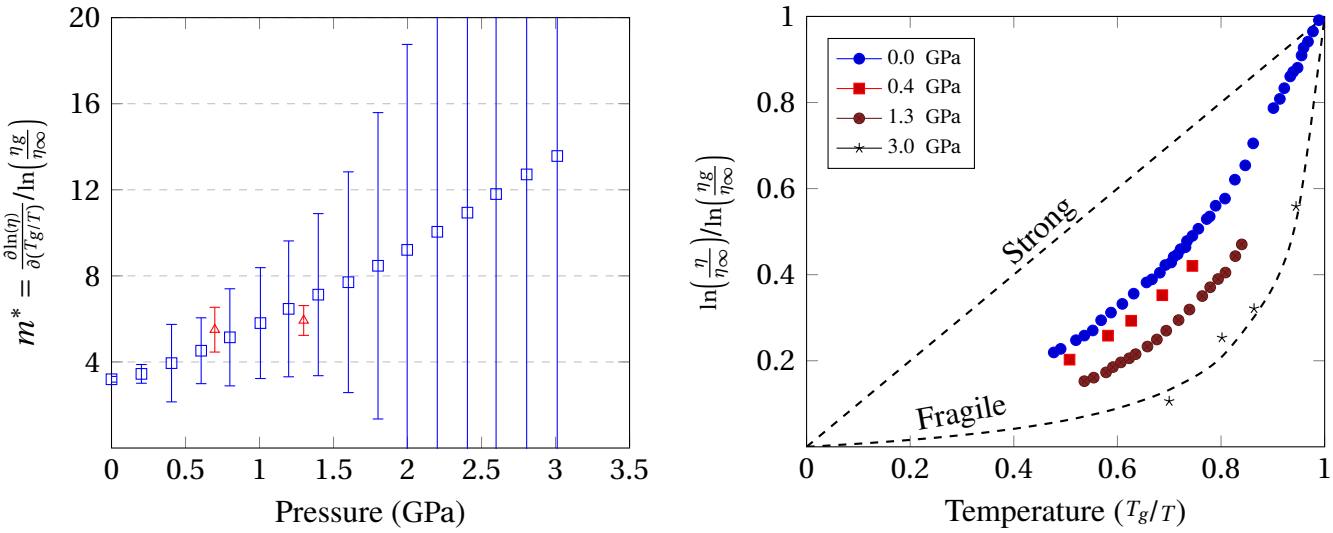


Figure 3.4: Left: Cook's calculations for the isobaric fragility of glycerol as a function of pressure with the glass transition defined at 10^{15} centipoise. Note that these values have been modified to reflect our redefinition of fragility as defined in Eq. 2.2. This scaling of fragility is trivially accomplished as Cook conveniently provides his fit parameters. Right: Angel plots for Cook's 0.4, 1.3, and 3.0 GPa corrected isobars [21]. The 0.0 GPa isobar data is from Laughlin's 1969 Ph.D. thesis although all values plotted here were directly extracted from Cook's published figures [62].

speaking this is a meaningless measurement. In truth, this point should likely have been omitted from the analysis of Cook *et al.* as it relies on only three reasonably reliable data points—the 50.0 °C data point requiring a free-volume model extrapolation of almost 1 GPa. However omitting the 50.0 °C data point would leave only the 22.5, 75.0, and 125.0 °C, two of which still require slight extrapolation to 3 GPa, and require one to perform a non-linear model fit with only three data points providing constraint to a three parameter VTF model—though technically possible, an arguably meaningless endeavor. This was the root cause of their exploding fragility estimations—simply a lack of data to properly constrain the VTF model fits and extreme extrapolation over orders of magnitude as none of this data is within the vicinity of the glass transition.

More recent work has made a well supported case that the fragility of glycerol increases with pressure due to the inhibiting of hydrogen bonding up to approximately 2 GPa at which point

it levels off becoming relatively constant. It would be reasonable to assert that the data of Cook *et al.* , as analyzed here, equally well supports this general trend of fragility. Arguably, the analysis supports a general increasing trend in fragility as pressure increases up to approximately 2 GPa even if little confidence can be put into the exact fragility values up to the point that the error begins to diverge rapidly. Beyond 2 GPa little can be confidently asserted from Cook's analysis however it does not contradict a constant fragility when the propagated errors are accounted for.

In order to assess the validity of Cook's high pressure data and the associated fragility, it is necessary to diminish these large error bars. This can be accomplished by addressing one of his two primary underlying issues—extreme data extrapolation. The TGP process developed by our group provides a direct measurement of the glass transition as a function of pressure allowing for a re-analysis of the cook data without the need for extrapolation of the VTF model over so many orders of magnitude. This TGP data is provided in App. C Tbl. B.1 along with the associated Anderson-Anderson regression fit. Using these values for the glass transition pressure on any given isotherm and assuming this transition to occur on an isochronal curve associated with a viscosity of $\eta = 10^{15} cps$, the data Cook *et al.* were refit to a classical VTF model.

Note that most of the data of Cook *et al.* were obtained over isotherms at varying pressures. In order to calculate fragility, they fit five sets of isothermal viscosity data to the classic free-volume model of Eq. 3.30. The Taite equation of state, fit to density data provided by Bridgman, was implemented to convert pressure to volume [11]. More details regarding the implementation of the Tait equation of state for glycerol are provided in App. E as Cook *et al.* and this work use this free-volume model to obtain (mostly) interpolated isobaric data as true isobaric data is difficult to obtain experimentally given that pressures will vary significantly within the DAC under thermal expansion. Thus it is this interpolated isobaric data that is used in the fits to the VTF model for

fragility analysis.

Figure 3.5 (left) provides a modified Angel plot both excluding the TGP glass transition point (solid) as well as the refit passing through the TGP point (dashed) for a 1.3 GPa isobar. Note that the lower five data points associated with both the dashed and solid regressions are in fact the same data from Cook *et al*. —scaled according to their associated VTF fits and glass transition temperatures. The TGP point is also provided and is, on a modified Angel plot, located at (0,0) by definition. It is immediately clear that both adequately describe Cook's data however omitting the TGP data leads to a significant over-estimation of fragility. This is associated with an over-estimation of the glass transition temperature. Figure 3.5 (right) shows eight isobaric data sets, consisting of four to five data points each, ranging left to right from 0 GPa to 3 GPa which are obtained from the before mentioned free-volume fits to Cook's published isothermal data. The VTF fits to this data exclude the TGP point and thus help to demonstrate the error that arises from the extrapolation of the VTF equation. Notice that at the lowest pressures (left most curves) there is a systematic under-estimation of the glass transition temperature due to model extrapolation. However, as we move up in pressure (to the right of the figure) we begin to increasingly over-estimate the glass transition temperature and thus over-estimate the fragility. This trend is what is giving rise to Cook's exploding fragility measurements.

From this reassessment of the data of Cook *et al*. , we see a very different behavior of fragility as a function of pressure. Though, as demonstrated in Fig. 3.7, the calculations of fragility attained from the Cook *et al*. data enhanced *viā* the TGP isochronal data were still not found to be in perfect agreement with more recent measurements, the general trend is found to be somewhat congruent [84, 87]. It should be noted that the actual values of fragility, apart from their trend with pressure, is somewhat dependent on the viscosity at the glass transition. We have chosen

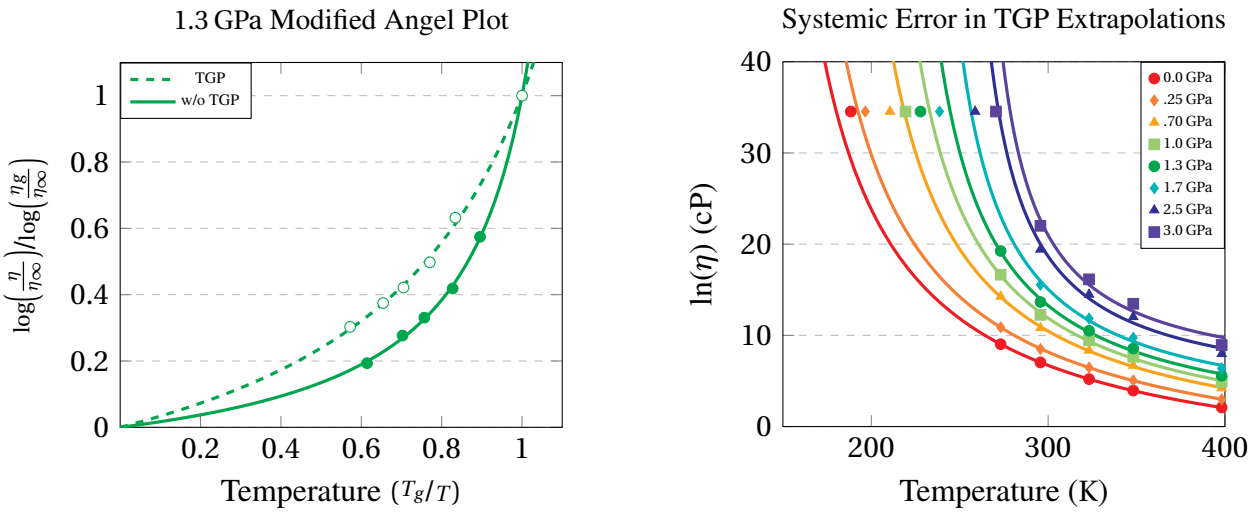


Figure 3.5: Left: A modified Angel plot of the data of Cook *et al*. clearly demonstrates the over-estimation of fragility occurring at 1.3 GPa due to extreme extrapolation up to the glass transition that is rectified *viā* the inclusion of TGP isochronal data. The same data is provided for both the solid and dotted modified VTF fits—scaled accordingly to their fit parameters and the glass transition temperatures they predict. Right: Isobaric data from Cook *et al*. showing the general trend of under-estimation followed by increasing over-estimation of the glass transition temperature with increasing pressure. The TGP method provides the isochronal data at the glass transition [21].

to assume the TGP transition correlates with a viscosity of $\approx 10^{15} cps$. A quick assessment of Fig. 3.6 might lead one to conclude that fragility is initially decreasing from ambient pressure up to approximately 1.5 GPa however, as evidenced by the error of the VTF fit parameters propagated through the fragility calculation as well as keeping in mind that our confidence in these values is still limited *viā* insufficient data particularly in the region of the glass transition, this trend should not be assumed statistically significant—the fragility is relatively constant up to around 1.3 GPa. It is worth noting here that, though cook's viscosity data was predominately taken along isotherms as this was experimentally favorable, he did manage to obtain two pseudo-isobaric runs at 0.7 and 1.3 GPa on which he then used interpolation between free-volume fits to his isothermal data to correct for slight deviations of pressure. Though this process was extremely time intensive, it did provide him with two isobaric runs over a large temperature range addressing the issue

of insufficient data points. It should be noted, however, that these data are still limited to well below the glass transition. Due to the larger number of data points associated with these pseudo-isobaric runs, the error associated with the fragility estimations are significantly smaller and these two points are included (△) in the left side of Fig. 3.4. These two higher-confidence points would alone imply a potentially constant fragility below 1.5 GPa—albeit at a significantly higher modified fragility ($m^* \approx 5.7$). Beyond 1.5 GPa the fragility steadily increases by $\Delta m^* \approx 7.8$ and then remain arguably constant beyond 2.5 to 3 GPa before rapidly diverging due to extrapolation errors again. This general trend of increasing fragility is, in the loosest sense, in "agreement" with subsequent published works as will be demonstrated. This leads to the conclusion that the viscosity data of Cook *et al*. is perhaps significantly more reliable than may have been assumed and much of its error, as perhaps expected, is predominately the result of these extrapolation issues. Our method of employing TGP data to diminish this extrapolation error significantly decreases the error associated with these fragility estimations however it is still limited—particularly at elevated pressures. More data is needed in the region of interest to further define the slope at the glass transition. It is possible that additional TGP runs with longer or shorter annealing periods could provide this however in the absence of a solid equation of state at high pressure and thermodynamic scaling to provide the effective viscosity at the glass transition to higher precision than the assumed 10^{15}cP, this is not yet a viable means of defining the slope near the glass transition. Other methods such as photon correlation spectroscopy and dielectric loss spectroscopy certainly exist and there is no shortage of data for glycerol dynamics within the literature around the vicinity of the glass transition, though generally limited to cryogenic measurements at low pressures. If the issues with these fragility calculations do in fact stem predominately from the lack of data in critical regions rather than the efficacy of the data itself, then supplementation of their data with other

literature data in critical regions should allow accurate determinations of fragility as a function of pressure. Glycerol data taken by several different groups extend over different pressure and temperature regions and include both viscosity and α-relaxation measurements, making glycerol a prime candidate for a surface fragility analysis, similar to what will also be done for cumene in Ch. 4.

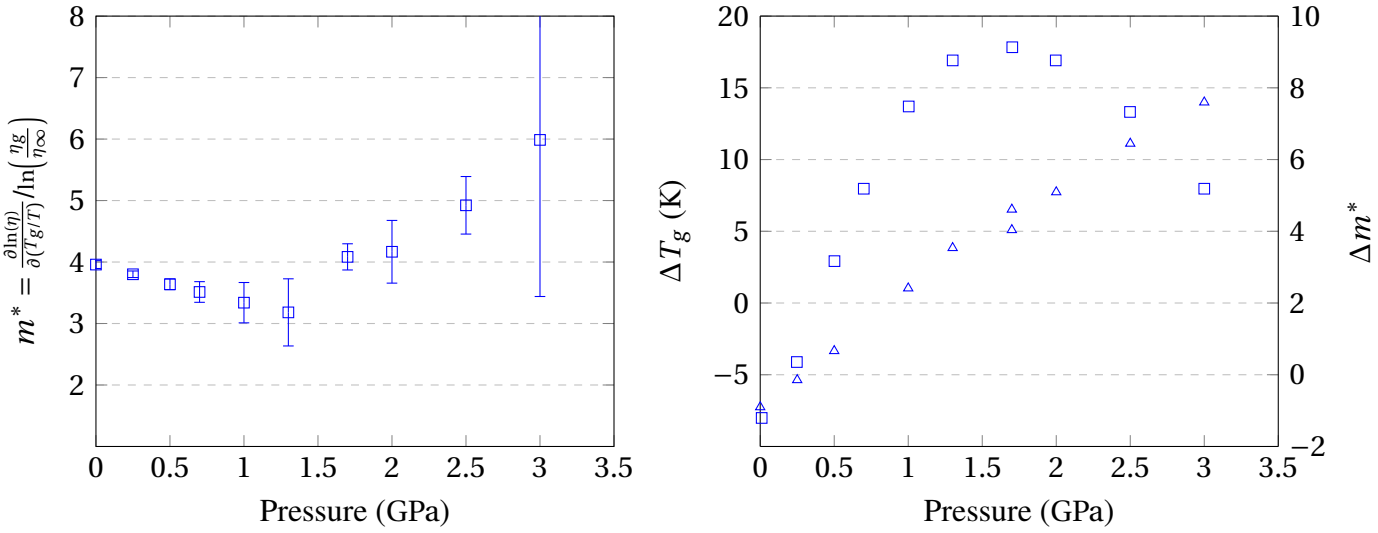


Figure 3.6: Left: Modified fragility as a function of pressure derived from classic VTF fits to the data of Cook *et al*. with the glass transition point included as determined *viā* our TGP isochron. Right: The error in terms of under or over-estimation of the glass transition temperature, ΔT_g, due to extreme extrapolation of the VTF model as demonstrated in Fig. 3.5, is provided (□) along with error in modified fragility (△) calculated from these extrapolated models.

The right side of Fig. 3.6 shows the under or over estimation of the glass transition temperature as a function of pressure derived from the classic VTF fits to the data of Cook *et al*. without the TGP isochronal data. The initial under-estimation of the glass transition temperature for the two lowest pressures is followed by a trend of increasing over-estimation up to about 1.7 GPa where we continue to over-estimate the transition temperature, albeit to an increasingly lesser degree. However the error in fragility associated with this over-estimation in glass transition temperature continues to increase in a fairly linear fashion leading to the excessively high fragilities at increased

pressure observed by Cook *et al*. in Fig. 3.4.

To add further validation to cooks data at lower pressures, the same analysis method of implementing the TGP data point to constrain the VTF fits in the vicinity of the glass transition was applied to the rolling ball data obtained by McDuffie and Kelly and the associated fragilities calculated [70]. Though these fragility values were much more in line with other works implementing dynamic light scattering (Brodin [42]), and dielectric spectroscopy (Reiser [87], Pronin [84], and McDuffie [70]), they all follow the same general trend of constant fragility below 1 GPa. The increasing trend mentioned by Pronin and Reiser is simply not statistically significant. This seems to suggest a strong increase in fragility does not occur until beyond 1 GPa. This may be due to the general expected trend of decreasing fragility with increasing pressure is being approximately balanced by the early onset of hydrogen bond breaking giving rise to a relatively constant fragility in this range.

3.4 Structural Relaxation Surface

Using the methods of Ch. 4, the data from these sources may be fit to a relaxation surface aimed at attaining a more detailed look at the behavior of the relaxation dynamics and fragility as a function of pressure, particularly in the region around 1.5 GPa where hydrogen bonding is presumed to break down.

Glycerol viscosity and relaxation data from several sources are shown, along with the Cook rolling ball data obtain from within a DAC, in Fig. **??** [42, 70, 79, 84, 87]. It is clear that Cook's rolling ball data is actually in excellent agreement with other literature data further bolstering the case that the vast majority of the error within the cook fragility measurements lies in the extreme extrapolation to the glass transition. The inclusion of the TGP isochronal data, though addressing

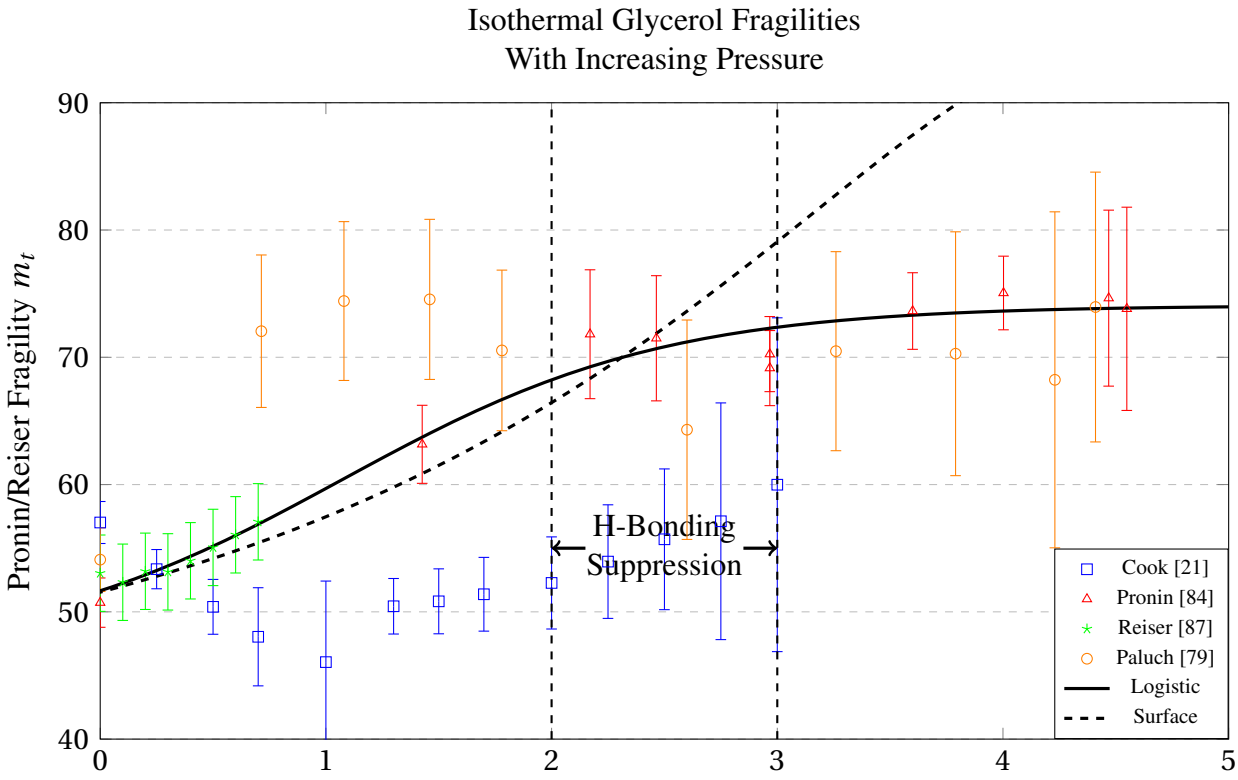


Figure 3.7: The fragility measurements calculated from classic VTF fits to Cook's viscosity data corrected *viā* the inclusion of our TGP glass transition point (□) is shown above to be no longer wholly incongruent with more recent work. A Logistic fit to Pronin (△) and Reiser's (⋆) data describes their fragility measurements well with the increase in fragility seeming to saturate between 2 and 3 GPa as originally asserted by Paluch [79] and Pawlus [82]. The Cook fragilities retain relatively large uncertainties, thus cannot reflect accurately the fine behavior of fragility as a function of pressure, yet do seem to reflect this fragility increase between 2 and 3 GPa. A reasonable trend-line is provided that falls within all error bars.

the issue of extrapolation to a significant extent, ultimately provides insufficient accommodation for an accurate determination of the rate of vitrification with increasing pressure in the vicinity of the glass transition, T_g. More data is needed to define the slope near T_g to obtain an isobaric fragility in consensus with other contributors to the literature, *i.e*. Pronin, Paluch, *etc*.

In short, there never was any issue with the validity of cook's data. As we will see, it is in excellent agreement with other measurements of vitrification or associated alpha relaxation behavior. It is only the lack of data to provide constraint to the VTF fit in key locations, specifically

in the vicinity of T_g, that gives rise to the incongruence of the cook fragility estimations with literature values, *i.e*. the slope in the vicinity of T_g is poorly constrained. The addition of the TGP isochronal data is not itself sufficient to provide the necessary constraint on this slope for reliable estimations of fragility. However, as we will here demonstrate, only one additional experimentally trivial, constraint is needed for cook's data to provide fragility values as a function of pressure in excellent agreement with the current literature consensus—an ambient isobar.

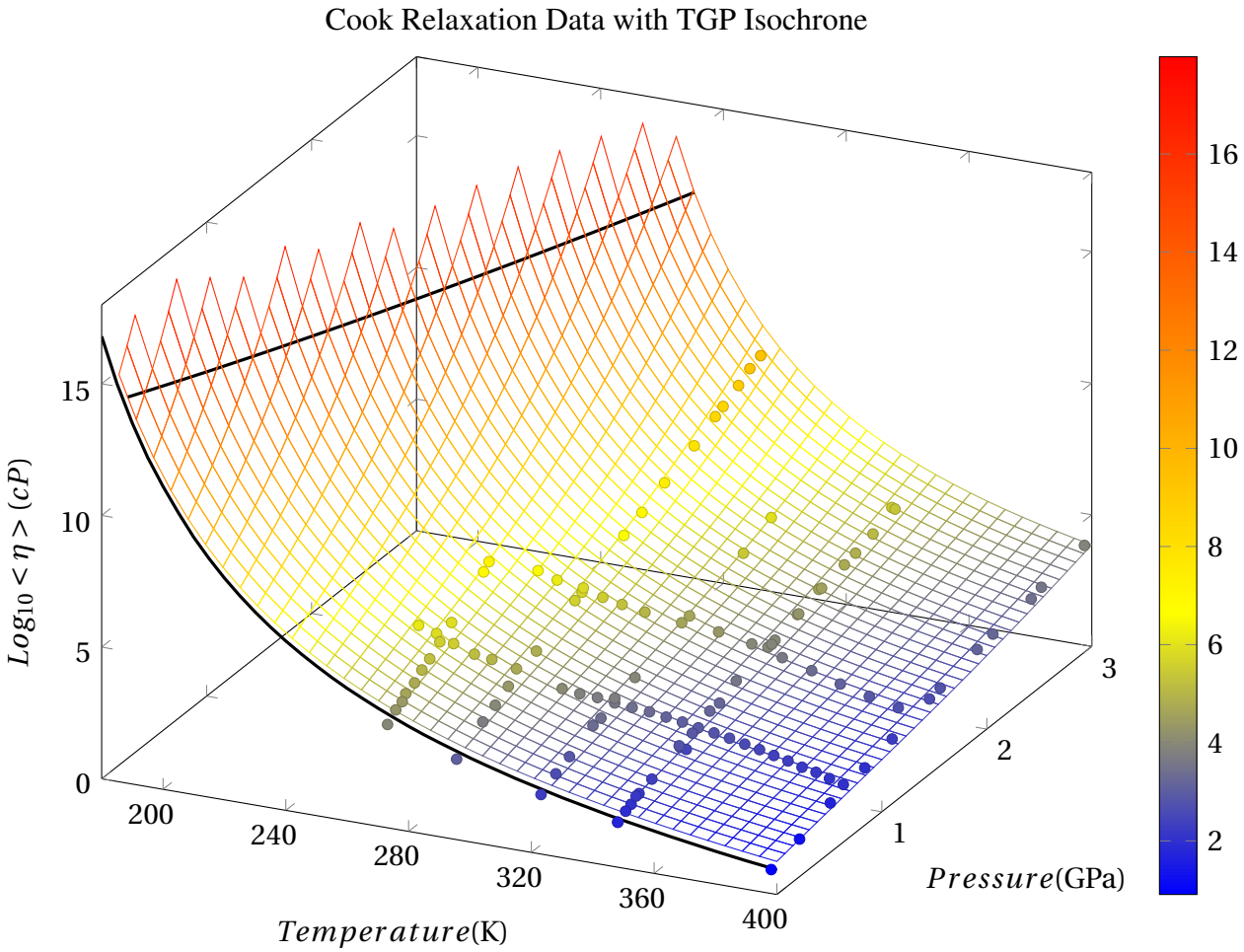


Figure 3.8: Cook's rolling ball viscosity data is here fit to the Andersson-Andersson constrained surface model described in App. A. The color scale, corresponding to pressure, facilitates visualization of how well the data conforms to the surface. The surface is extended to 3 GPa corresponding to the highest pressures for which cook obtained data [21]. It should be noted that, due to the relatively high degrees of freedom provided to the fragility vector, $\vec{d}$, the surface does not lend itself well to extrapolation beyond the domain of the data.

Figure 3.8 shows all of Cook's raw viscosity data provided by Ref. [21] fit to the Andersson-Andersson constrained surface model described in App. A. The Solid black isobaric line at atmospheric pressure, through which the surface is constrained, is well described in Fig. A, along with the supporting data. This 1-bar regression was also used to define the isochrone for the TGP curve—the second solid black line extending across the top of the surface in Fig. 3.8. The glycerol TGP data fit to a standard Andersson-Andersson model is provided in App. B and corresponds to an isochronal pressure-temperature relationship at $\tau \approx 10^{14}$ s through which the surface is also constrained.

This Andersson-Andersson constrained surface model, fit to the Cook viscosity data, allows for a determination of fragility as a function of pressure in similar fashion to that obtained for cumene in Fig. 4.8 previously. The resulting functional dependence of isobaric fragility is plotted as a solid black line in Fig. 3.7 above and found to be in significantly greater agreement with Pronin, Reiser, and other more recent works up to 2.5 GPa. This result, along with the excellent constrained surface fit, suggest that Cook's data could not be lacking significantly in quality.

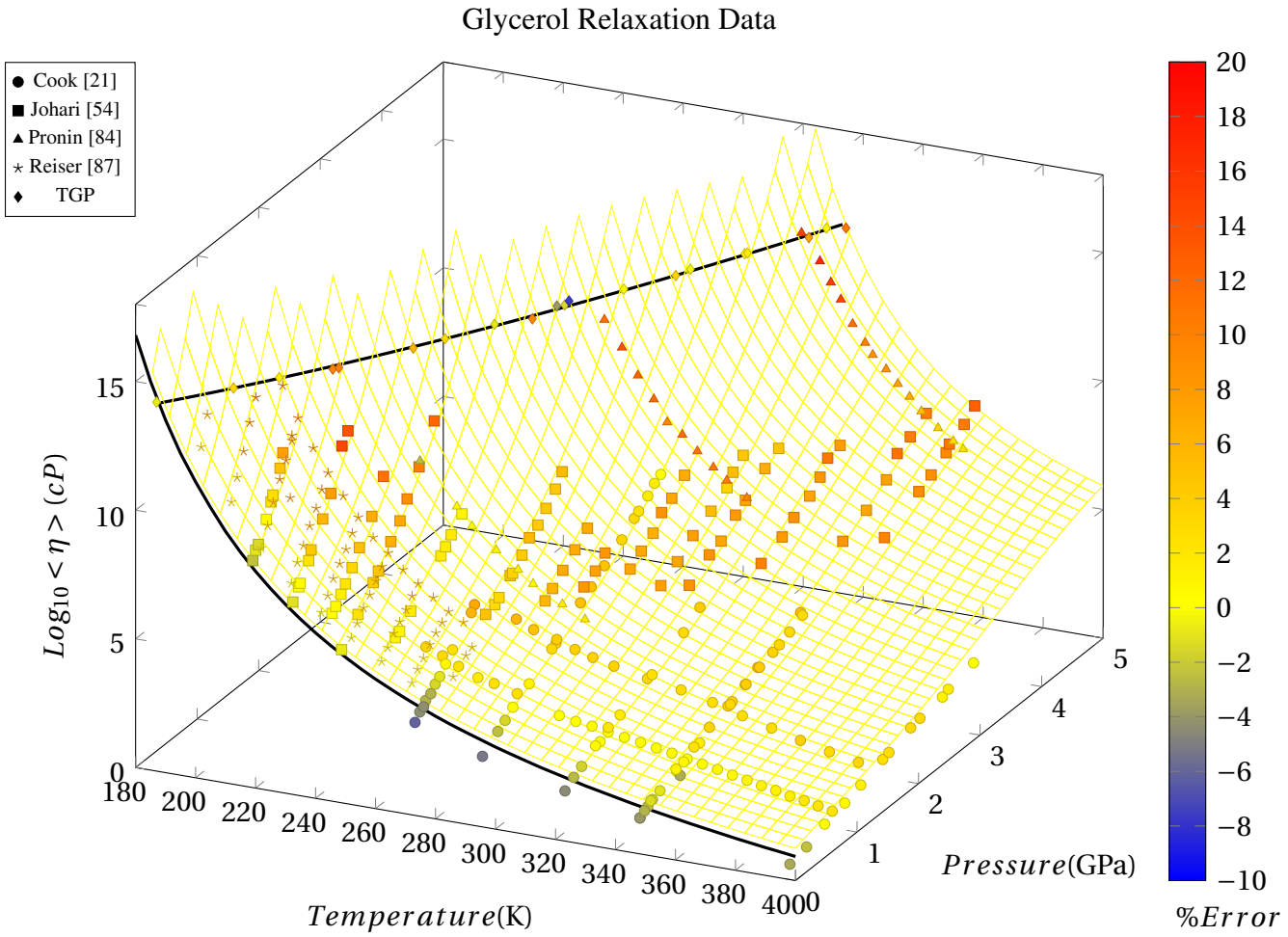


Figure 3.9: Here we have the model described in appendix A fit to a large selection of glycerol relaxation data from multiple contributers [19, 79, 84, 87, 42, 75]. The same TGP and 1-bar constraint was imposed here as was implemented in Fig. 3.8. The color scale now represents percent deviation of the data from the surface. Note how the high pressure Pronin data begins to diverge from the surface, approaching an error as high as 20%. This data was still included in the fit as it conforms well to the surface at higher temperatures and lower viscosities—only tending away from the surface in a region where the TGP constraint does not allow the data significant influence over the fit parameters.

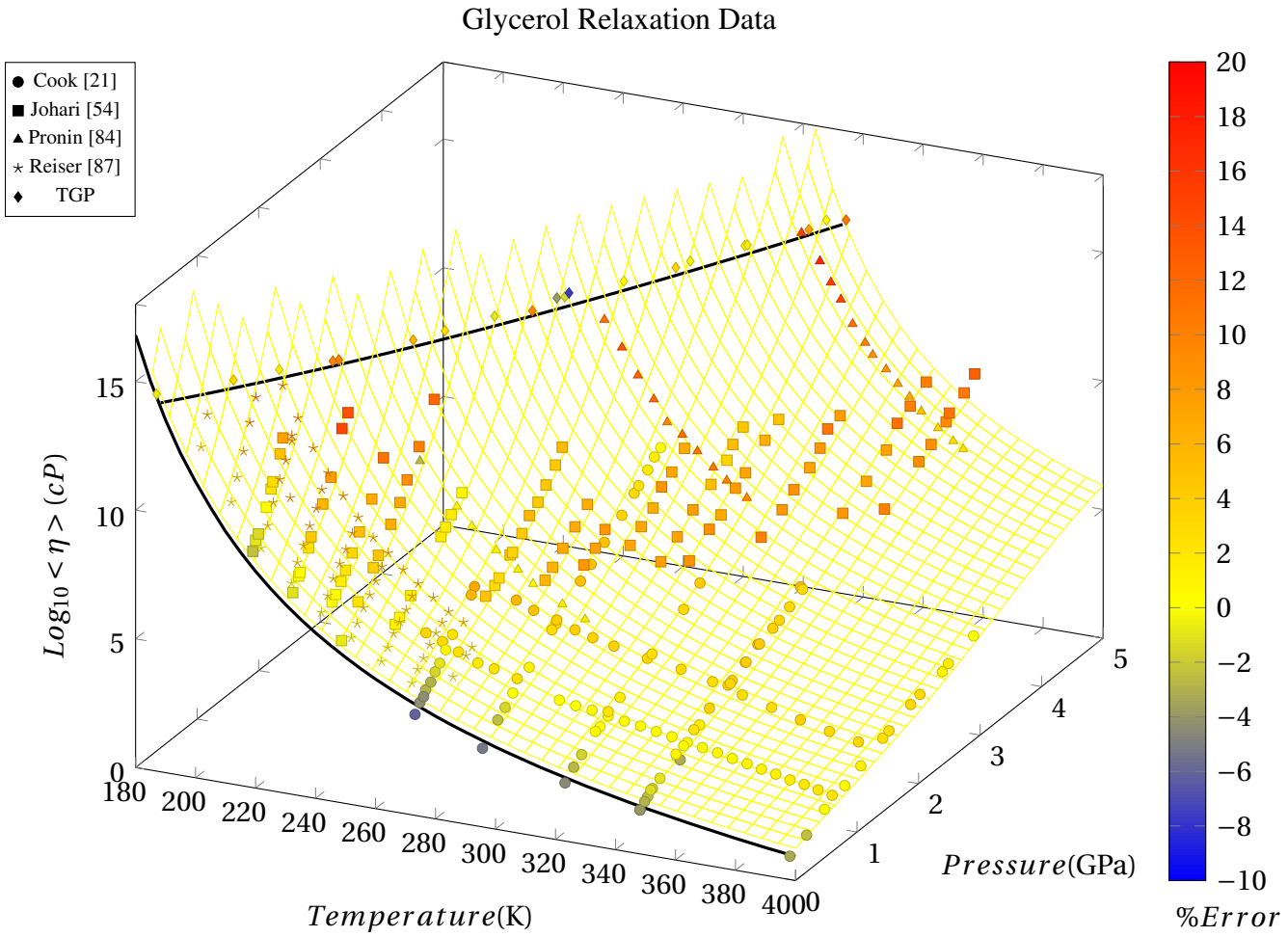


Figure 3.10: Here we have the model described in appendix A fit to a large selection of glycerol relaxation data from multiple contributers [19, 79, 84, 87, 42, 75]. The same TGP and 1-bar constraint was imposed here as was implemented in Fig. 3.8. The color scale now represents percent deviation of the data from the surface. Note how the high pressure Pronin data begins to diverge from the surface, approaching an error as high as 20%. This data was still included in the fit as it conforms well to the surface at higher temperatures and lower viscosities—only tending away from the surface in a region where the TGP constraint does not allow the data significant influence over the fit parameters.

Figure 3.10 shows the same Andersson-Andersson constrained surface model fit to a larger selection of literature data including Cook's rolling ball viscosity data. The same TGP and atmospheric constraints are applied. Here the actual TGP data are provided on the surface yet not included in the fit. It is found that Cook's rolling ball viscosity data is in excellent agreement with

other works including Johari, Pronin, Paluch, and Reiser thus we must conclude that the Cook data is very reliable—the issue of increasingly overestimated fragility arising predominately from extreme extrapolation [52, 79, 84, 87]. It should be noted that the four isobaric data sets extracted from Pronin's figure 3 in Ref. [84] do trend increasingly above the fit surface at higher pressures yet agree with Cook and others at lower pressures. The toroidal pressure cell used by Pronin *et al*. and described in Ref. [57] may not have provided a hydrostatic sample to the degree that was assumed—a scenario which would certainly explain the data's increasing divergence from the TGP isochrone at higher pressures and lower temperatures as the glass transition is approached. The Pronin isobars are still in reasonable agreement with Cook's data at higher temperatures where hydrostatic conditions are assured, even at the highest pressure. However this data was included as it does not heavily effect the surface at lower pressures where it is in reasonable agreement with the rest of the data yet allows us to extend this fit surface up to 5 GPa. Thus it should be assumed that our calculated fragility above 3 GPa may have reasonable error associated with it and any fine structure is not of high confidence. It should be noted that this Pronin data could be replaced with DPCS measurements at high pressure near the glass transition just as was done with cumene. Molecular simulations may provide high pressure fast dynamics at extreme temperatures[59].

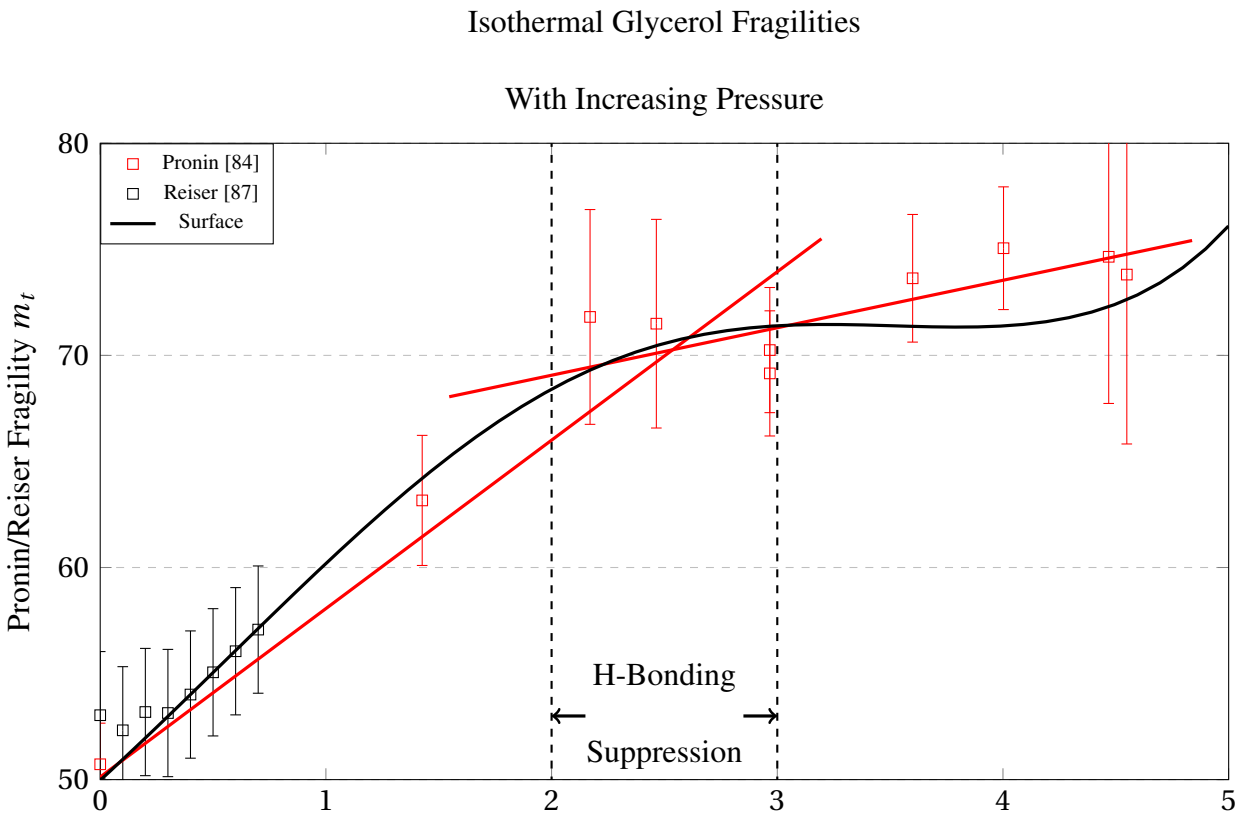


Figure 3.11: Fragility estimations from Pronin (□) and Reiser's (□) data are shown above. The pressure dependence of fragility derived from isobaric temperature derivatives of the Andersson-Andersson surface model described in App. A, fit to the data of Cook, Johari, Pronin, Paluch, and Reiser, is shown as a solid black line. The upward curve in fragility approaching 5 GPa is likely a relic due to the previously discussed issues with the Pronin data at high pressure as well as the surface being generally unconstrained by data beyond this pressure.

The calculated fragility as a function of pressure arising from the Andersson-Andersson surface model from App. A, including data from Cook, Johari, Pronin, Paluch and Reiser, is provided in Fig. 3.4—extending our fragility measurements out to 5 GPa. The red lines show Pronin's original estimation of the fragility behavior with increasing pressure as was lifted directly from his figure 4 in Ref. [84]. This result shows unequivocally Cook's rolling ball viscosity data to be in excellent agreement with other literature measurements of viscosity, η, and α-relaxation, τ. The Andersson-Andersson constrained surface model relieves us of the experimental burden of main-

taining isobaric conditions during data acquisition, something that is very difficult to accomplish in a diamond anvil cell, as well as avoids the reliance on free-volume models where a reliable equation of state may not be available or more dubious isothermal models such as the modified VTF equation.

3.5 Equation of State and Thermodynamic Scaling

In Ch. 4 Sec. 4.3.1 it is shown that the relaxation dynamics of a simple Van der Waals liquid such as cumene conforms very nicely to a single function of the form $\tau \propto \frac{1}{TV^{\gamma}}$, where γ is a constant scaling factor. This factor gamma has been shown to depend on the intermolecular potential energy landscape, $U(r)\sim\left(\frac{\sigma}{r}\right)^{3\gamma}$, where r is the spatial separation between any two molecules and σ is a material constant [90, 22]. In the case of a simple liquid such as cumene the potential is well approximated by the form of the Lennard-Jones model for which $3\gamma \approx 12$—consistent with the experimentally determined value of $\gamma \cong 4.77$. The dynamics of such liquids are strongly driven by simple free-volume considerations and thus we expect thermodynamic scaling to hold with increasing pressure and that is what we have observed to be the case.

However, in the case of strongly associated liquids such as glycerol, intra-molecular bonds influence the potential energy landscape. As these bonds are dependent on temperature and pressure, the energy landscape will be temperature and pressure dependent, thus one would not expect the emergence of thermodynamic scaling in these liquids. However several such liquids do exhibit thermodynamic scaling—at least over limited ranges of pressure and temperature. Glycerol is one such example, having a measured scaling parameter within the literature in the range of $1.4 \leq \gamma \leq 1.6$ [87, 60]. This lower value for γ is expected given that free-volume will play a reduced role in the dynamics of associated liquids as molecular mobility is no longer just a mater

of thermal diffusion past nearest neighbors exhibiting Van der Waals repulsion, but also will require energy to break hydrogen bonds with said neighbors. However these values for γ were all determined at relatively low pressures—on the order of 0.1 MPa. Yet we see in Fig. 3.4 a shift in the fragility behavior as a function of pressure occurring between 2 and 3 GPa which has been attributed to a suppression of hydrogen bonding. If hydrogen bonding between glycerol molecules is sufficiently suppressed at very high pressures, one might expect glycerol to behave much more like a simple Van der Waals liquid with a scaling parameter, γ, much closer to 4 as the potential energy landscape no longer is heavily influenced by these suppressed intra-molecular interactions.

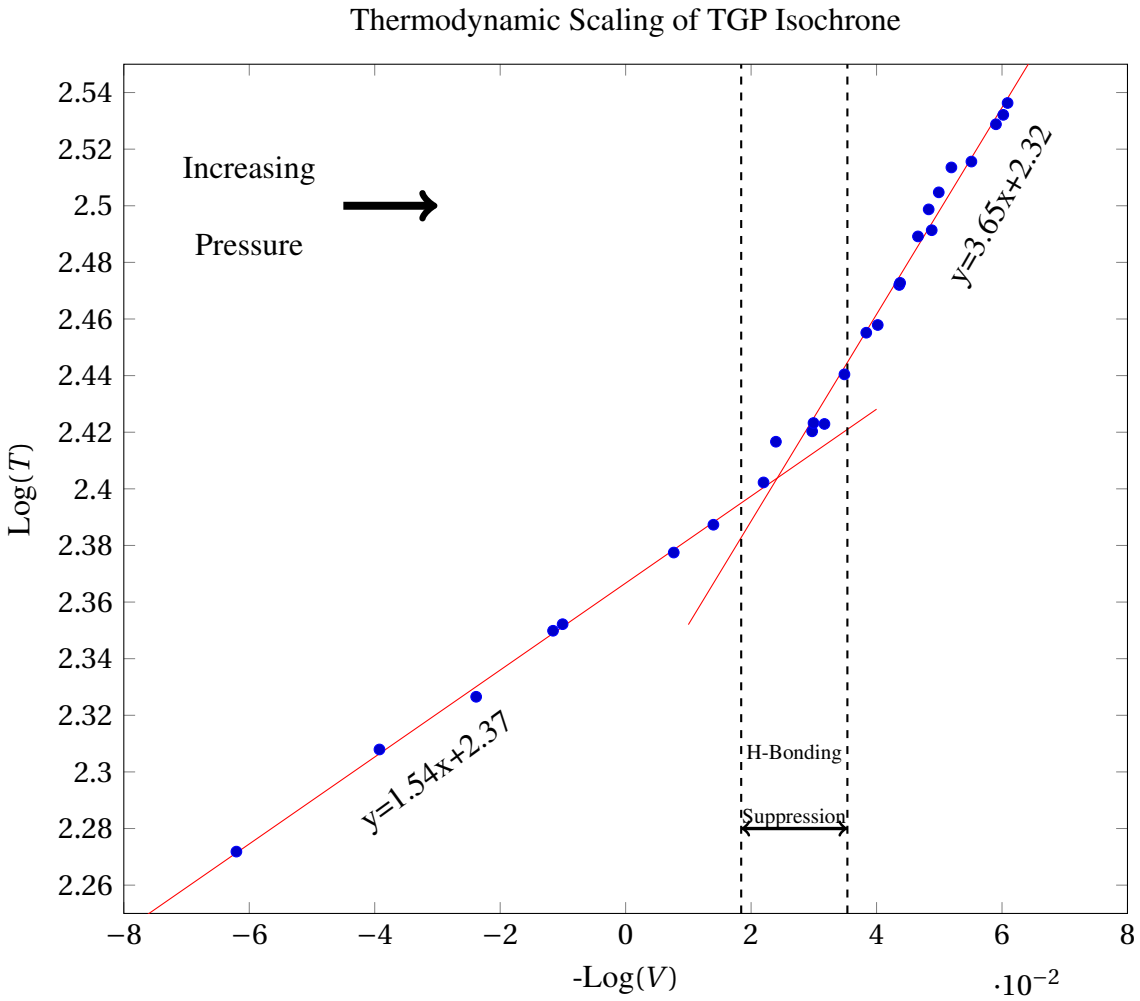


Figure 3.12: The linearized isochronal glycerol TGP data shows a transition from an H-bonded liquid at low pressure with a corresponding scaling factor of $\gamma \approxeq 1.54$ to a liquid which behaves more like one experiencing a simple Van der Waals potential at higher pressures, where H-boding is suppressed, exhibiting a scaling factor of $\gamma \approxeq 3.65$. The region between the dotted vertical lines corresponds the region indicated in Fig. 3.4 previously. The intersection of the two linear regimes is self consistent with the results of Pronin at approximately 2.5 GPa. It is also worthy of note that the average y-intercept of 2.35 implies a TGP isochrone of $\tau \approx 220$ s

In Fig. 3.5, the glycerol TGP data provided in App. B is linearized and plotted to determine the scaling parameter. The region between the dotted lines once again corresponds to that in Fig. 3.4—where pressure suppression of hydrogen bonding becomes prominent. The results are quite compelling. Two distinct regions emerge. At low pressure we find a scaling factor $\gamma = 1.54$ consistent with literature values. At high pressures, well within the region where hydrogen bonding

is assumed to be sufficiently suppressed, we find a significantly greater slope corresponding to $\gamma = 3.65$ consistent with what might be expected of a simpler Van der Waals liquid. A clear transition occurs between 2 and 3 GPa—consistent with our fragility results from our Andersson-Andersson contained surface model as well as Pronin's analysis.

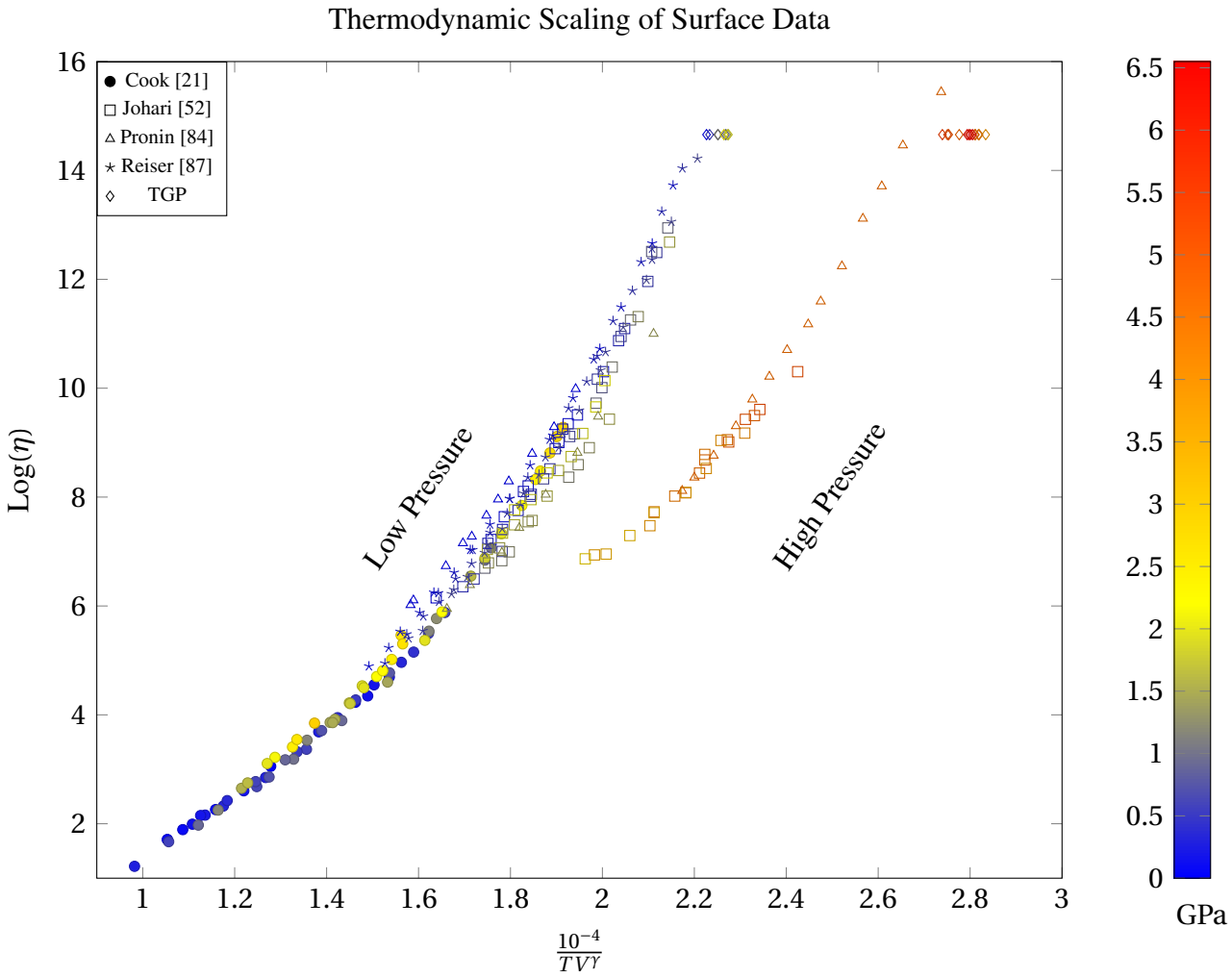


Figure 3.13: The results of thermodynamic scaling of glycerol for all low pressure data (< 2 GPa) with a scaling factor of $\gamma = 1.54$ and high pressure data (> 3 GPa) with a scaling factor of $\gamma = 3.65$.

All of the glycerol data that falls outside of the transition region between 2 and 3 GPa provided by Cook, Johari, Pronin, Reiser, and our TGP data exhibit a reasonable degree of thermo-

dynamic scaling along one of two curves. The results of this scaling are provided in Fig. 3.5. We see that all of the cook data scales nicely even well into the intermediate pressure regimen-perhaps due to the relative high temperatures allowing for molecular mobility which facilitates hydrogen bonding at these higher pressures. The Johari data, on the other hand, is all at relatively low to intermediate temperatures and a clear pressure dependent divergence from good thermodynamic scaling is observed as pressure increases and hydrogen bonding begins to be suppressed. All of this still relies on an extreme extrapolation of the Tait equation of state for which Bridgman only provides data up to approximately 1.2 GPa however thermodynamic scaling is possible, albeit in two distinct regimens each with different scaling factor, γ. Where hydrogen bonding is not being suppressed we find a scaling factor $\gamma = 1.54$ that reflects a liquid for which free-volume plays a diminished role in the dynamics as thermal activation is necessary to ionic bonds between neighbors. At very high pressures where hydrogen bonding is completely suppressed we find a scaling factor of $\gamma = 3.65$ that reflects a liquid governed much more strongly by free-volume and Van der Waals interactions.

Once gain we note that the high pressure Pronin data taken in the toroidal pressure cell seems to scale at lower pressures or higher temperatures however, as vitrification sets in at higher pressure the data peels away from the trend expected if thermodynamic scaling were to hold in precisely the same manner that it peeled away from the Andersson-Andersson constrained surface fit previously. It appears once again as if the reported pressures are lower than what the dynamics would imply. As this discrepancy is occurring only at the highest pressure and in the region where vitrification is rapidly increasing, it stands to reason that they have underestimated the prominence of pressure gradients within their toroidal cell. The lower pressure data, even in the vicinity of the glass transition, scales nicely just as it conformed very well to the relaxation surface. However as

pressures increase the odds of significant pressure gradients inside the cell increases greatly.

4 Photon Correlation Spectroscopy in a Diamond Anvil Cell

4.1 Justification

Section 1.3 describes several probes commonly implemented to measure molecular dynamics in glass forming systems as the timescale associated with structural the relaxation changes by as much as fifteen orders of magnitude while the liquid vitrifies into a glass. Wolynes and Lubchenko assert in their section on Dielectric Spectroscopy (DS) that this technique is uniquely suited to track molecular dynamics in glass forming systems over this large dynamic range and indeed, when taken as a whole, the multitude of methods that fall under the banner of DS provide an absolutely astonishing range of measurement for relaxation times from $\tau \approx 1$ nanosecond up to $\tau >$ 1000 seconds[112]. However many of these methods do not lend themselves well to the geometric constraints and experimental limitations of a diamond anvil cell. For example, reflection or transmission techniques require that the sample be arranged in a macroscopic coaxial configuration. The methods that may lend themselves well, notably frequency response and time domain analysis, have yet to gain significant implementation in the context of the diamond anvil cell, though though notable exceptions certainly exist [41, 45, 56].

As shown in Fig. 1.8, depolarized photon correlation spectroscopy covers a similar dynamic range as DS frequency response analysis, in principle from approximately 10 ns to beyond one thousand seconds, and provides a direct probe of the structural alpha relaxation. Though both DPCS and DS may provide similar windows for measuring molecular dynamics, DPCS does not require electrode structures within the sample volume; rather only optical access is required and

thus it is arguably a more accessible option for extending these measurements into the DAC. However it could be strongly argued that both methods are worthy of development as they have differing limitations. DS will only provide actual structural relaxation data if the dielectric susceptibility is representative of the orientational dynamics of the liquid under study. DPCS requires that the liquid under study have a healthy anisotropic component to its structure in order to give rise to a measurable depolarized signal in the first place. A good example of a molecule that would lend itself to DS, but not DPCS, would be n-propanol, which has a reasonably strong dipole moment of 1.68 Debye yet lacks anisotropy in its structure. However for most polymers, as the dipole moments are transverse to the chain and may be independently mobile, DS will often probe local segmental dynamics as well as the α-relaxation of the bulk material. Though a strong anisotropy may be present in the molecular structure, these local relaxations often make it difficult to sort out and analyze the bulk α-relaxation and other dynamic properties via DS. For such systems DPCS is an ideal probe of structural relaxation dynamics as it avoids these complications and provides a direct window on the stochastic thermal evolution of the system[89, 81].

There are significant experimental hurdles associated with extending photon correlation spectroscopy into the diamond anvil cell. Perhaps the most daunting hurdle is contending with the exceedingly small sample volumes–on the order of $\approx$ 10nl. In a typical light scattering experiment the scattered signal is collected from a region illuminated by a focused beam that may extend for as much as a couple centimeters however, in a DAC, this illuminated region is limited to on the order of hundreds of microns. At this point it become reasonable to check that there are enough scattering bodies to assure one is well within the stochastic limit. These issues, coupled the fact that stray spectral reflections from every surface interface as well as the inevitable splash from the gasket material is collected along with the desired signal, make isolating your scattering region

quickly become a challenge.

Herbst and King quite successfully tackled these hurdles in their work aimed at extending photon correlation spectroscopy into the diamond anvil cell to be used as a probe of viscosity at high pressures[45]. They were able to show reasonable agreement with known values of methanol viscosity at elevated pressures, up to 2.9 GPa, via analysis of the Brownian motion of polystyrene spheres added to the liquid. Though successful, it should be noted that this method of viscosity measurement has not shown up in the literature since its publication likely due to the tedium associated with the process not warranting its limited applicability–only probing viscosities up to around 90 centipoise. Though this method could certainly be extended to slightly higher viscosities via implementation of smaller particles and running longer timescale autocorrelations, it is clearly orders of magnitude from providing a probe at or around the glass transition of 10^{15} cP. In fact,during the same year, Richard Cook, working with Herbst and King, published a paper demonstrating centrifuge rolling ball measurements in a DAC reaching viscosities beyond 10^{9} cP[19]. Yet rolling ball viscometry is inherently limited to viscosities for which the macroscopic ball can progress through the fluid during a reasonable experimental timescale. This cannot, by definition, extend to the glass transition. However photon correlation spectroscopy within a DAC, in theory, could absolutely extend to the glass transition and beyond if, rather than scattering from nano-particles dissolved in the liquid, one could obtain the correlation spectra of light scattered from a pure liquid's density fluctuations. Perhaps the greatest challenge associated with this would be the fact that the intensity of light scattered from spherical particles decreases on the order of the fourth power of the particle radius[95]. As such, the intensity scattered from density fluctuations in a pure liquid will be several orders of magnitude less than that scattered from the 90nm polystyrene spheres used by Herbst and King. All of the issues arising from the small sample volume therefor

become tremendously amplified when attempting to extend their results to a pure liquid.

The goal of this chapter is to outline the process of extending DPCS into the diamond anvil cell thus providing a new probe for slow dynamics at very high pressures–up to and exceeding the glass transition. Isopropylbenzene, a simple van der Walls bonded glass-forming liquid, was chosen to be our proof of concept system both for its strong depolarized scattering signal as well as the fact that fast dynamic data from a concurrent Brillouin light scattering study was available allowing for characterization of the relaxation dynamics over a massive dynamic range–from tens of picoseconds to hundreds of seconds. A second concurrent study provided a direct measurement of the glass transition temperature as a function of pressure via the method described in chapter 6.2. Self consistency between these three methods will provide strong evidence as to the efficacy of DPCS as a probe of slow dynamics at high pressures in a diamond anvil cell.

4.2 Experimental Details

As previously mentioned it was expected that scattered light arising from density fluctuations within a pure liquid would be of exceedingly low intensity relative to that arising from undesired sources such as spectral reflections from the surface interfaces. An initial attempt was made to simply regard this additional light as a local oscillator–dumping all of the collected light onto the PMT. Neutral density filters were used to bring the PMT counts down to a safe level. Unsurprisingly, with so many coherence areas and the local oscillator undoubtedly many orders of magnitude more intense than the light scattered from density fluctuations, the correlation decay could not be resolved even after running for several hours. Thus it was clear that spacial filtering was in order as this would limit the number of coherence areas as well as reject much of the excessive local oscillator. However this now opened up a new problem–physically finding the scattering

volume within a exceedingly large area of optical splash. A telescopic viewfinder could be lowered into place to image the inside of the DAC for alignment purposes however this proved insufficient. One was still left to blindly raster the pinhole across the region of interest in an attempt to isolate the correlation decay signal. Having no idea how long we expected to have to signal average before a clear decay would present itself, as well as the tedious nature of rastering over a relatively large area blindly seeking this slow forming correlation decay signal. The solution was found in employing the already existing Brillouin setup to optimize the spacial filtering before redirecting the light on to the autocorrelator. The complete experimental layout is detailed in figure 4.2 below and will here be here described along with the process of obtaining an autocorrelation signal.

A Coherent Innova 300 actively locked single mode Ar^+ medium frame laser run at 514nm provided our light source. To obtain the highest level of power stability the laser would be allowed to warm up at a set current near where it would be operated for one to two hours. At this point manual adjustments were made to the rear mirror to maximize output. Next power and mode tracking would be enabled at 1 Watt output and the laser left again to warm up for an additional half hour making occasional adjustments to the alignment to keep the current minimized. Approximately once per month the mode tune algorithm would be run after this warm-up procedure. A set of reflective neutral density filters allows this 1 Watt output to be incrementally adjustable between 1 and approximately 30 milliwatts while maintaining the power stability of the laser. As our scattering signals are expected to be so excessively weak, laser stability is of the utmost concern. Future improvements on this system may include the implementation of cross-correlation in order to reduce the dependence on power stability and allow for the analysis of weaker scattering systems.

As shown in figure 4.2, the beam is focused through the DAC located at position (a). The

DAC is positioned on a four axis stage allowing for x, y, z, and θ translations of the cell. What we will refer to as the z axis is aligned along the incoming focused laser and is the primary axis that must be adjusted during signal optimization which will be discussed shortly. The θ axis is in the plane of the optical table and allows the DAC to be rotated as shown. The DAC is usually rotated away from the collection mirror in order to reduce the amount of spectrally reflected light mixed in with the signal. It should be noted that changes to θ do not effect the scattering vector $\vec{q}$ which is geometrically set by the collection mirror at location (b). This 1 cm diameter mirror is placed as close as possible to the incoming beam without occultation in order to closely approximate a 180°backscattering geometry. The actual external angle between the incoming beam and the path from the scattering volume to the center of the mirror is less than 3°. Due to the index of refraction of the diamonds and liquid sample, the actual scattering angle is significantly less than this–estimated to be less than 1°. The collimating optic that follows is preceded by an iris which is used in Brillouin studies to limit the scattering vector $\vec{q}$ collected however here is opened reasonably wide yet still used to limit the collection cone–effectively providing a degree of low resolution spacial filtering via rejection of any off axis light. The collimated light then proceeds through a Glan-Thompson polarizer at location (c) which may be removed for alignment purposes and replaced to select the depolarized component of the scattering signal. The beam hight is adjusted by the next two mirrors, intentionally after the polarizer, to match the input hight of the interferometer. A 200μm pinhole is used for spacial filtering and the flip mirror at location (d) is flipped down, out of the path of the incoming light, allowing for spectral analysis. Even at high pressure, near the glass transition, light scattering off of lateral acoustic modes gives rise to classic Brillouin peaks as shown in figure 4.2 below.

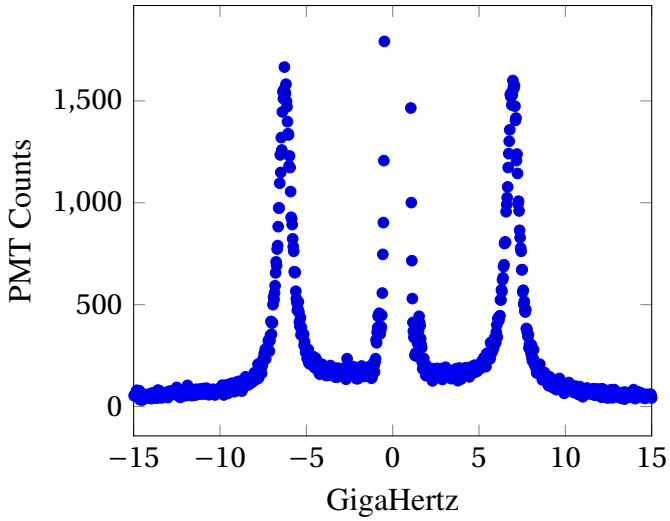


Figure 4.1: Typical Brillouin Spectrum with polarizer removed. The rapidly forming lateral acoustic (LA) modes are used to optimize the DAC position to obtain the strongest signal. The polarizer is then replaced and rotated until these LA modes are suppressed, ideally leaving only the depolarized signal for DPCS analysis.

Here the spacial filtering may be optimized via maximizing the intensity of these lateral acoustic mode peaks. This is done predominately by adjusting the before-mentioned z-axis which translates the DAC along the path of the incoming laser light such that the collection area being focused onto the pinhole at location (d) corresponds to the liquid sample between the diamond cutlets. The fact that this Brillouin spectra forms significantly more quickly than the much weaker depolarized spectrum allows for rapid optimization of the spacial filtering. Once this is achieved, the polarizer is replaced and oriented to suppress the LA-modes–passing only the depolarized component of the scattered light. The mirror at location (d) is then flipped up, redirecting the spatially filtered light towards the autocorrelator PMT. The next focusing optic re-collimates the light after the pinhole and is followed by two mirrors to once again adjust the hight of the beam to match the input hight of the autocorrelator optics. The autocorrelator setup itself was previously designed for use with a 1cm cuvette and it was desirable not to strip the system of this capability.

As such, the original PCS system was left in tact so that a cuvette and oven could be placed at location (e) or removed at will. To accommodate for PCS in the DAC, the lens following the hight adjustment mirrors creates an image of the DAC sample at location (e) which the PCS setup then further spatially filters before directing the light to the PMT.

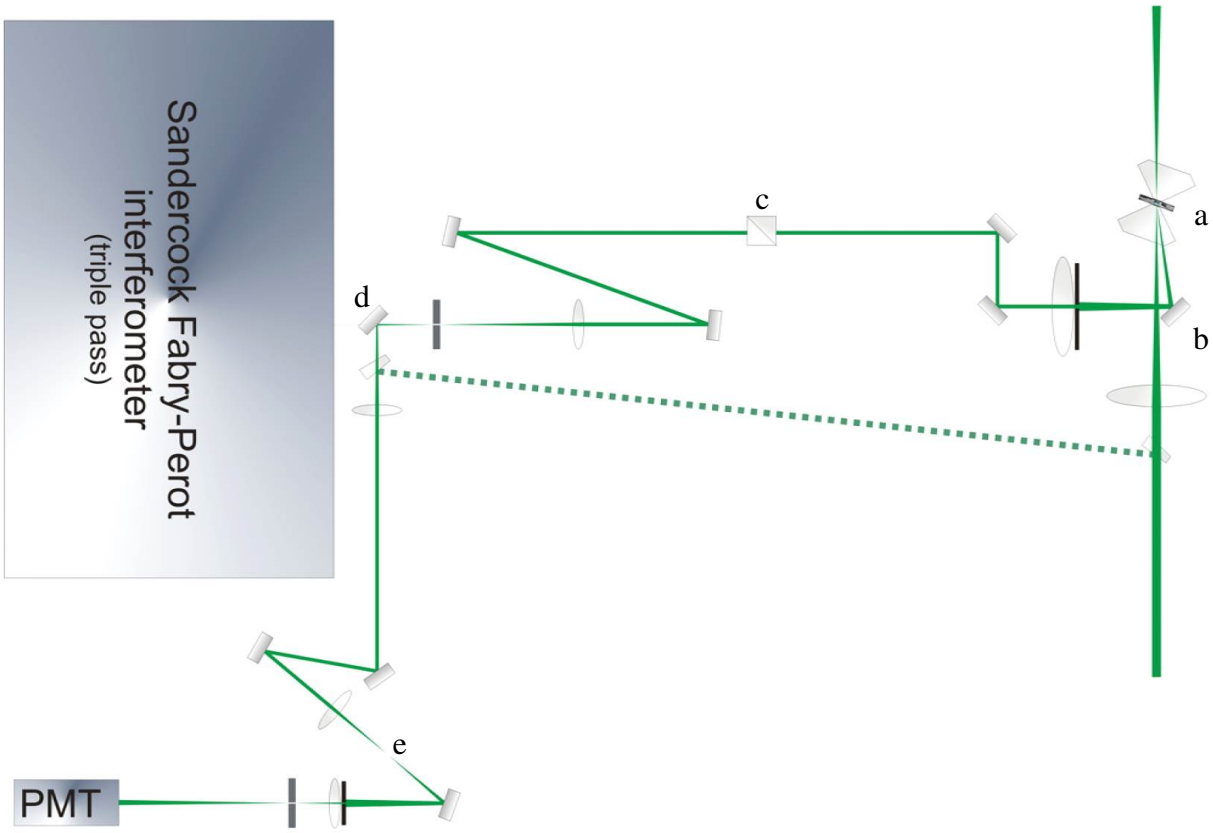


Figure 4.2: Major components of the experimental setup for DPCS are as follows: (a) Diamond Anvil Cell, (b) 1cm mirror to redirect scattered signal, (c) Glan-Thompson polarizer, (d) flip mirror to redirect signal from interferometer to autocorrelator. Irises and pinholes are shown as black and gray bars respectively. The dashed line represents an attempt to provide a local oscillator for heterodyne DPCS analysis.

4.2.1 Homodyne *vs*. Heterodyne

As mentioned in section 4.1, photon correlation spectroscopy may be performed in ether a homodyne (self-beating) or a heterodyne (local oscillator) arrangement. What was not mentioned

is the requirement that one is squarely within one of the two regimes in order to avoid prohibitively complex analysis. Elimination of the heterodyne component in the collected light signal is a significant hurdle to overcome when attempting PCS within a DAC as the presence of surrounding structure, the edges of the gasket, every surface interface, and even the inevitable dust on the diamond table, scatter light at significant intensity. This is further exacerbated by the fact that the scattering volume is excessively small (on the order of tens of nanoliters) thus it is very easy for this heterodyne component to be comparable to, or even significantly more intense than, light scattered from the sample volume. The majority of this heterodyne component is immediately rejected via the GT-Prism during depolarized studies however, due predominantly to birefringence within the diamonds at these high pressures, a reasonably large component of this local oscillator will always bleed through. Herbst and King expended a great deal of effort in addressing this issue when first achieving PCS in a DAC. [45] However they were able to achieve near-homodyne conditions through proper orientation of the diamond anvil cell to deflect spectral reflections away from the collection optics as well as spatial filtering of the collected light. Their method, however, proves insufficient if one wishes to extend this technique to directly probe density fluctuations in a pure liquid. Despite similar sample sizes, Herbst and King were measuring the thermal diffusion of 90nm polystyrene spheres, and thus had the advantage of orders of magnitude greater signal intensity. As Herbst's method did not completely eliminate the heterodyne component, it is unreasonable to assume that, when scattering from density fluctuations in a pure liquid, we would remain in near-homodyne conditions as some amount of spectrally reflected light from every surface of the gasket inevitably will be collected along with the desired signal despite the best attempts at spatial filtering. In fact, given that the signal arising from density fluctuations within a pure liquid from a few nanoliters of fluid is expected to be so minute, it is reasonable to suspect that pure heterodyne

conditions may be inevitable.

A simple means to test if conditions are reasonably well within the heterodyne regimen would be to simply increase the intensity of a local oscillator and see if the values obtained for the alpha relaxation times are affected. If one can significantly increase the ratio of idler to signal and the correlation function remains unaffected then it is reasonable to assume a heterodyne situation. To accomplish this, we effectively inverted Herbst and King's procedure–rotating the DAC to incrementally increase (as opposed to decrease) the amount of spectrally reflected light impinging on the collection optics thus providing a local oscillator of increasing intensity. Correlation functions were obtained at a set pressure and temperature while incrementally increasing the angle subtended between the diamond table and the incoming laser, the data for which is provided in table 4.1 below.

Angle (degrees)	Thousand Counts	$<\tau>$ (μs)	$\Delta<\tau>$ (μs)
0	200	1360	54
5	164	1188	57
10	156	1296	73
15	120	1232	33
20	250	1449	57

Table 4.1: PMT counts at various orientations of the diamond anvil cell and the associated alpha relaxation times $<\tau>$. Zero degrees corresponds to the incoming laser at normal incidence to the diamond table. The DAC was then rotated *away* from the collection mirror.

What is really of interest here is how the expectation value of the relaxation time changes with increasing intensity of any heterodyne component. It is reasonable to assume that the intensity

of the light scattered from the sample is not increased substantially simply by rotating the DAC twenty degrees. Thus the increase in PMT counts at certain angular locations is due predominately to varying degrees of scatter and specular reflections from the sources previously discussed. Though Figure 4.2.1 below does appear to demonstrate an increasing trend in $<\tau>$ as the heterodyne component is increased, it should be noted that the PMT counts more than double. Thus, as a bare minimum, we have doubled the intensity of the local oscillator and yet this only corresponds to an increase in $<\tau>$ of approximately 17 percent implying that we are near, though possibly not thoroughly within, the heterodyne regime. We will latter find this assessment consistent with the agreement of this data with dynamic data obtained via other probes–Brillouin, TGP, *etc*. .

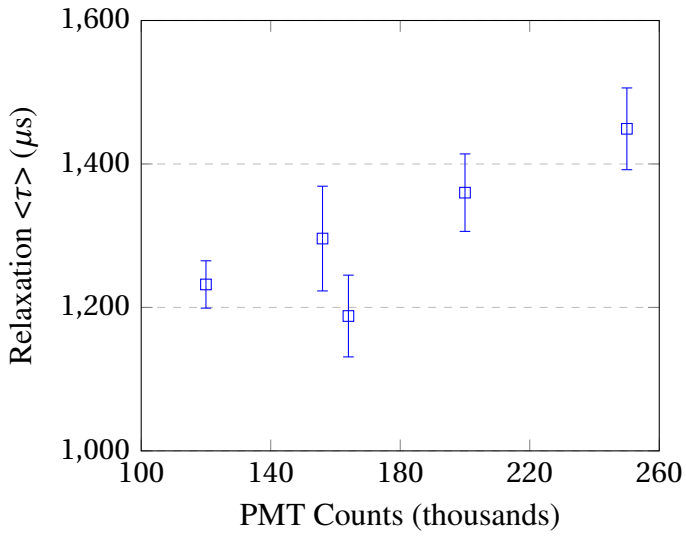


Figure 4.3: Structural relaxation times $<\tau>$ as a function of increasing scatter and specular reflections serving as a local oscillator. Confidence intervals at 95% are derived from the M.S.E. of the fit parameter τ.

4.2.2 Time-Correlation Functions

The correlation spectra arising from the Brownian motion of mono-disperse spheres in a liquid will have a single activation energy associated with any given particle breaking free of a cage

of its nearest neighbors. As such, one expects and finds a very simple single exponential decay of its correlation spectra with a characteristic time τ associated with this structural relaxation. However the average glass forming system proves to be of significantly higher complexity having a distribution of relaxation times. Historically speaking, these stretched exponential time-correlation spectra have been fit to the Kohlraush-Williams-Watts (KWW) function which takes the form of equation 4.1 below.

$$C(t) = e^{(-t/\tau)^{\beta}} \tag{4.1}$$

The use of the KWW fucntion to describe glassy systems is theoretically compelling for many reasons, not the least of which is that the Laplace transform of the time correlation function is related to the complex dielectric constant ϵ^*as follows:[13][110]

$$\frac{\varepsilon^* - \varepsilon_\infty}{\varepsilon_0 - \varepsilon_\infty} = \mathscr{L}\left[-\frac{\mathrm{d}C(t))}{\mathrm{d}t}\right] \tag{4.2}$$

In the simple case where $\beta = 1$, equation 4.1 reduces to a simple exponential decay which may be interpreted as a system driven by a single relaxation process, such as in the case of spherical particles breaking free of nearest neighbor cages within a viscous medium. When transformed via equation 4.2, this yields a simple Debye process with a characteristic relaxation time, τ[26].

$$\mathscr{L}\left[-\frac{\mathrm{d}C(t))}{\mathrm{d}t}\right]_{\beta=1} = (1 + i\omega\tau)^{-1} \tag{4.3}$$

When working in frequency space, such as with Broullioun light scattering or dielectric frequency response analysis, the relaxation dynamics of glass forming systems are routinely described in a

very similar fashion to equation 4.1 via the Cole-Davidson distribution:

$$\frac{\varepsilon^* - \varepsilon_\infty}{\varepsilon_0 - \varepsilon_\infty} \approx \left(\frac{1}{1 + i\omega\tau_{cd}}\right)^{\beta_{cd}} \tag{4.4}$$

The symmetry between equations 4.4 and 4.1, along the the fact that the simple Debye process described by equation 4.3 directly correlates to a pure exponential decay in the temporal domain, might lead one to conclude that the Cole-Davidson is the Laplace transform of the KWW stretched exponential function–it is not. Both may be interpreted as arising from a distribution of different relaxation processes, each independently describable by a simple exponential decay or the associated Debye relaxation in frequency space. However as they each assume a fundamentally different distribution function, the actual values for τ and β will not be identical. This is a concern as different probes which function in both frequency and temporal space are needed to track molecular dynamics of glass forming systems over such a massive dynamic range. In this study, we will be relating data fit to a Cole-Davidson to our time domain DPCS measurements. Lindsey and Patterson have addressed this issue via assuming that the KWW and Cole-Davidson distributions are arise from a distribution of simple exponential decays and Debye processes respectively as follows:[63]

$$e^{(-t/\tau_{kww})^\beta} = \int_0^\infty e^{-t/\tau} \rho_{kww}(\tau) d\tau \tag{4.5}$$

$$\left(\frac{1}{1 + i\omega\tau_{cd}}\right)^{\beta_{cd}} = \int_0^\infty \frac{1}{\tau}\left(\frac{1}{1 + i\omega\tau}\right) \rho_{cd}(\tau) d\tau \tag{4.6}$$

A great deal of effort was then put into solving for these distributions, ρ_{cd} and ρ_{kww}, analytically and via an expansion respectively. The differences between the two were then numerically assessed and a means for conversion between the τ and β parameters for the two distributions was

determined. Thus throughout the literature it has been the norm to fit frequency domain data to a Cole-Davidson distribution and time domain data to a KWW stretched exponential and convert one or the other via the relationship described by Lindsey and Patterson when they need to be related.

Of course another option, and arguably more physically correct, would be to relate the expectation values of the τ parameters. If the assumed distributions are valid, though the actual τ parameters may be different, one would expect that the expectation values, $<\tau>$ must agree. For a known distribution, $\rho(\tau)$, the expectation value $<\tau$ is defined as follows:

$$<\tau>=\int_0^\infty \tau\rho(\tau)d\tau \tag{4.7}$$

In the case of the KWW stretched exponential and the Cole-Davidson distributions, this yields:

$$<\tau>_{kww}=\frac{\tau_{kww}}{\beta_{kww}}\Gamma\left(\frac{1}{\beta_{kww}}\right) \tag{4.8}$$

$$<\tau>_{cd}=\beta_{cd}\tau_{cd} \tag{4.9}$$

However further results provided by Linsey and Patterson show that for $\beta_{cd} < 0.5$ the ratio of $\frac{<\tau_{kww}>}{<\tau_{cd}>}$ quickly becomes significantly greater than unity–*i.e.* the difference in the underlying distributions negates the validity of this assumption. For Cumene, $\beta_{cd} \approx 0.5$ thus it would be prudent to convert to one or the other parameter. However $\rho_{cd}(\tau)$, shown in equation 4.10 below, is also provided by Linsey and Patterson. [63]

$$\rho(\tau)_{cd}=\frac{sin(\pi\beta_{cd})}{\pi}\left(\frac{\tau}{\tau_{cd}-\tau}\right)^{\beta_{cd}} \tag{4.10}$$

This distribution, obtained from equation 4.6 and the normalization condition, $\int_0^\infty \frac{1}{\tau}\rho_{cd}(\tau)d\tau = 1$, allows one to define the Cole-Davidson function in the time domain. Recognizing the validity of equation 4.3, and that $\rho(\tau)_{cd}$ was obtained by assuming the Cole-Davidson is equivalent to a distribution of simple Debye processes, the Cole-Davidson can be expressed in the time domain as follows:

$$\left(\frac{1}{1+i\omega\tau_{cd}}\right)^{\beta_{cd}} = \mathscr{L}\left[-\frac{d}{dt}\int_0^{\tau_{cd}} \frac{1}{\tau}e^{-t/\tau}\rho(\tau)_{cd}d\tau\right] \tag{4.11}$$

The work of Linsey and Patterson has been available for almost four decades however it is evident why it was common to fit time domain data to a KWW stretched exponential and then convert the parameters, as opposed to fitting directly to the Cole-Davidson distribution. Requiring the nonlinear fit routine to perform this integration would have been quite resource prohibitive on 1980's hardware. It is non-trivial even now however it can be recast into a form involving one and two parameter gamma functions which, with the aid of modern lookup-tables, can be relatively quickly computed.

$$\int_0^{\tau_{cd}} \frac{1}{\tau}e^{-t/\tau}\rho(\tau)_{cd}d\tau = \frac{1}{\pi}\Gamma\left[1-\beta_{cd}\right]\Gamma\left[\beta_{cd}, \frac{t}{\tau_{cd}}\right] Sin\left[\pi\beta_{cd}\right] \tag{4.12}$$

Thus, in order to have values of τ that are directly comparable to those obtained from frequency domain Brillouin studies, all DPCS time domain correlation decays will be fit via Mathematica to a distribution of single exponential decays corresponding to a Cole-Davidson in the frequency domain as per equation 4.13.

$$C(t)_{cd} = \frac{1}{\pi}\Gamma\left[1-\beta_{cd}\right]\Gamma\left[\beta_{cd}, \frac{t}{\tau_{cd}}\right] Sin\left[\pi\beta_{cd}\right] \tag{4.13}$$

4.3 Results and discussions

4.3.1 Thermodynamic Scaling

Another means of checking the validity of the DPCS data, as well as an additional confirmation that the data was taken under heterodyne/homodyne experimental conditions, is to determine how well it conforms to established thermodynamic scaling for cumene. ?? add references to scaling ?? The cumene TGP data, provided here and in appendix C, allows us to determine the scaling factor. This is done by recognizing that liquids which conform to thermodynamic scaling relate temperature, volume and relaxation time as follows:

$$\tau \propto \frac{1}{TV^{\gamma}} \tag{4.14}$$

Thus, as the TGP data comprises an isochronous curve at $\tau \approx 100s$, taking the Log of both sides yields the linear expression provided in equation 4.15, the slope of which is the scaling factor γ and the y-intercept should give us an estimation as to the relaxation time associated with the TGP data. Experimentally, the TGP process provides the glass transition temperature as a function of pressure. In order to proceed it is necessary to convert these pressures to their associated specific volumes. For cumene, this is accomplished via the modified Tait equation of state provided in appendix D.

$$Log_{10}(T) = -\gamma Log_{10}(V) - Log_{10}(\tau_G) \tag{4.15}$$

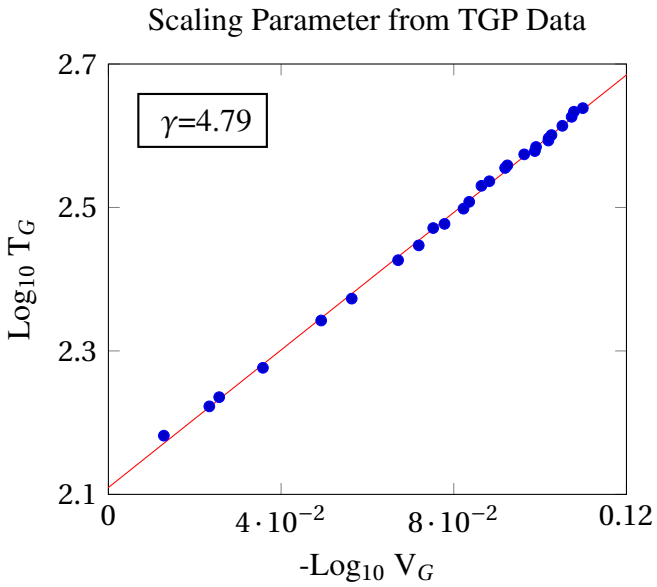


Figure 4.4: Linearized Cumene TGP data provided in appendix C has been fit to the regression model $Log_{10}(T) = -\gamma Log_{10}(V) - Log_{10}(\tau_G)$, the slope of which is the scaling parameter $\gamma = 4.77$ and the y-intercept of 2.11 implies an average TGP relaxation timescale of $\tau \approx 130$ s.

The value of the scaling factor having been obtained in this fashion, the DPCS data provided in appendix F may be checked to see if it scales in such a way as to agree with the Brillouin fast dynamic data and the TPG data for $\gamma = 4.77$. Again the specific volumes are obtained via equation D.1. The results of the thermodynamic scaling are given below in figure 4.3.1. With regards to the heterodyne/homodyne concerns addressed in section **??**, the great majority of data was taken with the DAC oriented such that a significant amount of local oscillator arising from the surfaces was collected along with the desired signal thus assuring we are well within the heterodyne regime. This data was fit to equation 4.13 to obtain the alpha relaxation parameter, τ_{cd}. These scaled values are labeled "DPCS" in figure 4.3.1 and are in excellent agreement with the scaled Bruillouin and TGP data. However, as the intensity of the light from the scattering volume is so much less than that of the local oscillator, it is often necessary to signal average for several hours when running in a heterodyne arrangement. As mentioned in section 4.2.1, it was generally assumed that we

are always solidly in the heterodyne regime due to the exceedingly small sample volume. Thus minimizing the splash off the DAC to decrease the acquisition time seemed reasonable. With this in mind, when the 75°isotherm was taken, the DAC was oriented to minimize the PMT counts. This resulted in relatively rapid data acquisition—in some cases as little as ten minutes. However, when fit to equation 4.13 and scaled, it was found that the points all fell systematically above the rest of the data that were known to be solidly heterodyne. This suggested that the data was perhaps closer to purely homodyne than was originally assumed. Having no reliable means to determine the extent of homodynicity, the data was fit to 4.16 which assumes a Cole-Davison distribution of purely heterodyne correlation decays.

$$\int_0^{\tau_{cd}} \frac{1}{\tau} e^{-2t/\tau} \rho(\tau)_{cd} d\tau = \frac{1}{\pi} \Gamma\left[1-\beta_{cd}\right] \Gamma\left[\beta_{cd}, \frac{2t}{\tau_{cd}}\right] Sin\left[\pi\beta_{cd}\right] \tag{4.16}$$

When refit assuming homodynicity, it was found that the scaled values were in agreement with the rest of the scaled data. Thus it appears that by orienting the DAC to reject as much non-target light as possible, very nearly pure homodyne experimental conditions were obtained. With further development it may be possible in many cases, particularly for liquids with a large depolarized scattering signal, to run confidently in a homodyne arrangement thus speeding up data acquisition. However, as the only means to be certain that an experiment is reasonably homodyne is to compare it with data known to be heterodyne, it was deemed best to intentionally orient the DAC such that sufficient splash from the surfaces was collected to assure the data was solidly within the heterodyne regime for data acquisition. This is how all data, with the exception of the 75 °C isotherm, was taken.

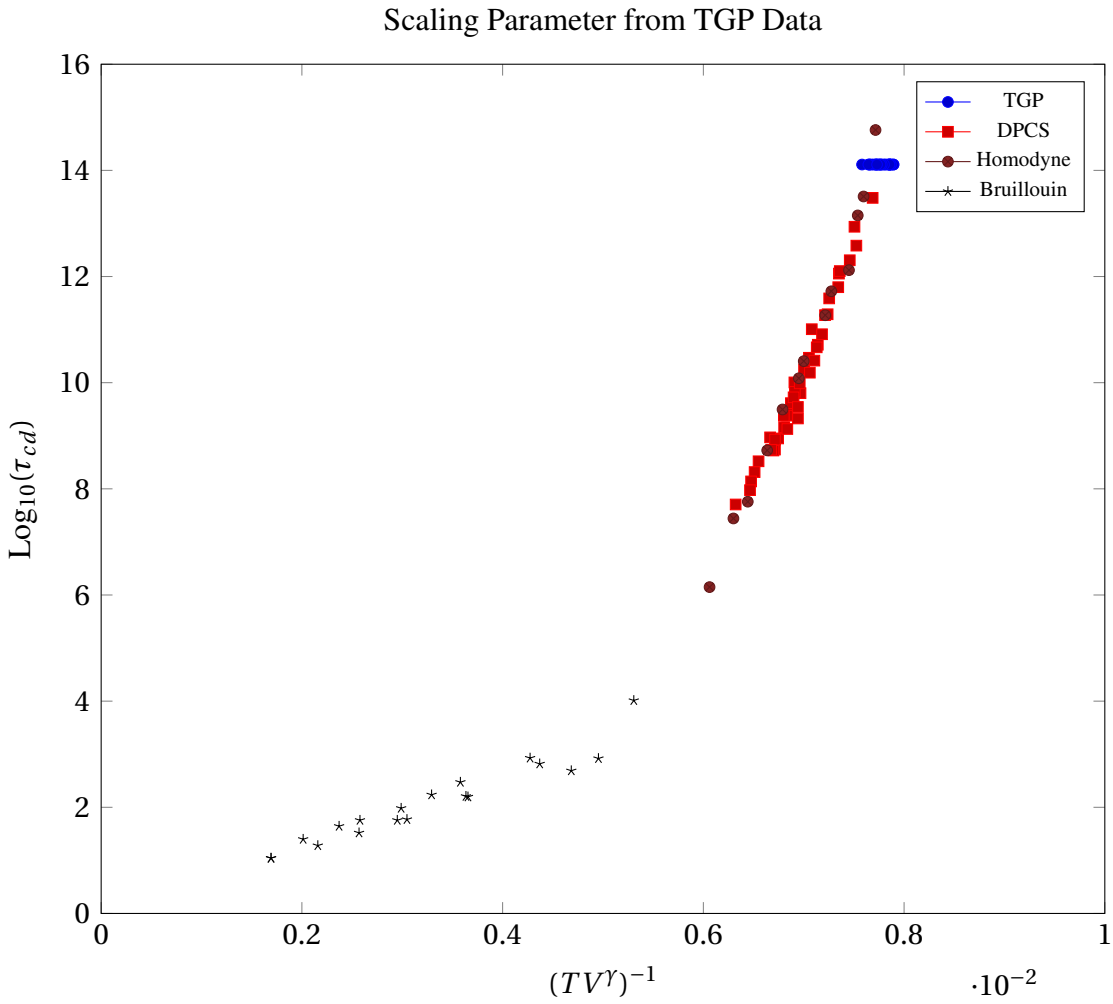


Figure 4.5: You momma

4.3.2 Alpha Relaxation Dynamics

To obtain a full description of the relaxation dynamics of cumene over the entire experimental range, data obtained from DPCS, and well as DLS and TGP methods, have been pooled together to fill in as much of the dynamic space as possible. A surface fit of this dynamic data was

performed implementing the Vogel-Fulcher-Tammon (VTF) equation,

$$\ln(\tau) = \ln(\tau_\infty) + \frac{D T_0}{T - T_0} \tag{4.17}$$

where τ_∞ is the infinite temperature relaxation time assumed to be constant for all pressures, T_0 is the zero mobility temperature at which the α-relaxation time asymptotically approaches infinity, and D is the fragility parameter. A simple power expansion of the fragility and zero mobility parameters allows this classically temperature dependent model to describe dynamics over a very large range of pressures–from zero to approximately 3 GPa in this case. The resulting pressure extended VTF model, as well as the values for these parameters obtained via a Levenberg-Marquardt fit routine, are described by equation 4.18.

$$\begin{aligned}
&\ln(\tau) = \ln(\tau_\infty) + \frac{D(P)\,T_0(P)}{T - T_0(P)} \\
&FOR \quad VECTORS: \\
&D(P) = \sum_{j=0}^{2} d_j P^j \quad AND \quad T_0(P) = \sum_{j=0}^{3} t_j P^j \\
&WHERE \quad \tau_\infty \equiv 1.2869 \text{ ps} \quad AND: \\
&\vec{d} = \begin{pmatrix} 3.61258 \\ 9.79785 \\ -0.193073 \\ 0.00162822 \end{pmatrix} \quad \vec{t} = \begin{pmatrix} 116.263 \text{ K} \\ 0.0354668 \text{ Kkbar}^{-1} \\ 0.00403396 \text{ Kkbar}^{-2} \end{pmatrix}
\end{aligned} \tag{4.18}$$

It should be noted that the values for τ_∞, d_0, and t_0, obtained from a separate VTF fit to the ambient pressure isobaric data, were set as constants for the surface fit thus forcing the surface through the

well established values at atmospheric pressure. This significantly increased the fidelity of the surface fit in the low pressure domain where there is significantly less data. The resulting surface is provided in figure 4.6. Partial derivatives of equation 4.18 with respect to temperature allow for the calculation of isobaric fragility directly as a function of pressure. From literature values and previous work it is expected that cumene have a fragility at atmospheric pressure of approximately $m_p = 70 \pm 5$, decreasing with pressure to a value of $m_p = 50 \pm 3$ at approximately 3 GPa. [108][86]

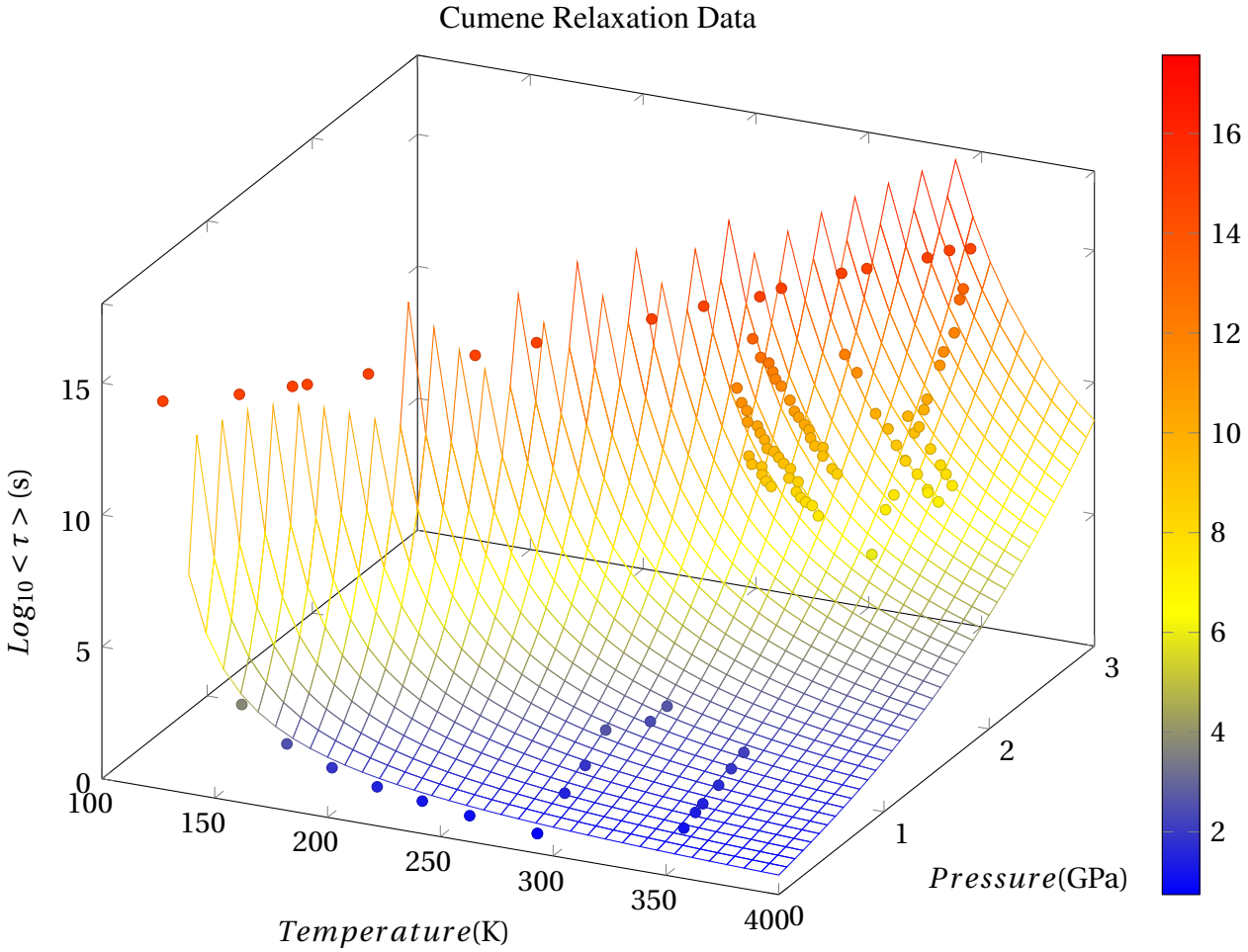


Figure 4.6: Stuff I wanna say about this shizzel

It is evident from quick examination of the surface in figure 4.6 that the isobaric fragility is

decreasing as pressure increases. The fragility my be expressed in terms of the surface fit parameters as provided in equation 4.19. The resulting fragility as a function of pressure, along with its slope, is plotted in figure 4.8.

$$m_p = (\log(\tau_\infty) - 14) * \left(\frac{\log(\tau_\infty) - 14}{D(P)} - 1\right) \tag{4.19}$$

It is found that fragility at 3 GPa, $m_p = 37 \pm 6$, is significantly less than previously determined in reference [86]. This is possibly due to the non-physical pseudo-Arrhenius assumption previously made to calculate the fragility from high pressure isothermal data near the glass transition. Most glassy system are approximately Arrhenius at high temperature and low pressure where the potential energy landscape is not so heavily influential. In this regimen, a single activation energy, or equivalently volume, can be assumed and the data aptly modeled as follows:

$$\tau_\alpha(P) = \tau_\infty \exp^{\frac{P\Delta V^\#}{RT}} \tag{4.20}$$

where τ_∞ is the infinite temperature relaxation time, $\Delta V^\#$ is the activation volume, and R is the universal gas constant. However, at higher pressures approaching the glass transition, this assumption of Arrhenius behavior becomes very non-physical as it fails to predict the asymptotic divergence of the structural relaxation time. Alternatively, one could fit the isothermal data to a modified VTF as follows:

$$\tau_\alpha(P) = \tau_\infty \exp^{\frac{DP}{P_0 - P}} \tag{4.21}$$

Relating equation 4.20 and 4.21 one can determine that, at the glass transition, the activation volume $\Delta V^\# = \frac{DRT}{P_0 - P_G}$. From this the fragility could be assessed in a more physically meaningful

manner. However this was not done as equation 4.21 does not aptly describe the pressure dependency of the structural relaxation, τ_α, over the entire pressure range as seen in figure 4.7 below and in reference [86].

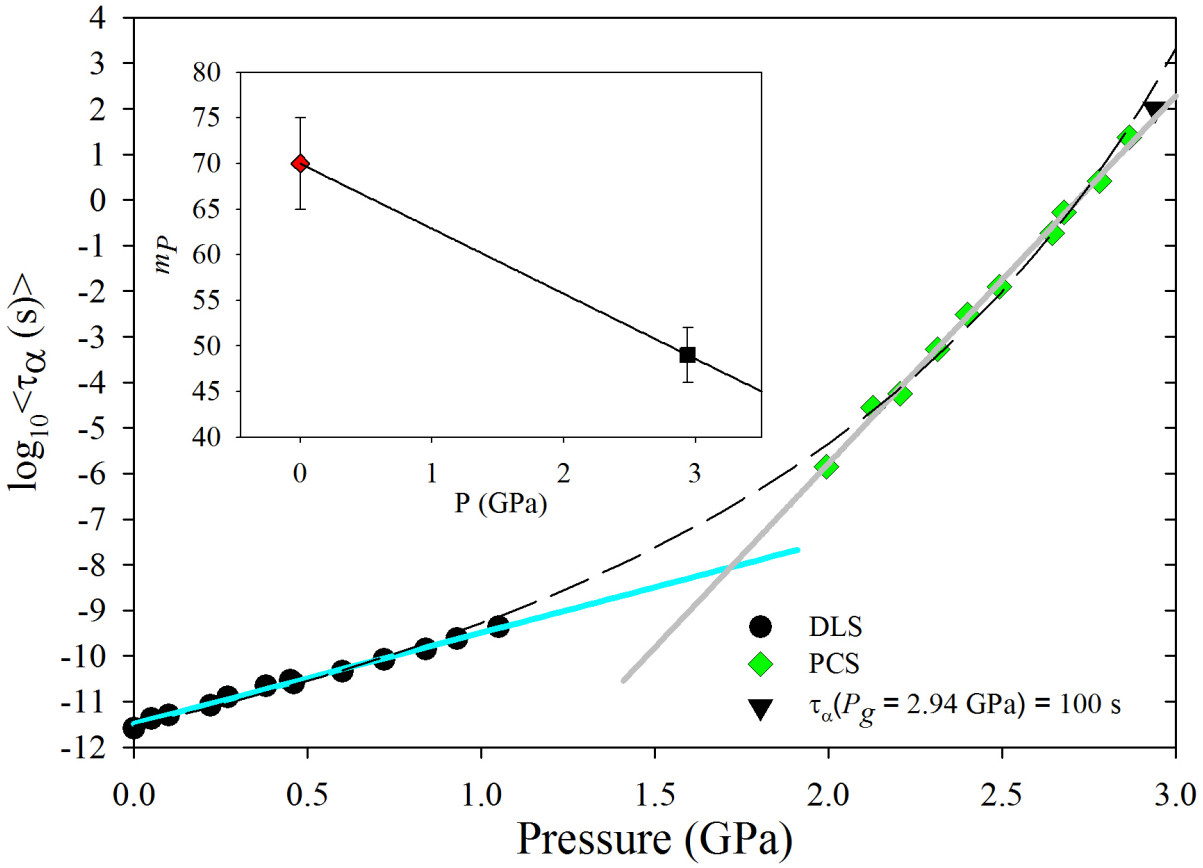


Figure 4.7: The pressure dependence of the α-relaxation time is shown above for the 75 °C isotherm from reference [86]. The data was obtained from three molecular probes, DLS (●), DPCS (♦), and TGP (▼), to cover the entire dynamic range from several pico-seconds to the glass transition. High pressure data was fit assuming pseudo-Arrhenius behavior to equation 4.20 (gray line) to determine the activation volume, $\Delta V^{\#}$ which, along with the isochronal data from TGP, allows the calculation of isobaric fragility as $m_p = \frac{\Delta V^{\#}}{ln(10)R}\left(\frac{\partial T_g}{\partial P}\right)^{-1}$ for this isotherm provided in the insert (■) along with the atmospheric pressure fragility (♦) obtained from references [108][64][6].

None the less, this method provided a means to obtain an estimation of fragility at approximately 3 GPa allowing for a reasonable initial assessment of the general trend of fragility in cumene as a function of pressure over this range even if the error associated with this measure-

ment is somewhat suspect. Lending further credence to these results is the fact that the slope of fragility with respect to pressure, albeit obtained from just these two data points, was comparable to other similar systems[15, 91]. However, in contrast, equation 4.19 is obtained from a classic temperature dependent VTF equation which fits the data much more aptly over this large dynamic range, does not require the non-physical pseudo-Arrhenius assumption near the glass transition, and provides significantly more detailed structure of the relationship of isobaric fragility as a function of pressure. Partial derivatives of equation 4.18 with respect to the fit parameters allow for the standard errors in these parameters to be propagated through to the uncertainty in the derived values of the isobaric fragility, m_p. The resulting isobaric fragility of cumene as a function of pressure, as well as the associated error window, is provided in figure 4.8 below. The fragility shows a smooth yet decidedly non-linear decreasing trend with increasing pressure. The uncertainty in fragility increases with pressure due to the lack of very high temperature fast dynamic data corresponding to the isotherms represented in the upper right corner of the surface in figure 4.6. However, at a maximum of $\Delta m_p \approx \pm 6$ comparable to the uncertainty in most fragility studies, these results constitute a non-trivial contribution with an unprecedented level of detail regarding the pressure dependence of fragility at high pressure. As seen in the inset plot, the rate at which fragility is decreasing increases, and then goes through an inflection point somewhere between 1 and 1.5 GPa at which point both the fragility and the slope of fragility continue to decrease as the model predicts an asymptotic approach to Arrhenius behavior at very high pressure beyond 3 GPa.

Though this is in agreement with the expectation that cumene should smoothly transition into a strong glass former at very high pressures, any prediction of where this transition to Arrhenius behavior occurs requires extrapolation of this model well beyond the available data. However the experimental cutoff of the DPCS and TGP data at approximately 350 K is somewhat arbitrary.

Though there is increasing uncertainty in pressure measurements as the ruby lines broaden at high temperatures, and cumene seems to exhibit increased absorption at high temperatures leading to damage, there is no reason to concluded that data could not be obtained at significantly higher temperatures. The VERY high temperatures needed for the fast dynamics might be obtained via molecular simulations. In principle, it may be possible to model cumene, or another similar glass forming system, up to the point where it becomes effectively Arrhenius. Future work may include dynamic simulation methods to fill in the very high temperature alpha relaxation data corresponding to DPCS and TGP data at higher pressures up to the Arrhenius regime.

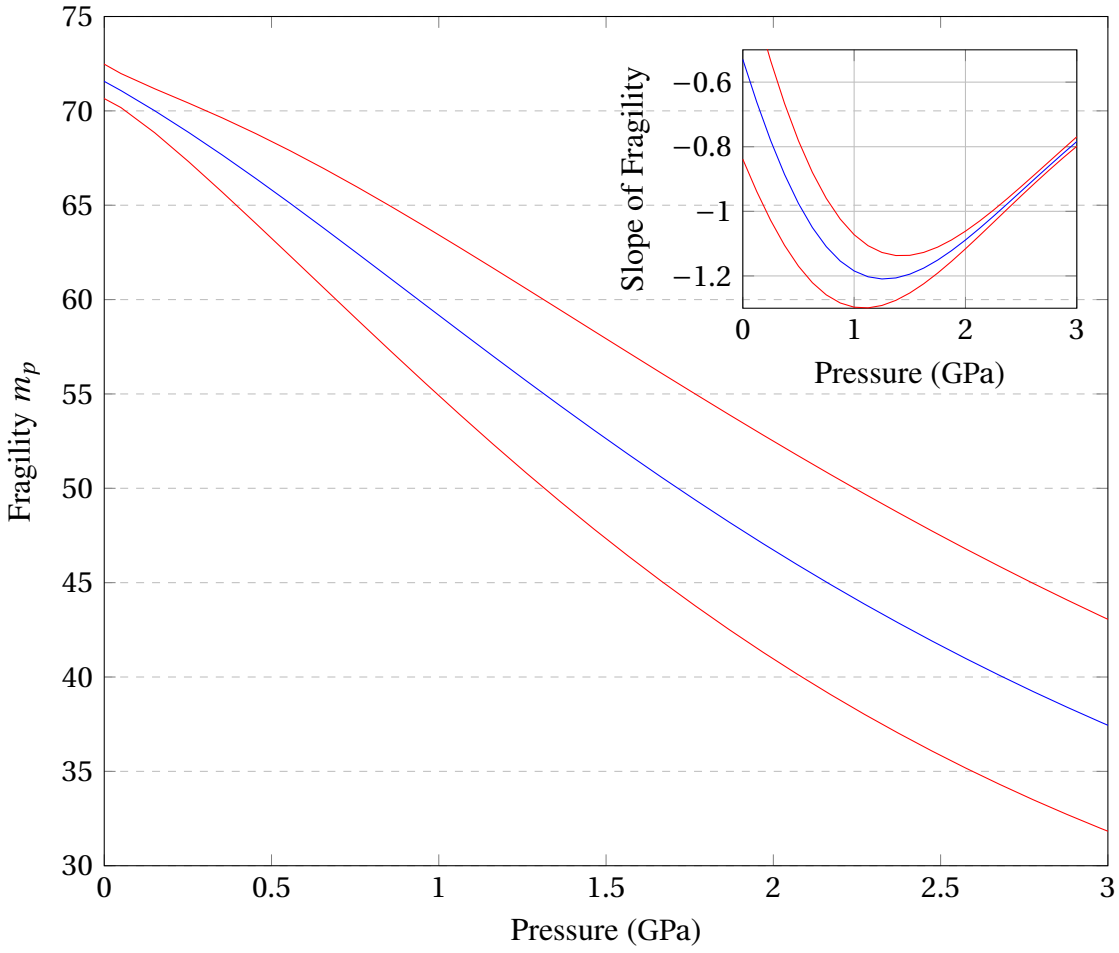


Figure 4.8: The Isobaric Fragility of Cumene, m_p, as calculated from equations 4.18 and 4.19 over a pressure range of 0 to 3 GPa corresponding to a glass transition set at $< \tau > \equiv 100$ s. The red curves represent the propagated uncertainty in the fragility calculation arising from the standard error σ reported for the surface fit parameters. The inset plot provides the slope of fragility, $\frac{\delta}{\delta P}\left(m_p\right)$, as a function of pressure.

5 Title of publication II or Result 2

5.1 Introduction

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5.2 Theoretical methods

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6 Instrumentation

6.1 Shutter Controller

It is often necessary or desirable to shutter out specific regions of the Fabry Perot spectrum in order to protect the PMT from excessive intensity or simply to mechanically remove features within a spectrum. This was previously accomplished via a LabVIEW program coupled with a National Instruments data acquisition card which counted the pulses sent by the JRS etalon controller to the multichannel analyzer and engaged a mechanical shutter at specific locations during each scan. A second dedicated computer was allocated for this task presumably in an attempt to address timing issues with the shutter, however the issues persisted. The first issue that was encountered with this solution was that the clock pulse from the JRS controller was pulled from TTL high down to around 2 volts due to the relatively low impedance of the clock input on the NI data acquisition card. The inability of the JRS controller to drive a decisive TTL high caused the LabVIEW program controlling the shutter to occasionally miss counts leading to shutter malfunctions. This ruined any data acquisition in progress as well as potentially left the PMT exposed to excessive counts. In an attempt to rectify this, six LM741 amplifiers were used to provide three isolated trigger and clock pulse outputs from the single JRS controller source. This solution required a power supply for the 741 amplification circuit and generally added to an already excessive level of complexity. Also, though this provided isolated trigger and clock sources for the multichannel scaler and other hardware, the issue with the NI data acquisition card clock's excessively low impedance remained along with the associated shutter malfunction, albeit with

diminished frequency, and a new issue arose as the 741 amplifier rise time is too slow to provide sharp TTL pulses for the multichannel card. It became apparent that a more thorough hardware solution was needed.

A PIC16F690 microcontroller based solution was designed to replace the dedicated computer, provide isolated trigger and clock pulses to all peripheral hardware, and provide a TTL output to signal the mechanical shutter. Additionally, monitoring of the PMT output was added so that the shutter could be closed in the event of excessive counts, providing more reliable protection. A LabVIEW program was still implemented to communicate with the controller via RS232, however only to set the parameters, such as the shutter locations and maximum PMT counts, which are stored on EEPROM-the computer is not required during normal operation or data acquisition. A breakdown of the architecture of this solution, useful schematics, as well as the code implemented on the micro-controller, will be provided here.

6.1.1 Hardware

The controller board is divided into four main functional blocks as seen in figure 6.1below–those being Power, PMT, CLK TRIG, and RS232. To alleviate the need for an external power supply, the board is powered by its own five volt linear regulated power supply. It should be noted that although the TTL inputs are isolated through a high impedance amplifier, they do share a common ground with the board power and as such this controller should always be plugged into a surge protector to protect the JRS controller and all other connected hardware in the event of excessive line voltage. AC input to the board is provided from a step-down transformer with a center tap to the connector labeled "AC" in the upper right corner of the POWER section of the board. Only the center and rightmost pin are used for 12V AC input–the leftmost pin is not

connected. The actual voltage on the AC input is irrelevant so long as the voltage provided to the LM7805 regulator is not in excess of its specifications and sufficient to account for its dropout voltage. A connection is provided for this power supply to drive a peripheral device if need be. It is clearly labeled +5V and GND.

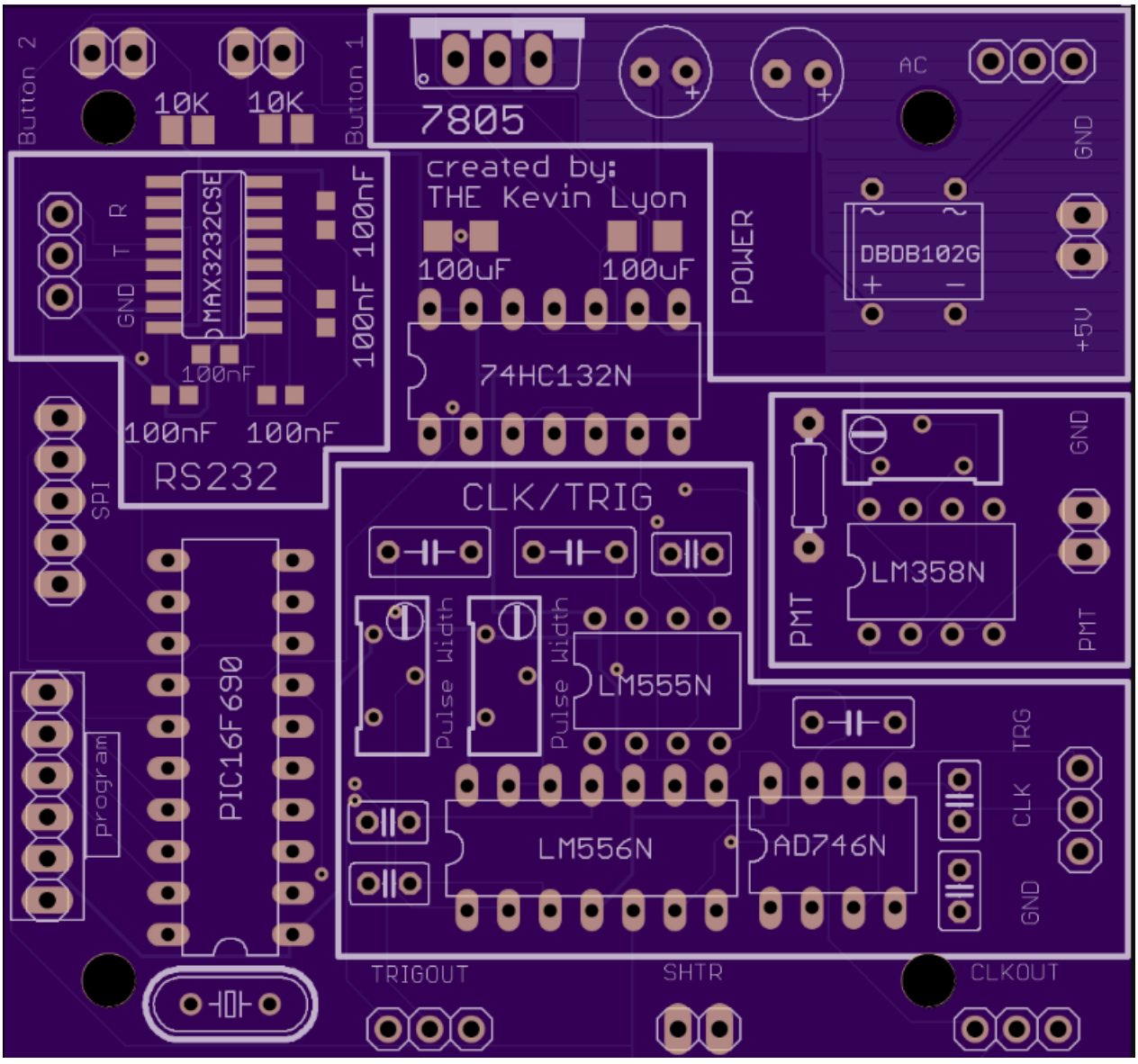


Figure 6.1: Control board provides TTL output for a shutter driver at preset scan locations, as well as two independently tunable clock pulses, and two repeated trigger outputs to signal the beginning of a new scan.

The TTL clock and trigger signals received from the JRS controller are repeated on two in-

dependent respective outputs. Both clock and trigger inputs are isolated via a unity gain AD746N fast rise time operational amplifier. This amplifer's 75V/μs slew rate and 500ns settling time maintains the fidelity of the input TTL pulses while requiring only a single 5V power supply making it a preferable choice to the 741 implemented previously. However, as this is a relatively expensive amplifier and the clock circuit does not technically require a sharp TTL signal, it likely could be replaced by the significantly less expensive LM358 used for the PMT monitor. The repeated trigger input from the AD746 isolating amplifier is connected directly to pin 19 of the micro-controller where any change in state is mirrored via an interrupt routine on pins 5 and 6 which are in turn connected to the TRIGOUT port at the bottom of the board providing two independent and isolated trigger outputs. The trigger line only changes state approximately once per second–at the beginning and end of every scan of the etalon plates. However, the clock signal changes state twice for every one of the 1064 channels of the multichannel analyzer during each scan. This would require an interrupt routine to change the state of two output pins approximately every half millisecond to provide for two isolated clock outputs. To free up the micro-controller, the mirrored clock outputs were instead provided via an analog circuit consisting of three 555 timers and a couple nand gates provided by the LM74132. As seen on the lower half of figure H.2, when the input clock swings high it raises the reset (pin 5) on the LM555 timer and momentarily lowers the trigger (pin 2). As this LM555 is configured in a bistable arrangement, the output (pin 3) swings to the negative rail (GND) which via a couple nand gates, results in the trigger pin being raised immediately back to Vcc. Thus the output of the LM555 provides a quick negative trigger pulse to the duel LM556 configured in a monostable arrangement which then, in turn, provides the two isolated clock outputs. The duration of these clock output pulses are independently tunable via the two potentiometers located on the middle right of the CLK/TRIG section of the board in figure 6.1.

RS232 communications with the board is facilitated via a standard MAX232 level compensator. Pins 12 and 10 of the micro-controller provide Rx and Tx respectively. A standard 9-pin D-Sub port has been provided for direct connection to a PC serial port. Though only one is currently being used, hardware denounce is provided for two buttons via the LM74132 nand gate with Schmitt triggering along with pin 15 and 14 (port C bits 1 and 2 respectively) on the micro-controller which may provide for future expanded functionality. An SPI bus is also provided, though not currently used, on the left side of the board. The SPI pin out is not clearly labeled on the silkscreen however one may test for ground continuity to find pin 1 on the provided connector. The pin out of the connector is as shown on the provided schematic (figure H.2). A program port is also provided and connected to a female 6-pin header to allow for updates to the firmware, the pinout for which matches that for the Microchip PICKIT3 programmer.

6.1.2 Software

A simple LabVIEW program has been written to communicate with the shutter controller however any serial terminal program will suffice. The serial specifications and command list are provided in table G.1. To facilitate RS232 communication, the 16F690 micro-controller's enhanced universal synchronous/asynchronous receiver transmitter (EUSART) module was implemented. This module has as single 8-bit received data register. This frees up the processor from the overhead associated with constantly checking the port and bit-banging out data, however it does require that each character of an incoming command sequence be individually handled by the processor as they are received. Each time a single character is received by the EUSART module an interrupt is triggered and the processor stores the received character into incrementing locations within a 10-byte buffer array. Commands are parsed and executed when a line feed (\n) character

is received or the buffer is overrun–whichever occurs first. Invalid commands invoke an "Unknown Command\n" response.

In order to further simplify communication with the shutter controller, the before mentioned Lab-View GUI interface was created. When the program is run, it initializes via reading all of the set parameters from the device EEPROM via sending five c#r, one for each channel #, and one pmt? command. **This takes approximately one second at 9600bps thus might interfere with proper shutter timing as well as significantly delay the closing of the shutter in the event of excessive pmt counts. It is therefor recommended that this Lab-view program not be started while the shutter is running.** However, pushing the button to close the shutter while the program is initialized will alleviate these concerns. Pushing the button again to return to shuttering mode, one can then adjust the location or width of an individual channel without adversely effecting the efficacy of the shutter. To change the start location or width of an individual channel from the Labview GUI, simply update the values for a particular channel and then click the associated green "Update #" button below it. To update the maximum pmt counts, adjust the slider at the far left which roughly corresponds to a fraction of the scale on the Mech-Tronics 777 rate-meter and then click the green Update PMT button below it.

Clicking the green AtoD button on the top left of the GUI will return the last acquired analog to digital conversion value. In theory, this is a value between 0 and 1024 resulting from a 10-bit analog to digital conversion corresponding to a voltage between 0 and 5 V. However, as previously mentioned in section 6.1.1, the LM358 amplifier will not swing all the way to the positive rail thus the max output voltage is limited to approximately 2.3 V.

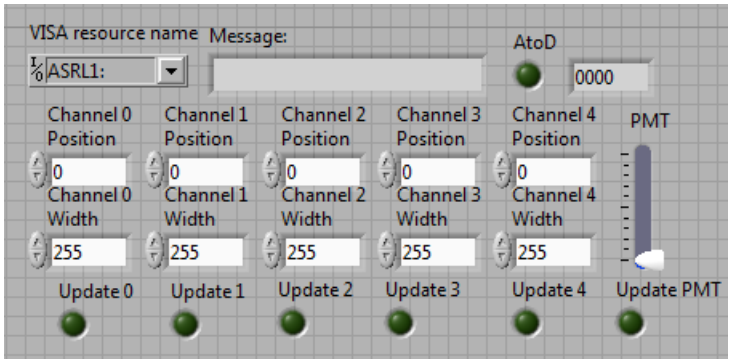


Figure 6.2: Lab-View Graphical User Interface for RS-232 communication to shutter controller.

6.2 TGP System

Two factors that are very important for obtaining reliable TGP data are maintaining consistent times between pressure measurements and assuring that the measurements are taken repeatably from the same locations within the DAC. The time given for thermalization between pressure measurements needs to correspond to the defined glass transition time of between 100 s and 1000 s if it is to serve as a measurement of the glass transition and it is desirable for all pressure measurements to be made as rapidly as possible. As pressure gradients may exist across single rubies, failing to take pressure measurements from repeatable locations in the DAC will potentially add a great deal of uncertainty in the data. Both of these concerns make this process a prime candidate for automation. Thus the following will describe the system developed to automate these measurements and alleviate these concerns.

6.2.1 Stage Controller

The stage controller requires two sources of power–a 15 to 25 volt, 1 amp D.C. supply to drive the stepper motors as well as 5V for the digital electronics. The digital supply may be

provided via the DC power jack (7–12 V) or simply via the USB. The advantage of implementing the DC power jack would be that the controller can remain on when the computer is shut down thus retaining ruby locations that are stored in RAM and are lost any time the controller looses power.

Once powered on, the LCD will displays the current motor location which will default to zero after a restart. To manually move the stage and/or save ruby locations, press the reset button on the controller box. This will pass control to the Nintendo control pad. Note that in this local control mode, all serial communications will be ignored. Pressing the D-pad will move the stage and the LCD will show the current x-axis and y-axis locations. Pressing the 'A' button will store the current location into one of 10 memory slots. The number of stored locations will appear in the upper right corner of the LCD display. When all ruby locations have been stored, pressing the reset button will return the controller to remote control mode. The controller will not respond to serial commands unless it is returned to remote mode. To clear all stored ruby locations, press and hold the reset button. The LCD display will indicate when the memory has been cleared.

Currently there is no means to edit individual ruby locations or manually cycle through them. This functionality could be trivially added at a future point if deemed necessary.

It is often desirable to change the speed of the translation stage. When selecting rubies in a DAC it may be easier to precisely position the laser with the stages moving at a slower speed. Alternatively, when characterizing rubies on a slide, one may need to move great distances thus a higher translation speed is desirable. Thus the motor speed is selectable over a fairly large range via a potentiometer shown to be wired in green into analog port 4 in figure 6.3. Adjusting this potentiometer will change the motor speed and briefly report the setting on the LCD screen.

To move to a specific saved ruby location one should first assure that the controller is in

remote control mode by pressing and releasing the reset button. At this point the controller will respond to several single character serial commands over the USB port. The serial specifications and recognized commands are provided in table 6.1.

Baud rate:	9600 bps
Data bits:	8
Stop bits:	1
Parity:	None
Flow control:	None
Termination:	None
Command Set:	
I	Returns list of recognized serial instructions and their function
#	Moves to ruby location for ruby # from 0 to 9
Z	Erases all stored ruby locations
S	Returns 0 to 9 corresponding to the last selected ruby
N	Returns 0 to 10 corresponding to the number of saved ruby locations

Table 6.1: RS232 serial specifications and command list

Hardware

The two axis stage controller needed to repeatably move between rubies during pressure measurements was designed and constructed by several students as part of an Upward Bound summer research project under my guidance. As such, an Arduino-Uno based solution was pursued to significantly reduce the learning curve associated with the electronics and embedded programming as several of the students involved in the project had previously attended my course on embedded programming implementing the Arduino-Uno. The result was a 2-D translation stage set up under a microscope that would store and save locations of rubies allowing the quick and repeatable repositioning of the stage for each pressure measurement.

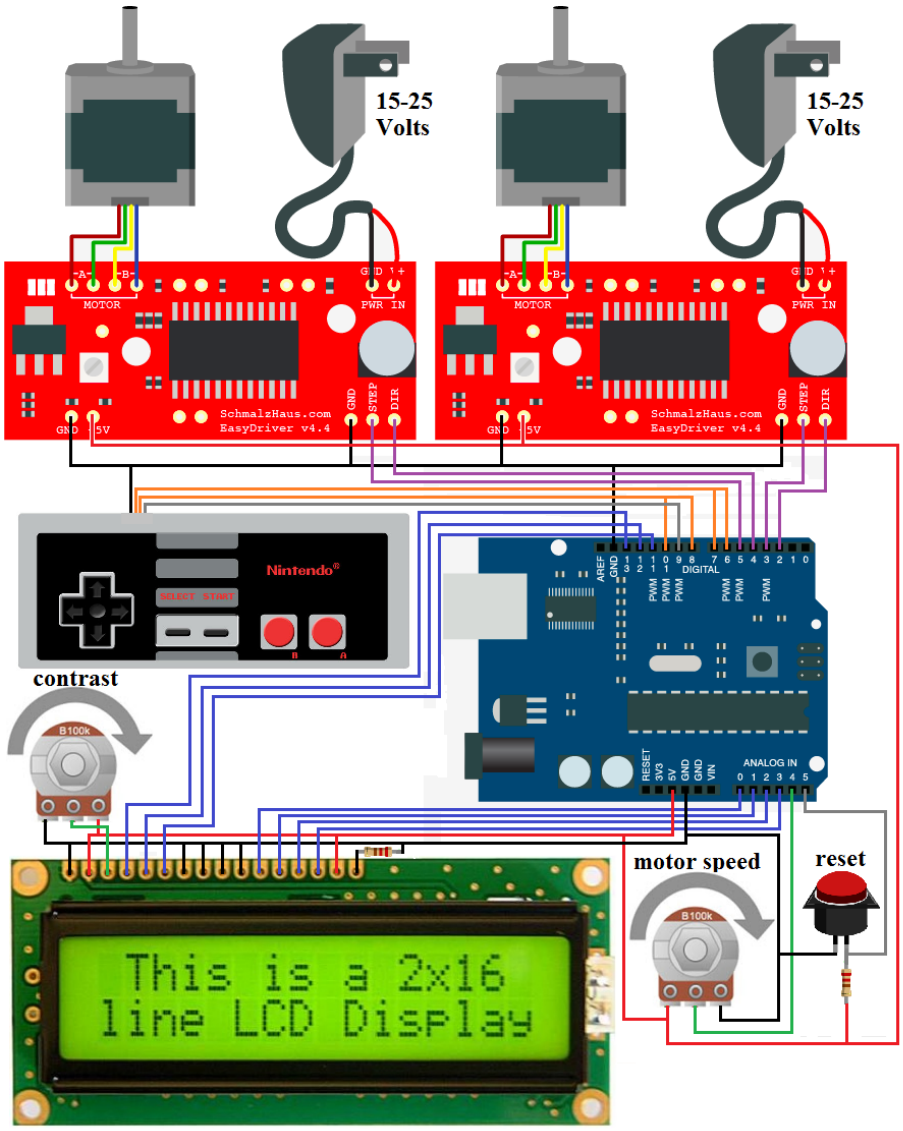


Figure 6.3: Wiring diagram for Arduino-Uno 2-D translation stage controller.

Two A3967 micro-stepping driver based breakout boards were used to physically drive the two bipolar stepper motors. These provide constant current control as well as electrical isolation between the digital control electronics and the high inductive load of the stepper motors. They are also the most probable points of failure and can be quickly replaced if repair is needed. The stepper motors themselves are connected via a single 9-pin D-sub connector which should always

be screwed firmly into place to avoid damage associated with inductive feedback that may occur if the motors are inadvertently unplugged while energized. Power is provided to the A3967 breakout boards via two banana plug ports color coded black for common and red for +15 to 25 V D.C. The direction and step control lines are shown in purple in figure 6.3 and are connected to Arduino digital pins 2, 3, 4, and 5. The state of pins 2 and 4 control the motor direction while pulses sent to pins 3 and 5 step the motor.

A Nintendo controller is used to set the ruby locations and store them into memory. The control lines, shown in orange in figure 6.3, are connected to digital pins 6, 7 and 8, 10 for the D-pad's x-axis and y-axis control respectively. The 'A' button is the only implemented button on the controller and is connected to digital pin 9–shown in gray. Though not shown in figure 6.3, all of these digital ports have pull-up resistors and drop low when activated.

A standard HD44780 compatible 2x16 character LCD display is provided for user interface. To save IO the display is run in 4-bit "nibble" mode thus the first four data bits are held low. All control and data lines implemented are shown in blue in figure 6.3. The contrast adjustment potentiometer does not need to be adjusted during normal usage thus is not accessible without opening the enclosure. An LCD back-lit screen was chosen so that it may be read while running in a a dark room thus the resistor on the LCD pin 16 limits the current to the led back-light.

Software

The embedded code for the stage driver was written in the Arduino EDU environment. This environment provides a limited C++ compiler which supports class structures. As there are two stepper motors to control, it was sensible to create a class structure to handle this. The resulting motor class object is defined in section I.2 followed by the main stage controller program. At

the beginning of the main program the Arduino-Uno pins associated with the D-pad, buttons, and motor speed control are defined so that they may be changed if necessary relatively easily. Immediately following, two motor class objects are defined as xmotor and ymotor as well as the LCD display object. It perhaps should be noted that if more IO-pins are needed in the future, the simplest means of accomplishing this may be to replace the LCD with an I^2C display as this would have minimal impact on the code–only requiring that the object be appropriately re-defined here as there exist an Arduino I^2C LCD library with compatible methods.

Following the object definitions, a 3 row by 10 column integer array is defined to store ruby locations. The 0th row stores a 0 if the corresponding ruby has not been set and is updated to a 1 when the values in the 1st and 2nd row correspond to stored x and y motor steps respectively. Beyond this the function needed by the main program loop are defined to save ruby locations into memory, manually move the stage to select ruby locations, write to the LCD, move to ruby locations, reset ruby locations, and adjust the the motor speed. The main program loop is at the very end of section I.2 and simply constantly polls for a received serial byte or if the reset button has been pressed. If it receives an ASCII character between ’0’ and ’9’ the controller will move to the saved position associated with that ruby if and only if that ruby has been stored. If that ruby has not been stored the received character will be ignored just as if it received an invalid character. When moving to a stored ruby location, appropriate backlash will be automatically accounted for via overshooting in one direction by the number of steps specified by the "motorSlack" variable.

This code was written with ease of modification in mind. Important parameters are defined at the beginning of the code and object oriented methods are employed wherever possible. The Arduino-Uno, with its bootloader program, also makes it very easy to update the code. However to allow for programming over the serial port, the atmega328 microprocessor has a bootstrap loader

in the upper memory. In order for this bootsrap which resides in high memory to facilitate serial programming the microprocessor must be reset. To accomplish this the Arduino-Uno has glue logic that drops the reset (pin-1) low whenever the USB to serial communication is initiated thus launching the bootstrap loader. Unfortunately this means that any time one attempts to initialize a serial link with the Arduino-Uno the microprocessor is reset which, for our application, implies that all stored ruby locations would be lost. To avoid this a trace connected to pin-1 of the atmega328 was cut on the PCB to inhibit this reset. Effectively this means one can not update the firmware on the stage controller via its USB port. One must pull the physical micro-controller from the Arduino-Uno and insert it into another for programming or resort to direct ISP programming.

6.2.2 Monochometer Controller

Hardware

The SPEX500M monochromator implemented in the TGP system was equipped with a stepper motor controlled main diffraction grating however the electronics to support the motor were no longer available. The project to build a new stepper driver for the monochromator fell to an unusually capable undergraduate student and her work is well described in her honors thesis.[99] She chose to use an Arduino-Uno to serve as an intermediary between a Labview control program and a stepper drive circuit of her own design. The hardware has only been slightly adapted from her original drive circuit. The LM741 operational amplifiers used to electrically isolate the Arduino--Uno from the motor's inductive load were replaced with LTV-826 opto-isolators which provide superior protection from inductive feedback as well as nullify the need for duel rail power supplies for the 741's. An input to accept discriminated TTL pulses was also added to facilitate PMT

photon counting via the Arduino's 16-bit timer. The final design was then laid out as an Arduino--Uno shield and the PCB professionally fabricated. The resulting PCB is shown in figure 6.4 and the associated schematic is provided in figure H.1. Four TIP29 NPN transistors are implemented

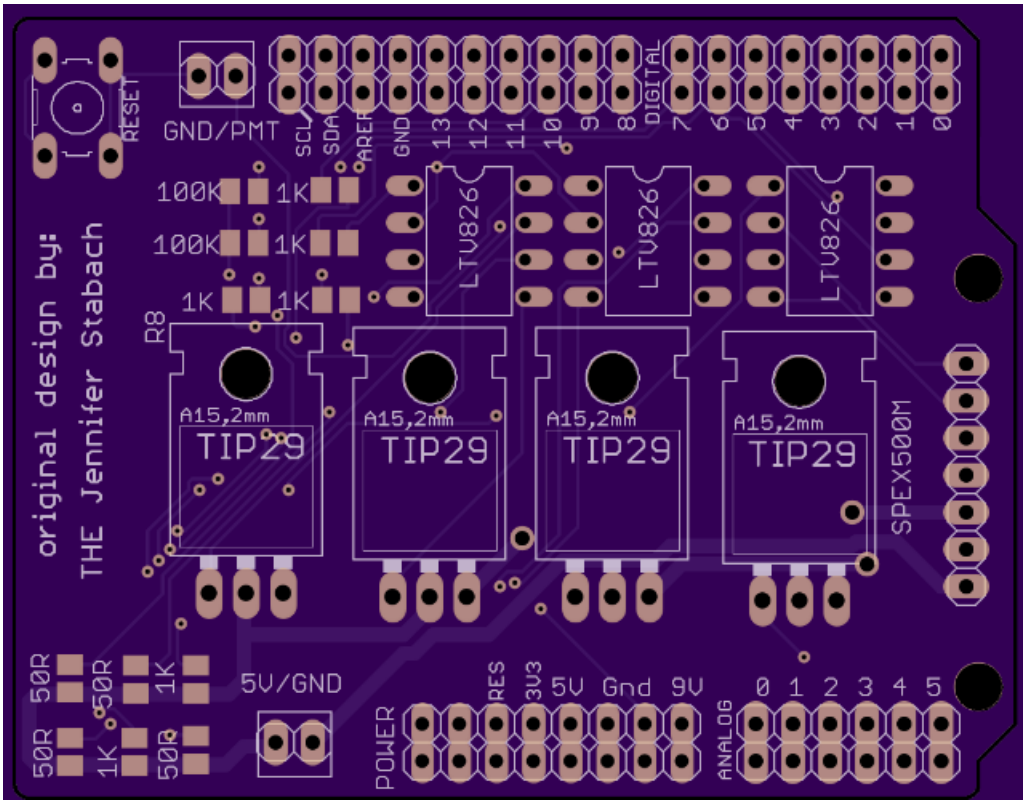


Figure 6.4: Spex500M Grating Stepper Motor Drive shield for an Arduion-Uno.

for high current switching to a unipolar wired motor. LTV-826 opto-isolators are used to isolate the high current circuit from the digital control electronics. The port on the right edge of the shield provides power to the motor as well as limit switch detection. The lower pin provides a common ground followed by three unipolar positive motor lines. The top two pins provides continuity to ground until the associated limit switch is activated. The limit switches are pulled up through on board 100K resistors connected to the high current motor power supply. Thus they are also electrically isolated from the digital electronics via a third LTV-826.

7 Conclusion

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Bibliography

[1] G. Adam and H. J. Gibbs. “On the temperature dependence of cooperative relaxation properties in glass-forming liquids”. In: *J. Chem. Phys.* 43 (1965), p. 139.

[2] S. Peter Andersson and Ove Andersson. “Relaxation Studies of Poly(propylene glycol) under High Pressure”. In: *Macromolecules* 31 (1998), p. 2999.

[3] C. A. Angell. “Formation of Glasses from Liquids and Biopolymers”. In: *Science* 267.5206 (1995), p. 1924.

[4] C. A. Angell. “Insights into Phases of Liquid Water from Study of its Unusual Glass-Forming Properties”. In: *Science* 319 (5863 2008), p. 582.

[5] M. F. Ashby and A. L. Greer. “Metallic glasses as structural materials”. In: *Scripta Materialia* 54 (3 2006), p. 321.

[6] A. J. Barlow, J. Lamb, and A. J. Matheson. “Viscous behaviour of supercooled liquids”. In: *Proc. R. Soc. A* 292.1430 (1966), p. 322.

[7] Z. Batschinski. In: *Phys. Chem.* 84 (1913), p. 644.

[8] Bruce J. Berne and Robert Pecora. *Dynamic Light Scattering*. Dover, Mineola NY, 2000.

[9] K. Binder and A. P. Young. “Spin glasses: Experimental facts, theoretical concepts, and open questions”. In: *Rev. Mod. Phys.* 58 (4 1986), p. 801.

[10] D. Bradley. “Glass Inequality”. In: *Materials Today* 21 (2018), p. 4.

[11] P.W. Bridgman. In: *Proc. Am. Acad. Arts. Sci.* 61 (1926), p. 57.

[12] F. Bueche. “Segmental mobility of polymers near their glass temperature”. In: *J. Chem. Phys.* 21.10 (1953), p. 1850.

[13] Manuel Cardona, Ralph V. Chamberlin, and Marx Werner. “Comment on the history of the stretched exponential function”. In: (2007).

[14] Chiara Caronna et al. “Dynamics of nanoparticles in a supercooled liquid”. In: *Phys. Rev. Lett.* 100 (2008), p. 55702.

[15] R. Casalini and C. Roland. “Scaling of the supercooled dynamics and its relation to the pressure dependences of the dynamic crossover and the fragility of glass formers”. In: *Phys. Rev. B* 71.1 (2005), p. 014210.

[16] Nian-Sheng Cheng. “Formula for the viscosity of a glycerol-water mixture”. In: *Ind. Eng. Chem. Res.* 47 (2008), p. 3285.

[17] I. Cibulka and T. Takagi. “P–t data of liquids: summarization and evaluation. 5. aromatic hydrocarbons”. In: *J. Chem. Eng. Data* 44 (1999), p. 411.

[18] Morrel H. Cohen and David Turnbull. “Molecular transport in liquids and glasses”. In: *J. Chem. Phys.* 31.5 (1959), p. 1164.

[19] Richard L. Cook, Chris A. Herbst, and H.E. King. “High-pressure viscosity of glass-forming liquids measured by the centrifugal force diamond anvil cell viscometer”. In: *J. Phys. Chem.* 97 (1993), p. 2355.

[20] Richard Lee Cook. “High pressure viscosity of simple and polymeric fluids”. PhD thesis. Harvard University, 1993.

[21] Richard L. Cook et al. “Pressure and temperature dependent viscosity of two glass forming liquids: Glycerol and dibutyl phthalate”. In: *J. Phys. Chem.* 100.7 (1994), p. 5178.

[22] D. Coslovich and C. M. Roland. “Thermodynamic Scaling of Diffusion in Supercooled Lennard-Jones Liquids”. In: *J. Phys. Chem. B* 112 (2007), p. 1329.

[23] D. W. Davidson and R. H. Cole. “Dielectric relaxation in glycerol, propylene glycol, and n-Propanol”. In: *J. Chem. Phys.* 19.12 (1951), p. 1484.

[24] Pablo G. Debenedetti and Gene Stanley. “Supercooled and Glassy Water”. In: *Physics Today* 56 (2003), p. 60.

[25] Pablo G. Debenedetti and Frank H. Stillinger. “Supercooled liquids and the glass transition”. In: *Nature* 410 (2001), p. 259.

[26] P. Debye. *Polar Molecules*. Dover, New York, 1945.

[27] C. Demoulin, C. J. Montrose, and N. Ostrowsky. “Structural relaxaton by digitalcorrelation spectroscopy”. In: *Phys. Rev. A* 9.4 (1974), p. 1740.

[28] R. Donald and Pardeep P. Fulay. *Essentials of Materials Science & Engineering*. Cengage Learning, 2008.

[29] Authur K. Doolittle. “Studies in Newtonian Flow. II. The dependence of the viscosity of liquids on free-space”. In: *J. App. Phys.* 22.12 (1951), p. 1471.

[30] Henry Eyring. “Viscosity, Plasticity, and Diffusion as Examples of Absolute Reaction Rates”. In: *J. Chem. Phys.* 4 (1936), p. 283.

[31] Hans Forsman, Per Anderson, and Gunnar Backström. “Dielectric relaxation of glycerol and n-propyl alcohol at high pressure”. In: *J. Chem. Soc.* 82 (1986), p. 857.

[32] J. Frenkel. *Kinetic Theory of Liquids*. Oxford University Press, 1946. Chap. 5.

[33] Raymond M. Fuoss and John G. Kirkwood. “Electrical Properties of Solids. VIII. Dipole Moments in Polyvinyl Chloride-Diphenyl Systems”. In: *J. Am. Chem. Soc.* 63 (1941), p. 385.

[34] R. H. Geils and R. C. Keezer. “Small-volume, inclined, falling-ball viscometer”. In: *Rev. Sci. Instrum.* 48.7 (1977), p. 783.

[35] J. H. Gibbs. “Nature of the Glass Transition in polymers”. In: *J. Chem. Phys.* 25 (1956), p. 185.

[36] J. H. Gibbs and Edmund A. DiMarzio. “Nature of the Glass Transition and the glassy state”. In: *J. Chem. Phys.* 28 (1958), p. 373.

[37] L.S. Grattan. *Optical Fiber Sensor Technology: Advanced Applications–Bragg Gratins and Distributed Sensors*. Springer Science & Business Media, 2013.

[38] A. Lindsay Greer. “Metallic Glasses”. In: *Science* 267.5206 (1995), p. 1947.

[39] The Elliot Group. *Ab initio study of vibrational and structural properties of polyol glasses*. 2013. URL: https://www.ch.cam.ac.uk/group/elliott/research/organic-glasses (visited on 02/19/2019).

[40] R.W. Hartel and A.K. Hartel. *Candy Bites: The Science of Sweets*. Springer New York, 2014. Chap. 10.

[41] Chunyuan He and Chunxiao Gao. “In situ electrical impedance spectroscopy under high pressure on diamond anvil cell”. In: *Appl. Phys. Lett.* 91 (2007), p. 92124.

[42] A. H.E.Brodin and E. A. Rossler. “Depolarized light scatting study of glycerol”. In: *Eur. Phys. J. B* 44 (2005), p. 3.

[43] D Heiman, D Hamilton, and R W. Hellwarth. “Brillouin scattering measurements on optical glasses”. In: *Phys. Rev. B* 19 (1979), p. 12.

[44] R. J. Hemley et al. “Pressure-induced amorphizaton of crystalline silica”. In: *Nature* 334 (1988), p. 52.

[45] C. A. Herbst et al. “Dynamic light scattering measurments of highpressure viscosity utilizing a diamond anvil cell”. In: *J. Appl. Phys.* 72 (1992), p. 838.

[46] Ian M. Hodge. “Adam-Gibbs formulation of enthalpy relaxation near the glass transition”. In: *J. Res. Natl. Inst. Stand. Technol.* 102.2 (1997), p. 195.

[47] Leon Van Hove. “Correlations in space and time and Born approximation scattering in systems of interacting particles”. In: *Phys. Rev.* 95.1 (1954), p. 249.

[48] R. M. Hubbard and G. G. Brown. In: *Ind. Eng. Chem.* 15 (1943), p. 212.

[49] John David Jackson. *Classical Electrodynamics*. Third. John Wiley & Sons, Inc., 1999.

[50] Gregg Jaeger. “The Ehrenfest classification of phase transitions: Introduction and evolution”. In: *Arch. Hist. Exact Sci.* 53 (1998), p. 51.

[51] Xin Jiang et al. “Investigation on germanium oxide-based glasses for infrared optical fibre development”. In: *Optical Materials* 31 (2009), p. 1701.

[52] Gyan P. Johari. “Introduction to the Glassy State in the Undergraduate Curriculum”. In: *J. Chem. Educ.* 51 (1974), p. 23.

[53] Gyan P. Johari and Martin Goldstein. “Viscous liquids and the glass transition. II. Secondary RElaxations in Glasses of Rigid Molecules”. In: *J. Chem. Phys.* 53 (1970), p. 2372.

[54] Gyan P. Johari and E. Whalley. “Dielectric properties of glycerol in the range 0.1-105 Hz, 218-357 K, 0-53 kb”. In: *Faraday Symp. Chem. Soc.* 6 (1972), p. 23.

[55] K. Jurkiewicz et al. “Modelling of glass-like carbon structure and its experimental verification by neutron and X-ray diffraction”. In: *J. Appl. Cryst.* 50 (2017), p. 36.

[56] Olga Kheifets et al. “Electric properties of AgPbAsSe3 at pressures up to 45Â GPa”. In: *High Pressure Research* 29.2 (2009), pp. 261–266.

[57] L. G. Khvostantsev, V. N. Slesarev, and V. V. Brazhkin. “Toroid type high-pressure device: history and prospects”. In: *High Press. Res.* 24 (2004), p. 371.

[58] L. I. Komarov and I. Z. Fisher. “Theory of Rayleigh scattering of light in liquids”. In: *JETP* 16.5 (1962), p. 1358.

[59] Pradeep Kumar. personal communication.

[60] Z. W. Kyaw and N. Menon. “Glass transition of glycerol in the volume-temperature plane”. In: *Phys. Rev. E* 73 (2006), p. 040501.

[61] L.D. Landau, E.M. Lifshitz, and L.P. Pitaevskii. *Electrodynamics of Continuous Media*. Second. Vol. 8. Elsevier Butterworth-Heinemann, Burlington MA, 2004.

[62] W. T. Laughlin. PhD thesis. Massachusetts Institute of Technology, 1969.

[63] C. P. Lindsey and G. D. Patterson. “Detailed comparison of the Williams-Watts and Cole-Davidson functions”. In: *J. Chem. Phys.* 73 (1980), p. 7.

[64] A.C. Ling and J.E. Willard. “Viscosities of some organic glasses used as trapping matrixes”. In: *J. Phys. Chem.* 72.6 (1968), p. 1918.

[65] Douglas Main. *Humankind’s most important material*. 2018. URL: https://www.theatlantic.com/technology/archive/2018/04/humankinds-most-important-material/557315/.

[66] Century Manufacturing. *The Lucite Story*. 2019. URL: https://centurymfg.com/the-lucite-story/ (visited on 02/19/2019).

[67] S. Martelli, N. Giovambattista Torquato, and R. Car. “Large-Scale Structure and Hyperuniformity of Amorphous Ices”. In: *Phys. Rev. Let.* 119 (2017), p. 136002.

[68] A. J. Matheson. “Role of free volume in the pressure dependence of the viscosity of liquids”. In: *Am. Inst. Phys.* 44 (1966), p. 695.

[69] Shiro Matsuoka. “Free Volume, Entropy, and Relaxation Phenomena”. In: *Journal of Rheology* 30.4 (1986), pp. 869–876.

[70] G. E. McDuffie and M. V. Kelly. “Effect of Pressure on the Viscosity and Dielectric Relaxation Time in Glycerol”. In: *J. Chem. Phys.* 41.9 (1964), p. 2666.

[71] O. Mishima, L. D. Calvert, and E. Whalley. “’Melting ice’ I at 77 K and 10 kbar: a new method of making amorphous solids”. In: *Nature* 310 (1984), p. 393.

[72] Raymond D. Mountain. “Spectral Distribution of Scattered Light in a Simple Fluid”. In: *Journal of Research of the National Bureau of Standards* 3 (1966), p. 207.

[73] Raymond D. Mountain. “Spectral Distribution of Scattered Light in a Simple Fluid”. In: *Rev. Mod. Phys.* 38.1 (1966), p. 205.

[74] J. Mydosh. *Spin Glasses*. CRC Press, 1993.

[75] K. L. Nagai, P. Lunkenheimer, and C. Leon. In: *J. Chem. Phys.* 116 (2002), p. 9839.

[76] Design News. *Use of Metallic Glass Simplifies 3D Printing of Metals*. 2018. URL: https://www.designnews.com/materials-assembly/use-metallic-glass-simplifies-3d-printing-metals/2046166359487 (visited on 02/19/2019).

[77] Michael I. Ojovan and William E. Lee. "Glassy Wasteforms for Nuclear Waste Immobilization". In: *Metallurgical and Materials Transactions A* 42.4 (2011), p. 837.

[78] M. J. O'Neill. "The Analysis of a Temperature-Controlled Scanning Calorimeter". In: *Anal. Chem.* 36.7 (1964), p. 1238.

[79] M. Paluch et al. "Effect of pressure on the α relaxation in glycerol and xylitol". In: *J. Chem. Phys.* 116.22 (2002), p. 9839.

[80] Adam Patkowski, Jacek Gapinski, and Gerhard Meier. "Dynamics of supercooled van der Waals liquid under pressure. A dynamic light scattering study". In: *Colloid & Polymer Science* 282 (Jan. 2004), p. 874.

[81] G. D. Patterson and D. J. Ramsay. "Depolarized light-scattering spectroscopy and polymer characterization". In: *Anal. Chim. Acta.* 189 (1986), p. 57.

[82] M. Pawlus, J. Paluch, and J. Ziolo. "On the pressure dependence of the fragility of glycerol". In: *J. Phys.: Condens. Matter* 21 (2009), p. 332101.

[83] John Plodinec. "Borosilicate Glasses for Nuclear Waste Immobilization". In: *Glass Technology - European Journal of Glass Science and Technology Part A* 41 (Dec. 2000), pp. 186–192.

[84] A. A. Pronin et al. “Pressure-Inuduced change in the relaxation dynamics of glycerol”. In: *JETP Letters* 92.7 (2010), p. 479.

[85] T. C. Ransom and W. F. Oliver. “Glass Transition Temperature and Density Scaling in Cumene at Very High Pressure”. In: *Phys. Rev. Lett.* 119 (2 2017), p. 025702.

[86] Tim Ransom. “Viscous Liquids and the Glass Transition at Extremely High Pressure: Optical Techniques Applied to Cumene in a Diamond Anvil Cell”. PhD thesis. University of Arkansas, 2016.

[87] Andreas Reiser, Gernot Kasper, and Siegfried Hunklinger. “Pressure-induced isothermal glass transition of small organic molecules”. In: *Phys. Rev. B* 72 (2005), p. 94204.

[88] S. A. Rice. In: *Phys. Rev.* 112 (1958), p. 804.

[89] C. M. Roland. *Dielectric Relaxation Spectroscopy*. 2014. URL: `http://polymerphysics.net/pdf/ImpedanceSpectroscopy%20chapter.pdf` (visited on 07/17/2017).

[90] C. M. Roland, S. Blair, and R. Casalini. “Thermodynamic scaling of the viscosity of van der Waals, H-bonded, and ionic liquids”. In: *J. Chem. Phys.* 125 (2006), p. 124508.

[91] C. M. Roland et al. “Supercooled dynamics of glass-forming liquids and polymers under hydrostatic pressure”. In: *Rep. Prog. Phys.* 68.6 (2005), p. 1405.

[92] David Sherrington and Scott Kirkpatrick. “Solvable Model of a Spin-Glass”. In: *Phys. Rev. Lett.* 35 (26 1975), p. 1792.

[93] Hiroshi Shintani and Tanaka Hajime. “Frustration on the way to crystallization in glass”. In: *Nature Physics* 2 (2006), p. 200.

[94] W. Shockley. *Imperfections in nearly perfect crystals (see chapter by C. Zener)*. John Wiley & Sons, Inc., New York, 1952.

[95] David Sinclair. "Light scattering by spherical particles". In: *J. Opt. Soc. Am.* 37 (1947), p. 475.

[96] R. N. Sinclair. "Inelastic neutron scattering from amorphous solids". In: *J. Non-Crystalline Solids* 76 (1 1985), p. 61.

[97] Anders Sodergard and Mikael Stolt. "Properties of lactic acid based polymers and their correlation with composition". In: *Progress in Polymer Science* 27.6 (2002), p. 1123.

[98] C. Soundrapandian et al. "Porous bioactive glass scaffolds for local drug delivery in osteomyelitis: development and in vitro characterization." In: *AAPS PharmSciTech* 11 (2010), p. 1675.

[99] THE Jennifer Stabach. "Retrofitting a Monochromator for High-Pressure Raman and Optical Studies of Glass-Forming Liquids". B.S. Thesis. University of Arkansas, 2013.

[100] Jonathan F. Stebbins, Paul F. McMilan, and Donald B. Dingwell. *Structure, Dynamics and Properties of Silicate Melts*. Walter de Gruyter GmbH & Co KG, 2018, p. 15.

[101] J. L. Sutterby. In: *Trans. Soc. Rheol.* 17 (1973), pp. 559, 575.

[102] Grzegorg Szklarz, Karolina Adrjanowicz, and Marian Paluch. "Cooling-rate versus compression-rate dependence of the crystallization in the glass-forming liquid, propylene carbonate". In: *Cryst. Growth Des.* 18 (2018), p. 2538.

[103] G. Tammann and W. Hesse. In: *Z. anorg. allgem. Chem.* 156 (1926), p. 245.

[104] Albert Tolle. “Neutron scattering studies of the model glass former ortho-terphenyl”. In: *Rep. Prog. Phys.* 64 (2001), p. 1473.

[105] P.L. Travers. *Mary Poppins*. New York: Harcourt, Brace & Co., 1934.

[106] G. H. Vineyard. “Liquid Metals and Solidification”. In: *Am. Soc. Metals, Cleveland, Ohio* (1958), p. 1.

[107] Kauzmann Walter. “The nature of the glassy state and the behavior of liquids at low temperatures”. In: *Chemical Reviews* 43.2 (1948), p. 219.

[108] Li-Min Wang, C. Austen Angell, and Ranko Richert. “Fragility and thermodynamics in nonpolymeric glass-forming liquids”. In: *J. Chem. Phys.* 125.7 (2006), p. 74505.

[109] Ronald P. White and Jane E. G. Lipson. “Polymer free volume and its connection to the glass transition”. In: *Macromolecules* 49.11 (2016), p. 3987.

[110] G. Williams and David C. Watts. “Non-symmetrical dielectric relaxation behavior arising from a simple empirical decay function”. In: *Trans. Farad. Soc.* 66 (1970), p. 80.

[111] Malcom L. Williams, Robert F. Landel, and John D. Ferry. “The temperature dependence of relaxation mechanisms in amorphous polymers and other glass-forming liquids”. In: *J. Am. Chem. Soc.* 77 (1955), p. 3701.

[112] Peter G. Wolynes and Vassiliy Lubchenko. *Structural glasses and supercooled liquids*. John Wiley & Sons, Inc., Hoboken New Jersey, 1991.

[113] S. Yasutomi, S. Bair, and W. O. Winer. “An application of a free volume model to lubricant rheology I–Dependence of viscosity on temperature and pressure”. In: *J. Tribology* 106 (1984), p. 291.

[114] I. A. Zaliznyak and S. H. Lee. *Magnetic neutron scattering. Modern techniques for characterizing magnetic materials*. Springer, Heidelberg, 2005.

[115] Edgar Dutra Zanotto. “Do cathedral glasses flow”. In: *Am. J. Phys.* 66 (1998), p. 5.

[116] J. Zarzycki. *Glasses and the vitreous state*. Cambridge University Press, Cambridge New York, 1991.

A Andersson-Andersson Constrained Surface

Equation 4.18 in Sec. 4.3.2 extends the classic isobaric VTF model to account for variation in pressure simply by allowing the constants $\ln(\tau_\infty)$, D, and T_0 to be polynomial functions of pressure. In light of our TGP process providing an extremely well defined isochronal curve corrosponding to the glass transition temperature, T_g, via the Andersson-Andersson equation, it makes sense to constrain this surface model to this isochronal curve[2]. The standard isobaric VTF equation may be solved for the zero mobility temperature, T_0, in terms of the glass transition temperature, T_g, as follows:

$$\begin{aligned} \ln(\tau_g) &= \ln(\tau_\infty) + \frac{DT_0}{T - T_0} \\ \Rightarrow T_0 &= \frac{\left(\ln(\tau_\infty) - \ln(\tau_g)\right) T_g}{\ln(\tau_\infty) - \ln(\tau_g) - D} \end{aligned} \tag{A.1}$$

Substituting this back into the original VTF model for an arbitrary temperature, T, yields the following model in which all parameters are again functions of pressure and the glass transition temperature T_g is provided by the Andersson-Andersson equation from appendix B. This model was fit to both alpha relaxation data, τ, all measured in picoseconds and viscosity data, η, all measured in centipoise extracted from the work of Cook, McDuffie, Paluch, Pronin, Reiser, Broden, and Johari[21, 70, 79, 84, 87, 42, 52].

$$\ln(\tau) = \ln(\tau_\infty) + \frac{\left(\ln(\tau_\infty) - \ln(\tau_g)\right) D T_g}{\left(\ln(\tau_\infty) - \ln(\tau_g) - D\right) T - \left(\ln(\tau_\infty) - \ln(\tau_g)\right) T_g}$$

$$FOR:$$

$$D = \sum_{j=0}^{3} d_j P^j \quad AND \quad \ln(\tau_\infty) = \sum_{j=0}^{1} t_j P^j$$

$$WHERE \quad \ln(\tau_g) \equiv 32.489 \quad AND \quad VECTORS: \tag{A.2}$$

$$\vec{d} = \begin{pmatrix} 19.0557 \\ -0.690394 \\ 0.0115522 \\ -0.0000719137 \end{pmatrix} \quad \vec{t} = \begin{pmatrix} -6.73079 \\ 0.258835 \end{pmatrix}$$

To further constrain the surface, the values of d_0 and t_0 were not included in the Levenberg-Marquardt surface fit routine, but rather obtained from a standard VTF fit to ambient pressure isobaric data from Brodin, Pronin, and Nagai shown below[84, 42, 75]. This atmospheric isobar was also used to determine the characteristic time/viscosity for the glycerol TGP isochronal curve via assuring that both curves cross on the surface at zero pressure.

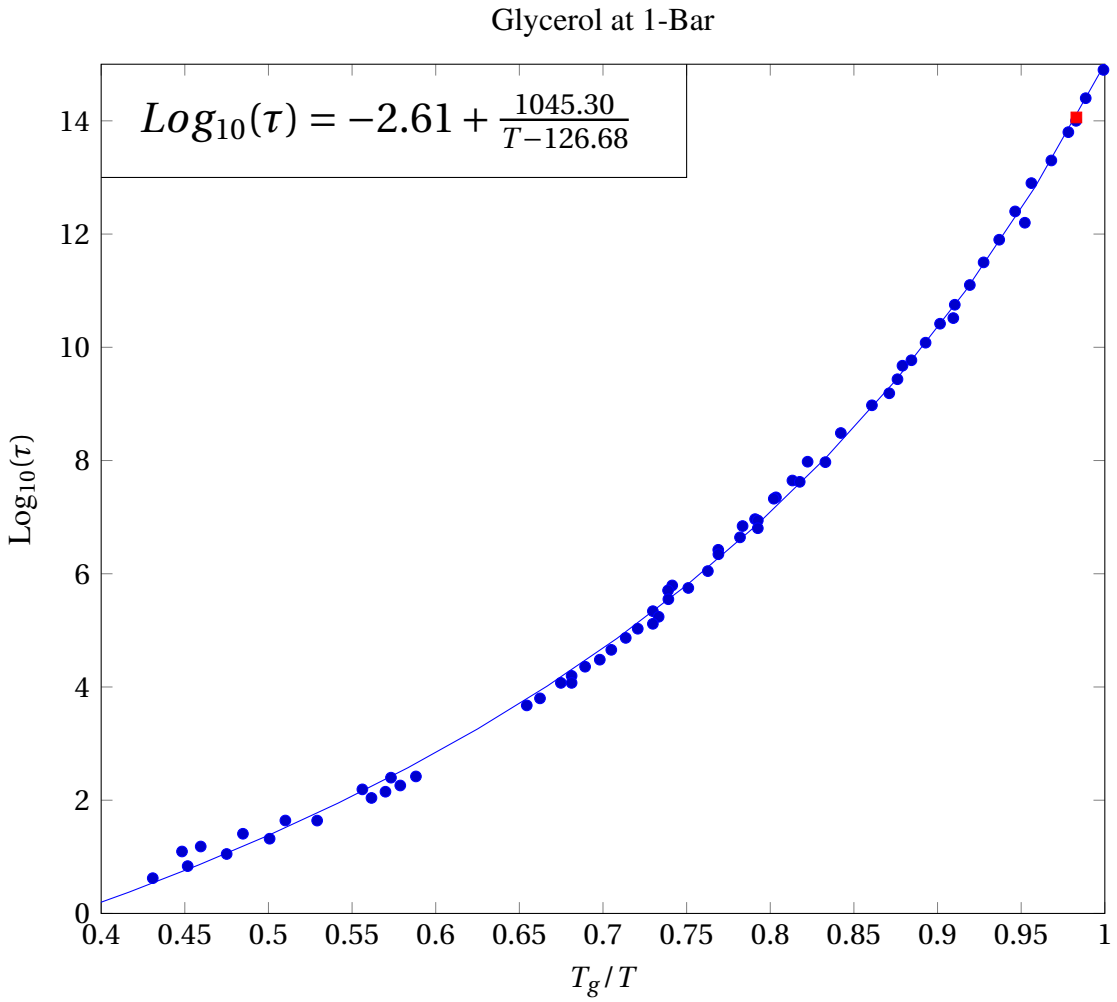


Figure A.1: Shown is the classic Angel Plot for glycerol at atmospheric pressure resulting in a glass transition temperature of T_g = 185.14 K defined at $\tau_g \equiv 100$ s. From the associated VTF model fit we may define the zero order coefficients of the Andersson-Andersson surface model as $d_0 = 19.06$ and $t_0 = -6.73$ leaving the nonlinear surface fit with only four remaining degrees of freedom. The average relaxation time associated with the isochrone obtained from the glycerol TGP experiments at atmospheric pressure is also shown above (■) corresponding to an α-relaxation time or viscosity of $\approx 10^{14}$ seconds or centipoise respectively. This TGP isochrone was latter determined to greater precision via the linearizion of the TGP data as reported in figure 3.5. This more precise value of $\tau = 220$ s was implemented when providing constraint to the Andersson-Andersson surface model.

B TGP Data (Glycerol)

Table B.1: Glycerol TGP Data

Pressure GPa	Temperature K		Volume $cm^3\,g^{-1}$
0	187		1.154
0.460	203		1.095
0.770	212		1.056
1.070	224		1.027
1.110	225		1.023
1.660	239	*	0.982
1.910	244		0.968
2.300	253		0.951
2.500	261		
2.790	263		
2.830	265		
2.910	265		
3.270	276		
3.650	285	**	0.915
3.800	287		0.912
4.220	297		0.904
4.240	297		0.904
4.680	308		0.898
4.890	310		0.894
4.950	315		0.895
5.170	320		0.891
5.470	326		0.887
5.800	328		0.881
6.380	338		0.873
6.550	341		0.871
6.680	344		0.869

* Specific volumes corresponding to this pressure and above were obtained with moderate extrapolation of the Tait EOS beyond Bridgman's data

$$T_G(P) = 188.376\left(1 + 2.046\frac{P}{5.596}\right)^{\frac{1}{2.046}} \tag{B.1}$$

C TGP Data (Cumene)

Table C.1: Cumene TGP Data

Pressure GPa	Temperature K	Volume $cm^3\,g^{-1}$
0.198	152	0.971
0.367	167	0.948
0.411	172	0.943
0.620	189	0.921
0.985	220	0.893
1.215	236	0.878
1.645	267	0.857
1.860	280	0.847
2.050	296	0.841
2.170	300	0.836
2.420	315	0.827
2.510	322	0.825
2.720	339	0.820
2.830	344	0.816
3.090	359	0.809
3.130	362	0.808
3.410	375	0.801
3.570	379	0.797
3.610	384	0.796
3.820	392	0.791
3.840	395	0.791
3.900	399	0.790
4.120	411	0.785
4.320	423	0.781
4.390	430	0.780
4.550	435	0.776

$$T_G(P) = 131.091\left(1 + 1.642\frac{P}{1.2177}\right)^{\frac{1}{1.642}} \tag{C.1}$$

D Equation of State (cumene)

All conversions from pressure to density or specific volume for cumene in this work were obtained via the modified Tait equation of state developed by Tim Ransom as shown in equation D.1 below [86]. The density as a function of temperature at atmospheric pressure, $\rho_0(T)$, was provided by Cibulka [17].

$$\rho(T,P) = \frac{\rho_0(T)}{1 - C(T)\,ln\left(\frac{B(T)+P}{B(T)+0.0001GPa}\right)}$$

$$FOR \quad VECTORS:$$

$$B(T) = \sum_{j=0}^{4} b_j \left[\frac{T - T_0}{100}\right]^j \quad AND \quad C(T) = \sum_{j=0}^{1} c_j \left[\frac{T - T_0}{100}\right]^j$$

$$WHERE \quad T_0 \equiv 298.15\ \mathrm{K} \quad AND: \tag{D.1}$$

$$\vec{b} = \begin{pmatrix} 0.11110\ \mathrm{GPa} \\ -0.080954\ \mathrm{GPa\,K^{-1}} \\ 0.0226\ \mathrm{GPa\,K^{-2}} \\ -0.0034\ \mathrm{GPa\,K^{-3}} \\ 0.00028\ \mathrm{GPa\,K^{-4}} \end{pmatrix} \quad \vec{c} = \begin{pmatrix} 0.093736 \\ -0.0040032\ \mathrm{K^{-1}} \end{pmatrix}$$

E Equation of State (glycerol)

To facilitate the conversion of glycerol pressure data (GPa) to specific volume (Å^{-3}) the Tait equation of state, fit to Bridgman's glycerol data, was implemented [11]. It should be noted that Bridgman only obtained data up to just under 1.2 GPa yet Cook *et al*. extrapolates this equation of state up to 3 GPa [21]. This work extrapolates this equation of state up to 6.68 GPa to allow for the thermodynamic scaling assessment of Sec. 3.5. Clearly, experimental measurement of the glycerol equation of state at these elevated pressures is desperately needed yet remains absent within the literature. However, despite dependence upon these extreme extrapolations, thermodynamic scaling of glycerol data implementing this equation of state yields self-consistent results as was demonstrated in Fig. 3.5. The isothermal Tait equation of state is given as follows:

$$\frac{V}{V_0} = 1 - c \times Ln\left(\frac{b+P}{b+P_0}\right), \tag{E.1}$$

where P_0 and V_0 are simply arbitrarily chosen reference values of pressure and volume respectively at a specific temperature. For this work P_0 was set to atmospheric pressure and V_0 was found to be sufficiently well modeled as a linear function of temperature as follows:

$$V_0(T) = 0.065T + 119.7, \tag{E.2}$$

where temperature, T, is expected in (°C) and the resulting specific volume, V_0, is returned in units of ($\text{Å}^3/_{molocule}$). The density data for glycerol to which Eq. E.2 was fit was taken from Cheng [16].

The remaining fit parameters c and b may be trivially related to the bulk modulus K_0 and its pressure derivative $K_0^{'}$ via Eq. E.3.

$$K_0 = \frac{P_0}{c} + \frac{b}{c} \quad and \quad K_0^{'} = \frac{1}{c} - 1 - Log\left(\frac{b}{b+P_0}\right) \tag{E.3}$$

The reported values of the bulk modulus, K_0, and its pressure derivative, $K_0^{'}$, calculated from five separate isothermal Tait equations of state by Cook *et al*. was fit to a third order polynomial model shown in Fig. E. This allows for the accurate interpolation of K_0 and $K_0^{'}$ at any arbitrary temperature within this range.

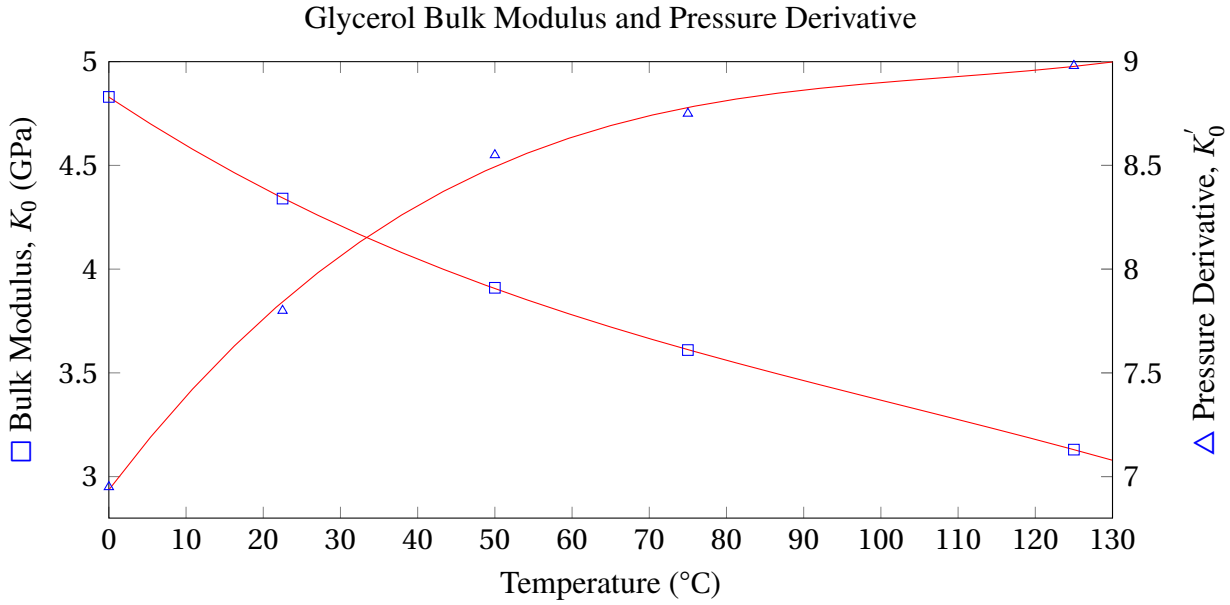


Figure E.1: The glycerol bulk modulus, K_0, and its pressure derivative, $K_0^{'}$, as extracted from reference [21] are provided above. Each are fit to a third order polynomial to allow for interpolation to arbitrary temperatures within the range of 0 °C to 130 °C.

From the interpolated values of the glycerol bulk modulus and its first pressure derivative, the c and b constants of the Tait model provided in Eq. E.1 can be obtained via the simultaneous solution of Eq. E.3.

F Cumene Data

Table F.1: Cumene α-relaxation data

Pressure GPa	Temperature K	Volume $cm^3\,g^{-1}$	$Log_{10}(\tau_{cd})$ psec
1.81	302.41	0.855	9.501
1.80	304.34	0.856	9.266
1.85	306.16	0.854	9.025
1.81	307.98	0.856	8.859
1.81	309.87	0.857	8.649
1.82	311.65	0.857	8.439
1.86	294.55	0.850	11.754
1.86	296.95	0.851	11.287
1.87	298.80	0.851	10.972
1.83	300.65	0.853	10.706
1.89	302.50	0.851	10.414
1.88	304.35	0.852	10.168
1.88	306.15	0.852	9.953
1.86	308.05	0.854	9.704
1.89	309.77	0.852	9.527
1.90	311.55	0.852	9.313
1.90	313.35	0.852	9.078
1.93	315.15	0.851	8.827
1.88	317.05	0.855	8.669
1.92	319.05	0.853	8.421
1.87	320.59	0.856	8.218
1.88	322.33	0.856	8.015
1.88	324.15	0.856	7.839
1.90	326.05	0.855	7.674
1.88	329.73	0.857	7.404
2.01	294.57	0.843	13.181
2.00	298.81	0.844	12.637
2.04	300.70	0.843	12.281
2.03	302.51	0.843	12.006
2.02	304.45	0.844	11.802
2.04	306.15	0.844	11.498
2.04	309.73	0.844	10.988
2.04	311.61	0.845	10.612
2.05	313.41	0.845	10.365
2.06	315.22	0.844	10.115
2.07	317.01	0.845	9.887
2.04	318.77	0.846	9.688
2.04	320.73	0.847	9.423
2.09	322.26	0.845	9.241
2.05	324.15	0.847	9.078
2.05	327.81	0.848	8.627
2.06	329.70	0.848	8.435

G Serial Specifications

Baud rate:	9600 bps
Data bits:	8
Stop bits:	1
Parity:	None
Flow control:	None
Termination:	Line feed (\n)
Command Set:	

atd	Returns last analog to digital conversion from 777 rate-meter as number between 0 and 1024 (technically between 0 and 615 due to limitations of the LM358N)
max	Returns atod value corresponding to max output of the LM358N amplifier as it is not capable of swinging to the positive rail. This value is set in firmware and can only be updated by re-flashing the firmware.** Its current value is 615 corresponding to approximately 3 volts.
pmt?	Returns current setting for maximum PMT counts as number between 0 and 1024 (technically between 0 and 615 due to limitations of the LM358N)
c#r	Channel (# number between 0 and 4) read. Returns both the stored channel position and width as follows: Channel # Position: 0450 Width: 050
c#s#	Channel (# number between 0 and 4) set (# number between 0 and 65536). Sets channel start position at which shutter will be closed. Channel is disabled by setting this to location 0, thus it is not possible to shutter on the 0th channel.
c#w#	Channel (# number between 0 and 4) width (# number between 0 and 255). As these are only stored in an 8-bit location, it is only possible to shutter a maximum width of 255 with a single channel. However, if a larger shutter is needed, multiple channels may be used sequentially.
pmt#	pmt (# number between 0 and 1024) This will set the value of the pmt max counts at which point the shutter will be held closed to protect from excessive counts. Again, technically this value must be less than 615 as per the firmware set max value corresponding to the upper limit of the LM358N. To offer the highest possible efficacy of this protection, this value should be set as low as possible without effecting data collection.
** It should be noted that the max value stored in firmware is not actually implemented by the system. It was included as a scale factor for the LabVIEW program in the case that it was decided to use a different amplifier. The controller absolutely will accept a pmt max value up to 1024 (5V) however, as the current amplifier is unable to swing closer than $\frac{2}{3}$ of the way to its positive rail, the shutter will never close if a value greater than 615 is stored via pmt#.	

Table G.1: RS232 serial specifications and command list

H Schematics

H.1 SPEX500M Grating Drive

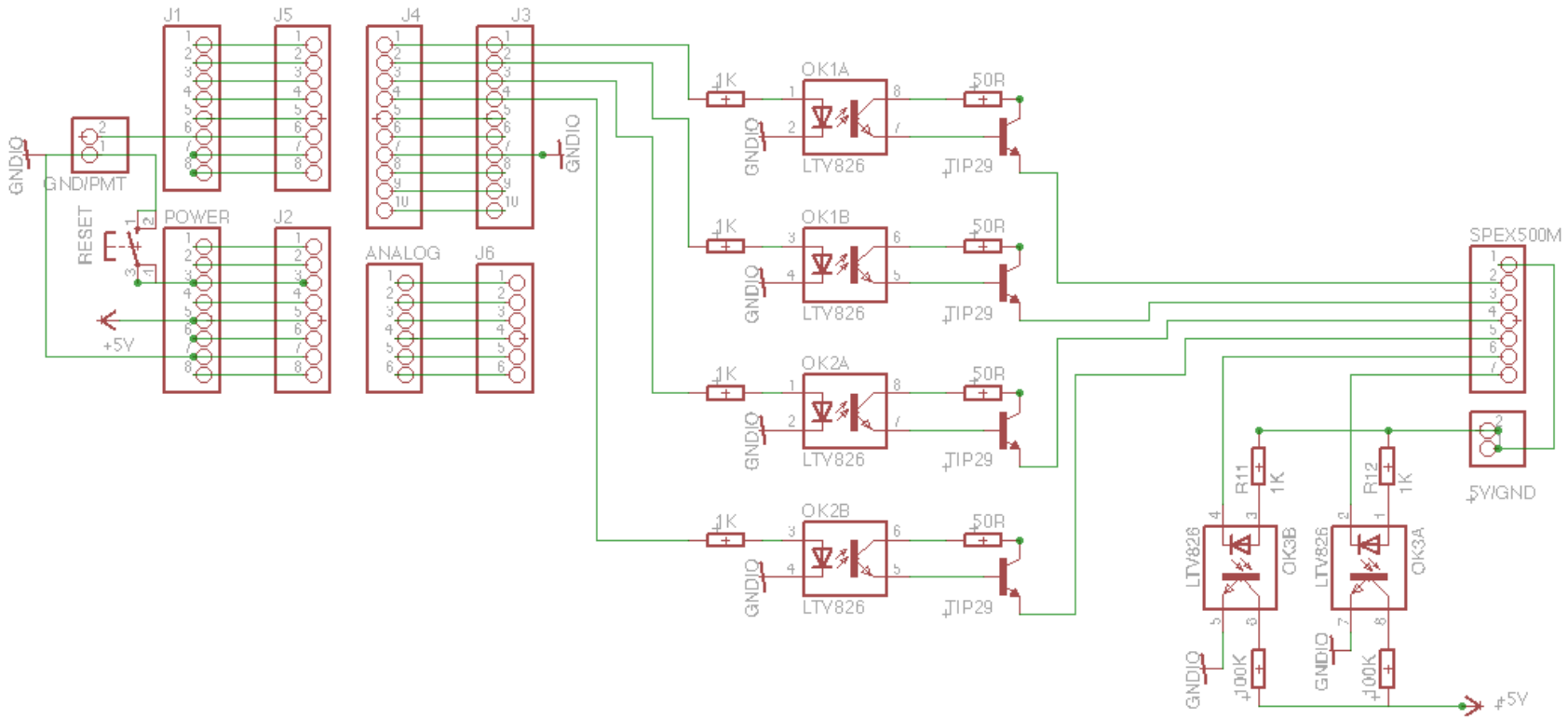


Figure H.1: Spex500M Grating Drive Schematic

H.2 Shutter Control Schematic

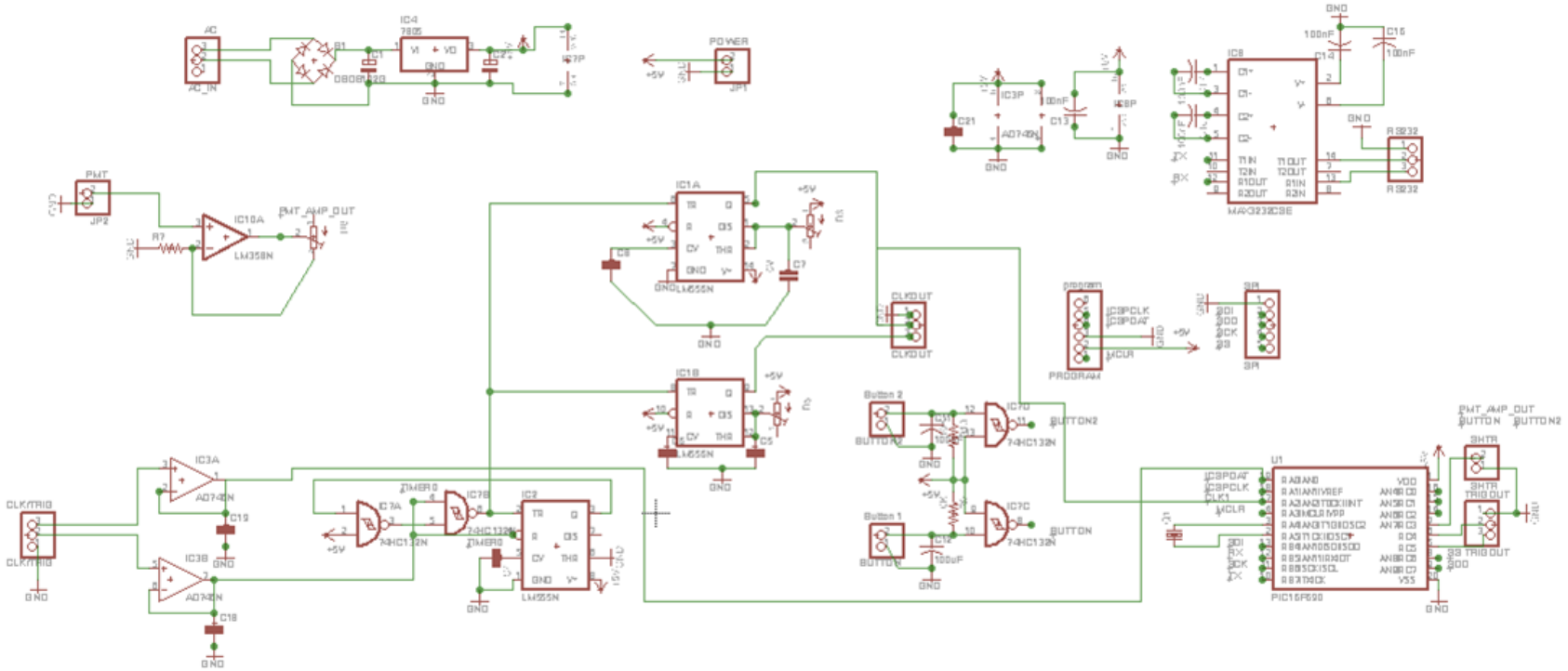


Figure H.2: Shutter Controller Schematic

I Embedded Code

I.1 Shutter Controller Code

File: main.c
Created: June 30th, 2013
Compiler: Microchip Hi-Tech C
Description: Microchip PIC16F690 shutter control main program (v5.2)

```
#include <stdio.h>
#include <stdlib.h>
#include <xc.h>
#include <htc.h>
#include "usart.h"
#include "main.h"

// PIC16F690 Configuration Bit Settings
#pragma config FOSC = HS
#pragma config WDTE = OFF
#pragma config PWRTE = OFF
#pragma config MCLRE = ON
#pragma config CP = OFF
#pragma config CPD = OFF
#pragma config BOREN = ON
#pragma config IESO = ON
#pragma config FCMEN = ON

//IO definitions
```

```
#define trigger PORTAbits.RA0          //define trigger pin
#define button PORTCbits.RC1           //define button pin
#define shutter PORTCbits.RC3          //define shutter pin
#define trigger_out_1 PORTCbits.RC4    //define trigger output
#define trigger_out_2 PORTCbits.RC5    //define trigger output
#define PMT_channel 4                  //Analog pin
#define PMT_AMP_MAX 615                //Max value of PMT op-amp

//Definitions to make the code that follows more readable.
#define time TMR0
#define time_reset TMR0=0; count=0;
#define CMD_Length 10                  //Max length of USART commands
#define debounce 200                   //debounce wait
#define close shutter=1                //define shutter close
#define open shutter=0                 //define shutter open
#define button_pressed button==1
#define button_released button==0
#define trigger_high trigger==1
#define trigger_low trigger==0

//Global Variables to handle RS232 Communications
char usartdata;                        //Last Received Byte of RS232 data
char usartcount=0;                     //Number of received bytes
char usartarray[CMD_Length];           //String received via RS232
unsigned int count=0;                  // value of the MCP Clock
char button_time=0x0;

// Global variables for channel positions, widths, and PMT max value
unsigned int max_ADC=0xFE;
unsigned int Last_ADC=0;
unsigned int Channel_0_Position;
unsigned int Channel_0_Width;
unsigned int Channel_1_Position;
unsigned int Channel_1_Width;
unsigned int Channel_2_Position;
unsigned int Channel_2_Width;
unsigned int Channel_3_Position;
unsigned int Channel_3_Width;
```

```
unsigned int Channel_4_Position;
unsigned int Channel_4_Width;

//Initialize Port I/O directions
void port_init() {
    TRISB = 0b11111111;                 // configure port B as input
    TRISA = 0b11111111;                 // configure port A as input
    TRISC = 0b00000111;                 //C0 through C2 for PMT and buttons.
}

//Load saved values from EEPROM
void init() {
    Channel_0_Position = eeprom_read(0x0)*256+eeprom_read(0x1);
    Channel_0_Width = eeprom_read(0x2);
    Channel_1_Position = eeprom_read(0x3)*256+eeprom_read(0x4);
    Channel_1_Width = eeprom_read(0x5);
    Channel_2_Position = eeprom_read(0x6)*256+eeprom_read(0x7);
    Channel_2_Width = eeprom_read(0x8);
    Channel_3_Position = eeprom_read(0x9)*256+eeprom_read(0xA);
    Channel_3_Width = eeprom_read(0xB);
    Channel_4_Position = eeprom_read(0xC)*256+eeprom_read(0xD);
    Channel_4_Width = eeprom_read(0xE);
    max_ADC=eeprom_read(0xF)*256+eeprom_read(0x10);
}

//Returns power of 10
unsigned int power_10(int power) {
    unsigned int value=1;
    for(int i=1;i<power+1;++i) value=value*10;
    return value;
}
```

```
//Error handeling for the program
void error_handel(int error_code) {
    switch (error_code) {
        case 0:
            T0IE=0;                         //disable the button pressed timer interrupt
            break;
        case 2:
            close;
            USARTWriteString("PMT Overflow Error\n");
            break;
        case 3:
            USARTWriteString("Unknown Command\n");
            usartcount=0;
            break;
        default:
            close;
            USARTWriteString("Unknown Error\n");
            break;
    }
    if(error_code!=3) {                     //UnShuttered Mode
        while(button_released) {
            CLRWDT(); //Wait for reset button to be pressed
            if(shutter == 0) Last_ADC = ADCRead(PMT_channel);
            if(Last_ADC > max_ADC) close;
        }
        for(char i=0;i<debounce;) {         //Software debounce for button
            CLRWDT();
            if (button_released) ++i;
            else i=0;
        }
        USARTWriteString("Shutter Enabled\n");
        while (trigger_high) CLRWDT();      //Resume shuttering at next trigger
        open; time_reset;
    }
}
```

```
void usart_parse() {                            //Parse out RS232 Commands
    usartcount=0; //reset bytes received
    char arraysize=0; //size of received string
    int j; //the test parameter
    for(arraysize=0;usartarray[arraysize]!='\n';++arraysize);   //determine size of string

    if(usartarray[0]=='c' && usartarray[2]=='s') {              //Syntax c#s####
        unsigned int Channel_Set_Value=0;
        for(int i=0;i<arraysize-3;++i) {
            Channel_Set_Value=Channel_Set_Value+(usartarray[arraysize-1-i]-0x30)*power_10(i);
        }
        eeprom_write(0x0+(usartarray[1]-0x30)*0x3,Channel_Set_Value » 8);
        eeprom_write(0x1+(usartarray[1]-0x30)*0x3,Channel_Set_Value & 0x00ff);
        init();
    }

    if(usartarray[0]=='m' && usartarray[1]=='a' && usartarray[2]=='x') {
        USARTWriteInt(PMT_AMP_MAX,4);
        USARTWriteByte('\n');
    }

    if(usartarray[0]=='a' && usartarray[1]=='t' && usartarray[2]=='d'){
        if(usartarray[3]=='+') USARTWriteInt(Last_ADC,4);
        else USARTWriteInt(ADCRead(PMT_channel),4);
        USARTWriteByte('\n');
    }

    if(usartarray[0]=='p' && usartarray[1]=='m' && usartarray[2]=='t') {
        if(usartarray[3]=='?') {
            USARTWriteInt(max_ADC,4);
            USARTWriteByte('\n');
        }
        else {
            unsigned int pmt_value=0;
            for(int i=0;i<arraysize-3;++i) {
                pmt_value=pmt_value+(usartarray[arraysize-1-i]-0x30)*power_10(i);
            }
```

```
        if(pmt_value > PMT_AMP_MAX) pmt_value = PMT_AMP_MAX;
        if(eeprom_read(0xF) != (pmt_value»8)) eeprom_write(0xF,pmt_value»8);
        if(eeprom_read(0x10) != (pmt_value & 0x00ff)) {
            eeprom_write(0x10,pmt_value & 0x00ff);
        }
        init();
    }
}

if(usartarray[0]=='c' && usartarray[2]=='w') {
    unsigned int Channel_Width=0;          //syntax: c#w###
    for(int i=0;i<arraysize-3;++i) {
        Channel_Width=Channel_Width+(usartarray[arraysize-1-i]-0x30)*power_10(i);
    }
    eeprom_write(0x2+(usartarray[1]-0x30)*0x3,Channel_Width);
    init();
}

if(usartarray[0]=='c' && usartarray[2]=='r') {
    unsigned int Channel_Set_Value=0;    //syntax c#r
    unsigned int Channel_Width_Value=0;
    switch (usartarray[1]){
        case '0':
            Channel_Set_Value=eeprom_read(0x0)*256+eeprom_read(0x1);
            Channel_Width_Value=eeprom_read(0x2);
            break;
        case '1':
            Channel_Set_Value=eeprom_read(0x3)*256+eeprom_read(0x4);
            Channel_Width_Value=eeprom_read(0x5);
            break;
        case '2':
            Channel_Set_Value=eeprom_read(0x6)*256+eeprom_read(0x7);
            Channel_Width_Value=eeprom_read(0x8);
            break;
        case '3':
            Channel_Set_Value=eeprom_read(0x9)*256+eeprom_read(0xA);
            Channel_Width_Value=eeprom_read(0xB);
            break;
```

```
            case ’4’:
                Channel_Set_Value=eeprom_read(0xC)*256+eeprom_read(0xD);
                Channel_Width_Value=eeprom_read(0xE);
                break;
            default:
                USARTWriteString("Select Channel 1 - 5\n");
                break;
        }
        USARTWriteString("Channel ");
        USARTWriteByte(usartarray[1]);
        USARTWriteString(" Position: ");
        USARTWriteInt(Channel_Set_Value,4);
        USARTWriteString(" Width: ");
        USARTWriteInt(Channel_Width_Value,3);
        USARTWriteByte(’\n’);
    }
}
unsigned int ADCRead(unsigned char ch) {
    unsigned int ADC_result=0;
    if(ch>13) return 0;                     //Invalid Channel
    ADCON0=ADCON0 & 0b11000011;
    ADCON0=ADCON0 | (ch«2);                 //Select ADC Channel
    GO_DONE=1;                              //Start conversion
    while(GO_DONE);                         //wait for the conversion to finish
    ADC_result = ADRESH;
    ADC_result = (ADC_result«8)|ADRESL;
    return ADC_result;
}
void interrupt tc_int(void) {
    if (RCIE && RCIF) {
        usartdata=USARTReadByte();
        if(usartcount==CMD_Length) error_handel(3);
        else {
            usartarray[usartcount]=usartdata;
            ++usartcount;
            if(usartdata==’\n’) usart_parse();
        }
```

```
        RCIF=0;
        return;
    }
    if(RABIF && RABIE) {
    trigger_out_1=trigger;
    trigger_out_2=trigger;
    time_reset;
    RABIF=0;
    return;
    }
    if(T2IF && T2IE) {                  //Enter open shutter mode
    if(button_time!=0xFA) ++button_time;
    T2IF=0;
    return;
    }
}

void main(int argc, char** argv) {
    USARTInit();                        //Initialize RS232
    ADON=1;                             //switch on the ADC module
    RCIE=1;                             //Enable USART Rx Interrupt
    port_init();
    ANSEL=0b00010000;                   //AN4 analog pin
    ANSELH=0b00000000;                  //ADConverter options
    ADCON0=0b10010001;
    ADCON1=0b00100000;                  //TAD for minimum for 20Mhz
    IOCA=0b00000001;                    //enable A0 interrupt for trigger
    IOCB=0b00000000;                    //disable port B interrupt
    RABIE=1;                            //Enable RB6 interrupt for trigger
    INTCONbits.GIE=1;                   //enable global interrupts
    INTCONbits.INTE=0;
    INTCONbits.PEIE=1;
    TMR1CS=1;                           //Set TMR1 as counter
    T1OSCEN=0;
    T1CON=0b00000111;
    time_reset;
    T0IE=0;                             //Timer 0 interrupt disabled
    T0CS=1;                             //External Pin Counter Mode
```

```
OPTION_REGbits.T0SE=0;
T2CON=0b01111011;                   //Prescaler 1:16 Postscaler 1:16 Timer OFF
T2IE=1;                             //Timer 2 interrupt enabled
init();                             //read eeprom values

close;                              //Shutter Closed on Startup
while(button_released) CLRWDT();
for(char i=0;i<debounce;) {
   CLRWDT();
   if (button_released) ++i;
   else i=0;
}
while (trigger_high) CLRWDT();

for(;;) {                           //Begin Main Loop
   CLRWDT();
   count = count + TMR0; TMR0=0;
   Last_ADC = ADCRead(PMT_channel);
   if(Last_ADC>max_ADC) error_handel(2);
   if(button_pressed) {             //for unshuttered mode
      close;                        //Close Shutter
      T2CONbits.TMR2ON=1;
      button_time=0x0;
      for(char i=0;i<debounce;) {
         CLRWDT();
         if(button_time==0xFA) open;    //reopen shutter
         if (button_released) {
            ++i;
            button_time=0;
         }
         else i=0;
      }
      T2CONbits.TMR2ON=0;
      error_handel(0);              //runs unshuttered
   }
```

```
        if(count>=Channel_0_Position && count<=Channel_0_Position+Channel_0_Width
        && Channel_0_Position!=1) close;
        else if (count>=Channel_1_Position && count<Channel_1_Position+Channel_1_Width
        && Channel_1_Position!=1) close;
        else if (count>=Channel_2_Position && count<Channel_2_Position+Channel_2_Width
        && Channel_2_Position!=1) close;
        else if (count>=Channel_3_Position && count<Channel_3_Position+Channel_3_Width
        && Channel_3_Position!=1) close;
        else if (count>=Channel_4_Position && count<Channel_4_Position+Channel_4_Width
        && Channel_4_Position!=1) close;
        else open;
    }
}
```

I.2 TGP Stage Driver Code

File: TGP_Stage_Driver.ino
Created: July 6th, 2016
Compiler: Arduino IDE 1.0.5-r2
Description: Arduino-Uno embedded code for stage driver

```
#include <LiquidCrystal.h>
#include <MSTimer2.h>

class motor {
    public:
    void initialize(int d, int s);          //d = direction pin and s = step
    int location();                         //return current location
    void zero();                            //Set zero at current location
    void go(long int);                      //goto specified location
    int dir();                              //return direction pin
    void setStepDelay(int);                 //set the delay between steps in ms
    int stepDelay();                        //return current step delay
    void setSlack(long int);                //set motor slack

    private:
    long int x;                             //Stepper Location
    int dirPin;                             //Direction selection input pin
    int stepPin;                            //Step input pin
    int step_Delay;                         //delay between steps
    int slack;                              //backslash overshoot
    void over();                            //overshoot to adjust for backlash
    void oneStep();                         //motor takes one step
    void slackAdjust();                     //adjust for mechanical slack
};

void motor::setSlack(long int x) {
    slack = x;
}

int motor::stepDelay() {
```

```
    return step_Delay;
}

void motor::initialize(int d, int s) {
    dirPin = d;
    stepPin = s;
    x = 0;
    step_Delay = 1000;
    slack = 0;
    pinMode(dirPin, OUTPUT);
    pinMode(stepPin, OUTPUT);
    digitalWrite(dir(), LOW);
    digitalWrite(stepPin, LOW);
}

long int motor::location() {
    return x;
}

void motor::zero() {
    x = 0;
}

int motor::dir() {
    return dirPin;
}

void motor::over() {
    if(slack > 1) digitalWrite(dir(), HIGH);
    else digitalWrite(dir(), LOW);
    for(long int y; y < slack; ++y) {
        oneStep();
    }
}

void motor::go(long int y) {
```

```
    bool slackflag = 0;
    if(x < y) {
        digitalWrite(dir(), HIGH);
        if(slack > 0) slackflag = 1;
    }
    else {
        digitalWrite(dir(), LOW);
        if(slack < 0) slackflag = 1;
    }
    while(x != y) {
        if(x > y) ++x;
        else --x;
        oneStep();
    }
    if(slackflag) slackAdjust();
}

void motor::setStepDelay(int del) {
    digitalWrite(stepPin, HIGH);
    digitalWrite(stepPin, LOW);
    delayMicroseconds(step_Delay);
}

//**************************************
// Main Program Section
//**************************************
//Define Buttons
#define xup 8
#define xdown 10
#define yup 6
#define ydown 7
#define button 9
#define reset A5
#define motor_speed A4

motor xmotor;
motor ymotor;
```

```
LiquidCrystal lcd(13, 11, A0, A1, A2, A3);  //lcd(RS,E,D4,D5,D6,D7) RW-Gnd

//store location of up to 10 rubies in 3x10 array
//0th row 0 for unused or 1 for used
//1st row x location
//2nd row y location
long int rubys[3][10];
int stepDelay = 200 + 1024;                 //min step delay for motor 532 added to slow motor
int setFlag=0;                              //flag 1 to print "SET" on LCD or 2 print current ruby
const int motorSlack = 1000;                //motor slack
char currentRuby=0;

void setRuby(long int x, long int y) {
int i = 0;
while(rubys[0][i] == 1 && i < 10) ++i;      //find non-activated ruby location
while(digitalRead(button) == 0);
rubys[0][i] = 1;                            //ruby set for use
rubys[1][i] = x;                            //set x location
rubys[2][i] = y;                            //set y location
}

void set() {
    setFlag = 1;
    xmotor.setSlack(0);
    ymotor.setSlack(0);
```

```
    while(digitalRead(button) == 0);            //wait for button release
    delay(500);                                 //debounce
    while(digitalRead(reset) == 1 && rubys[0][9] == 0) {
        //Directional Control from Joystick
        if(digitalRead(xup) == 0) {
            xmotor.go(xmotor.location() + 1);
        }
        if(digitalRead(xdown) == 0) {
            xmotor.go(xmotor.location() - 1);
        }
        if(digitalRead(yup) == 0) {
            ymotor.go(ymotor.location() + 1);
        }
        if(digitalRead(ydown) == 0) {
            ymotor.go(ymotor.location() - 1);
        }
        //Button to set Ruby position
        if(digitalRead(button) == 0) {
            setRuby(xmotor.location(), ymotor.location());
            while(button==0);
            delay(500);                         //debounce
        }
    }
    setFlag = 0;
    xmotor.setSlack(motorSlack);
    ymotor.setSlack(motorSlack);
}

void motorLCD() {
    int percentMotor = 100 - (int)((double)(xmotor.stepDelay() - stepDelay)/1024*100);
    lcd.clear();
    lcd.print("Motor Speed: ");
    lcd.print(percentMotor);
}

void stepTo(char x) {
```

```
    if(x > 47 && x < 58) {                  //character between 0 and 9
        x -= 48;                            //convert from ASCII to numerical value binary
        if(rubys[0][x] == 1) {              //check to see if location is set
            xmotor.go(rubys[1][x]);
            ymotor.go(rubys[2][x]);
            currentRuby = x + 48;
            setFlag = 2;
        }
    }
}

void zeroRubys() {
    for(int i = 0; i < 10; ++i) {
        for(int k = 0; k<3; ++k) {
            rubys[k][i] = 0;
        }
    }
}

void motorSpeed() {
    int currentDelay = xmotor.stepDelay();
    int newDelay = analogRead(motor_speed) + stepDelay;
    if(abs(currentDelay-newDelay) > 10) {
        xmotor.setStepDelay(newDelay);
        ymotor.setStepDelay(newDelay);
        motorLCD();
        delay(1000);
    }
    else
        lcd.clear();
        lcd.print("X: ");
        lcd.print(xmotor.location());
        lcd.setCursor(0,1);
        lcd.print("Y: ");
        lcd.print(ymotor.location());
```

```
        if(setFlag == 1) {
            lcd.setCursor(11,0);
            lcd.print("*SET*");
            lcd.setCursor(11,1);
            int i = 0;
            while(rubys[0][i] == 1 && i < 10) ++i;
            lcd.print(i);
        }

        if(setFlag == 2) {
            lcd.setCursor(15,0);
            lcd.print(currentRuby);                 //setflag set to ruby location
        }
    }
}

void setup() {
    //Initialize Serial and LCD;
    Serial.begin(9600);
    pinMode(12,OUTPUT);
    digitalWrite(12,LOW); //ground R/W
    lcd.begin(16, 2);

    //TIMER2 USED FOR MOTOR SPEED CONTROL ANALOG READ
    MsTimer2::set(200, motorSpeed);
    MsTimer2::start();

    //intialize ruby locations
    zeroRubys();

    //Initialize Motor pin9 Direction pin8 Step for x-axis
    xmotor.initialize(2,3);
    xmotor.setSlack(motorSlack);
    ymotor.initialize(4,5);
    ymotor.setSlack(motorSlack);
```

```
    //Set Button Pin modes
    pinMode(reset,INPUT);                   //A0 Set Button
    pinMode(button,INPUT);                  //A1 Button on Controller
    pinMode(xup,INPUT);                     //A2 X-axis Positive
    pinMode(xdown,INPUT);                   //A3 X-axis Negative
    pinMode(yup,INPUT);                     //A4 Y-axis Positive
    pinMode(ydown,INPUT);                   //A5 Y-asix Negative
    pinMode(A5,INPUT);                      //A5 analog input for speed control
}

void printIDN() {
    Serial.println("TGP microscope stage driver");
    Serial.println("————————————————");
    Serial.println("0-9 Moves to Ruby Location");
    Serial.println("Z Erases stored Ruby Locations");
    Serial.println("S Returns 0-9 for ruby last selected");
    Serial.println("N Returns Number of Set rubies");
}

void returnstate() {
    Serial.print(currentRuby);
}

void returnNumber() {
    int i = 0;
    for(int k = 0; k < 10; ++k) {
        i += rubys[0][i];
    }
    Serial.print(i);
}

void loop() {
```

```
    //serial handler
    if(Serial.available()) {
        byte character=Serial.read();
        if(character==’I’ || character==’i’) printIDN();
        else if(character == ’Z’ || character == ’z’) zeroRubys();
        else if(character == ’S’ || character == ’s’) returnstate();
        else if(character == ’N’ || character == ’n’) returnNumber();
        else stepTo(character);
    }

    //Button Handler
    if(digitalRead(button) == 0) set();

    if(digitalRead(reset) == 0) {
        MsTimer2::stop();
        lcd.clear();
        lcd.print(" Hold Button");
        lcd.setCursor(0,1);
        lcd.print(" to Clear");

        //time button for reset
        long int i=0;
        while(digitalRead(reset) == 0 && i < 1000000) ++i;
        if(i == 1000000) {
            lcd.clear();
            lcd.print("Ruby Locations");
            lcd.setCursor(0,1);
            lcd.print(" Cleared");
            zeroRubys();
        }
        while(digitalRead(reset) == 0);          //hold till release
        lcd.clear();
        MsTimer2::start();
    }
}
```

I.3 SPEX 500M Controller Code

File: Spex500M_v2.ino
Created: October 31st, 2014
Compiler: Arduino IDE 1.0.5-r2
Description: Grating motor control and PMT counter

```
#include <EEPROMex.h>
#include <EEPROMVar.h>
#include <EEPROMex.h>                    //EEPROM extended functionality

#define serial_speed 9600                //RS232 Serial Communication Speed

//Arduino Uno digital ports for Travel Limit Switches
#define high_limit 6
#define low_limit 7

//Port Values for Stepper Positions
#define motor_port PORTB
char position_1=B00000101;
char position_2=B00000001;
char position_3=B00001001;
char position_4=B00001000;
char position_5=B00001010;
char position_6=B00000010;
char position_7=B00000110;
char position_8=B00000100;
```

```
char step_delay_ms=2;                   //2ms delay between motor steps
boolean echo=false;                     //Initialize with Echo Off for Communication
boolean limit=true;                     //Initialize with limit switches enabled
boolean stringComplete = false;         //indicate line feed for end of command
String serialRecieved = "";             //a string to hold incoming commands
long int location;                      //Current Location of Motor in steps
long int destination;                   //Current Destination in steps
char current_step = 0;                  //Value 0-7 holds current step location
char slk = 0;                           // 1 if moving in negative direction for slack adjust
long int slk_steps = 1000;              //Number of steps for slack adjust (About 15 Angstroms)
unsigned long int counts=0;             //PMT Counts
String inputString= "";                 //received serial commands
char incomingByte=0;                    //length of inputString

void SetupTimer1(){
   TCCR1A = 0;                          // normal counting mode
   TCNT1 = 0;                           // clear the timer count
   TCCR1B = TCCR1B | B00000111;         //Set for external clock
   TIMSK1=1;                            //Enable Timer1 interrupt for overflow handling
}

//Print Instruction and ID when prompted or on startup
void idn() {
```

```
    Serial.println("");
    Serial.println("SPEX 500M");
    Serial.println("Stepper Control Board");
    Serial.println("Original Design by:");
    Serial.println("THE Jennifer Stabach");
    Serial.println("");
    Serial.println("Command Summary");
    Serial.println("———————");
    Serial.println("IDN?: Menu");
    Serial.println("?: poll current location");
    Serial.println("STP?: poll motion status");
    Serial.println("STP: stop motion");
    Serial.println("S,#: move to #");
    Serial.println("DLY,#: step delay # ms");
    Serial.println("E: toggle echo");
    Serial.println("0: release motor");
    Serial.println("+#: move # steps");
    Serial.println("-#: move # steps");
    Serial.println("ZR: Set Location Zero");
    Serial.println("ZR,#: Set Location #");
    Serial.println("EP: Save Location to EPROM");
    Serial.println("V?: Read Analog Voltage");
    Serial.println("LMT: Enable/Dissable Limit Switches");
    Serial.write(0x4);                          //Labview Termination (EOT)
}

void set_step(char current_step) {
```

```
    switch(current_step) {
        case 0:
            motor_port=position_1;
            break;
        case 1:
            motor_port=position_2;
            break;
        case 2:
            motor_port=position_3;
            break;
        case 3:
            motor_port=position_4;
            break;
        case 4:
            motor_port=position_5;
            break;
        case 5:
            motor_port=position_6;
            break;
        case 6:
            motor_port=position_7;
            break;
        case 7:
            motor_port=position_8;
            break;
    }
    delay(step_delay_ms);
}

void setup() {
```

```
    SetupTimer1();
    pinMode(high_limit,INPUT);              //limit switch pin set to input
    pinMode(low_limit,INPUT);               //limit switch pin set to input
    if(position_1 = B00010100) DDRD = B00111110;    //Old Board
    else DDRD = B00000010;                  //Board updated for TMR1 Counter
    DDRB = DDRB | B00001111;                //Motor drive pins for board with TMR1 Counter
    Serial.begin(serial_speed);             //init serial port
    serialRecieved.reserve(20);             //reserve 20 bytes for the serialRecieved

    //Load Last Known Location from EEPROM
    int i = EEPROM.read(0);                 //Location of first good EEPROM block
    location = (signed long int) EEPROM.readLong((i+1)*sizeof(long));
    destination = location;
    current_step = location % 8;            //set current step
    set_step(current_step);                 //Grab Motor on current step
}

void respond()
                                            //degbug operations
    if(serialRecieved=="1\n") motor_port=B00000101;
    if(serialRecieved=="2\n") motor_port=B00001001;
    if(serialRecieved=="3\n") motor_port=B00001010;
    if(serialRecieved=="4\n") motor_port=B00000110;
    if(serialRecieved=="0\n") motor_port=B00000000;

    //return Counts
    if(serialRecieved=="k\n") Serial.println(counts + TCNT1);
```

```
//Returns count rate in counts/second
if(serialRecieved=="RT\n") {
    unsigned long int rate;
    TCNT1=0;
    counts=0;
    delay(100);
    rate = (counts + TCNT1)/.1;
    Serial.println(rate);
    Serial.write(0x4);                  //Labview Termination
}

//Respond IDN?
if(serialRecieved=="IDN?\n" || serialRecieved=="idn?\n") idn();

if(serialRecieved=="0\n") motor_port=0;

//Enable/Dissable Limit Switches
if(serialRecieved=="LMT\n") {
    Serial.print("Limit Switches ");
    if (limit) {
        limit=false;
        Serial.println("Dissabled");
        Serial.write(0x4);              //Labview Termination
    }
    else {
        limit=true;
        Serial.println("Enabled");
        Serial.write(0x4);              //Labview Termination
    }
}

//PMT Counts over provided time in ms
if(serialRecieved[0]=='C' && serialRecieved[1]==',') {
    serialRecieved[0]='0';
    serialRecieved[1]='0';
    int x = serialRecieved.toInt();
    TCNT1=0;
```

```
    counts=0;
    delay(x);
    Serial.println(counts + TCNT1);
    Serial.write(0x4);                  //Labview Termination
}

//Analog Voltage Return from pin 0
if(serialRecieved=="V?\n") {
    Serial.println(analogRead(0));
    Serial.write(0x4);                  //Labview Termination
}

if(serialRecieved=="VP\n") {
    Serial.print(analogRead(0));
    Serial.print(",");
    Serial.println(location);
    Serial.write(0x4);                  //Labview Termination
}

//Respond with Current Position
if(serialRecieved[0]=='?') {
    Serial.println(location);
    Serial.write(0x4);                  //Labview Termination
}

//Set Current Location to Zero
if(serialRecieved=="ZR\n") {
    while(current_step != 0) {
        ++destination;
        step_motor();
    }

    destination=0;
    location=0;
}

//Set Location to Zero
```

```
if(serialRecieved[0]=='Z' && serialRecieved[1]=='R' && serialRecieved[2]==',') {
    long int x;
    for(char i=0;i<3;++i) serialRecieved[i]='0';
    //Prep string for conversion to integer
    if(serialRecieved[3]=='-') {
        x=-1;
        serialRecieved[3]='0';
    }
    else x = 1;
    x = x*serialRecieved.toInt();
    location = x;
    destination = x;
}

//Save Location to EEPROM
if(serialRecieved=="EP\n") {
    int i=EEPROM.read(0);                   //starting memory location used
        if(i > 240) {                       //warning if EEPROM gets old
        Serial.println("EEPROM APPROACHING FAILER");
        Serial.println("Replace ATmega328");
        Serial.write(0x4); //Labview Termination
    }
    for(char k=0;k==0;i++) {                //This will store long in EEPROM
        if(i < 255) {
            EEPROM.updateLong((i+1)*sizeof(long),location);
            if(EEPROM.readLong((i+1)*sizeof(long))==location) ++k;
        }
        else {
            Serial.println("Memory Write Error");
            Serial.println("!!!REPLACE EEPROM!!!");
            Serial.write(0x4);              //Labview Termination
            ++k;
        }
    }
    –i;                                     //i updated in loop thus decrement back

    //Return Error code over RS232 if EEPROM Bad Write
```

```
    //And update The EEPROM 0 location
    if(EEPROM.update(0,i)!=0) {
        Serial.println("EEPROM LOCATION BAD");
        Serial.println("LOCATION POINTER ITERATED");
        Serial.write(0x4);                  //Labview Termination
    }
    //Return EEPROM Value and Location Written
    Serial.print("EEPROM Location: ");
    Serial.println((signed long int) EEPROM.readLong((i+1)*sizeof(long)));
    Serial.print("Memory Chunk: ");
    Serial.println(i);
    Serial.write(0x4);                  //Labview Termination
}
//move steps
if(serialRecieved[0]=='+')
    serialRecieved[0]='0';
    destination += serialRecieved.toInt();
}
if(serialRecieved[0]=='-') {
    serialRecieved[0]='0';
    destination -= ((serialRecieved.toInt()+slk_steps));
    slk=1;
}
//Set destination
if(serialRecieved[0]=='S' && serialRecieved[1]==',') {
    serialRecieved[0]='0';              //replace first two non-numerical values
    serialRecieved[1]='0';              //so toInt parse will function
    if(serialRecieved[2]=='-') {
        serialRecieved[2]='0';
        destination = -1;
    }
    else destination = 1;
    destination = destination*serialRecieved.toInt();
    if((destination - location) < 0) {
        destination -= slk_steps;
        slk=1;
    }
```

```
}

//Set Scan Speed
if(serialRecieved[0]=='D' && serialRecieved[1]=='L' && serialRecieved[2]=='Y' &&
serialRecieved[3]==',') {
    for(char i=0;i<4;++i) serialRecieved[i]='0';          //replace first 3 non-numerical values
    int spd = serialRecieved.toInt();          //scan out speed multiplyer
    if((spd < 100) && (spd > 0)) {
        step_delay_ms = spd;
        Serial.print("Delay = ");
        Serial.print(spd);
        Serial.println(" ms");
        Serial.write(0x4);                     //Labview Termination
    }
    else Serial.println("Range 3 to 99");
    Serial.write(0x4);                         //Labview Termination
}
//Respond to motion query
if(serialRecieved=="STP?\n") {
    if((location==destination) && (slk==0)) {
        Serial.println('Y');
        Serial.write(0x4);                     //Labview Termination
    }
    else {
        Serial.println('N');
        Serial.write(0x4);                     //Labview Termination
    }
}
if(serialRecieved=="STP\n") {
    destination = location;
    slk=0;
}
//Respond to Set Echo
if(serialRecieved=="E\n") {
    if(echo) {
        echo=false;
        Serial.println("ECHO OFF");
```

```
            Serial.write(0x4);                  //Labview Termination
        }
        else {
            echo=true;
            Serial.println("ECHO ON");
            Serial.write(0x4);                  //Labview Termination
        }
    }
    //Last thing to do is reset the incoming string to NULL and reset the flag
    serialRecieved = "";
    stringComplete = false;                     // reset the string complete
}
```